



Cochrane
Library

Cochrane Database of Systematic Reviews

Vitamin B12 supplementation for growth, development, and cognition in children (Review)

Larvie DY, Güitrón Leal CE, Venkatramanan S, Palma Molina XE, Steel LB, Kuriyan R, Finkelstein JL

Larvie DY, Güitrón Leal CE, Venkatramanan S, Palma Molina XE, Steel LB, Kuriyan R, Finkelstein JL.
Vitamin B12 supplementation for growth, development, and cognition in children.
Cochrane Database of Systematic Reviews 2026, Issue 6. Art. No.: CD015264.
DOI: [10.1002/14651858.CD015264.pub2](https://doi.org/10.1002/14651858.CD015264.pub2).

www.cochranelibrary.com

Vitamin B12 supplementation for growth, development, and cognition in children (Review)

Copyright © 2026 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

WILEY

TABLE OF CONTENTS

ABSTRACT	1
PLAIN LANGUAGE SUMMARY	3
SUMMARY OF FINDINGS	5
BACKGROUND	12
OBJECTIVES	13
METHODS	13
RESULTS	16
Figure 1.	17
Figure 2.	20
Figure 3.	21
Figure 4.	22
Figure 5.	23
Figure 6.	24
Figure 7.	25
Figure 8.	26
Figure 9.	27
Figure 10.	28
DISCUSSION	28
AUTHORS' CONCLUSIONS	30
SUPPLEMENTARY MATERIALS	31
ADDITIONAL INFORMATION	31
REFERENCES	33
ADDITIONAL TABLES	42

[Intervention Review]

Vitamin B12 supplementation for growth, development, and cognition in children

Doreen Y Larvie^{1,2}, Cristina E Güitrón Leal¹, Sudha Venkatramanan^{1,2}, Ximena E Palma Molina^{1,2}, Lili B Steel^{1,2}, Rebecca Kuriyan³, Julia L Finkelstein^{1,2,3,4a}

¹Division of Nutritional Sciences, Cornell University, Ithaca, New York, USA. ²Cornell Joan Klein Jacobs Center for Precision Nutrition and Health, Cornell University, Ithaca, New York, USA. ³Division of Nutrition, St John's Research Institute, Bengaluru, India. ⁴Cochrane Center, Cornell University, Ithaca, New York, USA

^aORCID iD: <https://orcid.org/0000-0001-9512-5559>

Contact: Julia L Finkelstein, jfinkelstein@cornell.edu.

Editorial group: Cochrane Central Editorial Service.

Publication status and date: New, published in Issue 6, 2026.

Citation: Larvie DY, Güitrón Leal CE, Venkatramanan S, Palma Molina XE, Steel LB, Kuriyan R, Finkelstein JL. Vitamin B12 supplementation for growth, development, and cognition in children. *Cochrane Database of Systematic Reviews* 2026, Issue 6. Art. No.: CD015264. DOI: [10.1002/14651858.CD015264.pub2](https://doi.org/10.1002/14651858.CD015264.pub2).

Copyright © 2026 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

ABSTRACT

Rationale

Vitamin B₁₂ deficiency is an important public health problem in children, and has been associated with anaemia and impaired growth, cognition, and development. Vitamin B₁₂ deficiency may develop in children due to lower content in breast milk, low dietary intake, and as requirements increase with age. Vitamin B₁₂ supplementation may confer benefits to child growth and development, due to its role in DNA synthesis and methylation, red blood cell formation, and folate metabolism. However, no systematic reviews have been conducted to examine the effects of vitamin B₁₂ supplementation in children.

Objectives

To determine the effects of vitamin B₁₂ supplementation on growth, development, and cognition in children less than 12 years of age.

Search methods

We searched CENTRAL, MEDLINE, Embase, 12 other databases, and two trial registries on 23 September 2025. We screened reference lists of studies.

Eligibility criteria

We included randomised controlled trials evaluating the effects of oral vitamin B₁₂ supplementation, alone or in combination with other micronutrients, in children < 12 years of age. We excluded cross-over trials, trials targeting pregnant or breastfeeding women, or participants with critical illnesses, and interventions delivered by intravenous/intramuscular injections.

Outcomes

Critical outcomes were vitamin B₁₂ status, anaemia, haemoglobin, growth, cognitive function, developmental outcomes, and any adverse effects. Important outcomes included megaloblastic anaemia, body composition, health-related quality of life, morbidity, and all-cause mortality.

Risk of bias

We used Cochrane RoB 2 to evaluate bias.

Vitamin B12 supplementation for growth, development, and cognition in children (Review)

Copyright © 2026 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Synthesis methods

We conducted meta-analyses using fixed-effect models. If there was substantial heterogeneity, we used random-effect models. When meta-analyses were not conducted, we described findings from individual trials, and considered synthesis and presentation of findings using other methods.

We used GRADE to evaluate the certainty of evidence for critical outcomes.

Included studies

We included 16 trials (4083 children; 2 days to 11 years) in 10 countries, including India, Japan, Thailand, Nepal, United Kingdom, Italy, Spain, the United States, El Salvador, and Guatemala. Follow-up periods for outcomes in most studies ranged from five weeks to 31 months, with one study extending follow-up to nine years. Duration of vitamin B₁₂ supplementation varied, ranging from 28 days to 31 months, with doses from 0.9 µg/day to 150 µg/day.

Nine trials, including 2391 children, contributed data to meta-analyses.

Synthesis of results

Supplementation with vitamin B₁₂ alone versus placebo

There may be little to no difference between groups in height/length (mean difference (MD) -0.19 cm, 95% confidence interval (CI) -2.44 to 2.06; 2 trials, 143 children; low-certainty evidence).

There may be little to no difference between groups in cognitive function (e.g. attention) (standardised mean difference (SMD) 0.09, 95% CI -0.10 to 0.28; 2 trials, 438 children, low-certainty evidence).

There may be little to no difference between groups in motor skill development (6 to 9 years, NEUROPSYchological assessment, Second Edition (NEPSY-II); 1 trial, 403 children, low-certainty evidence).

Vitamin B₁₂ supplementation in children reduces vitamin B₁₂ deficiency (absolute risk difference (ARD) 56 fewer per 1000, 95% CI 68 fewer to 24 fewer; 1.8% versus 7.6%; risk ratio (RR) 0.27, 95% CI 0.11 to 0.69; 3 trials, 642 children; high-certainty evidence), and probably increases total vitamin B₁₂ concentrations (MD 68.53 pmol/L, 95% CI 41.83 to 95.23; 7 trials, 705 children; moderate-certainty evidence), compared to placebo, with an average increase of 68.5 pmol/L.

There may be little to no difference between groups in anaemia (ARD 21 more per 1000, 95% CI 26 fewer to 191 more; 9.1% versus 4.5%; RR 1.48, 95% CI 0.41 to 5.27; 2 trials, 144 children; low-certainty evidence).

No trials reported the occurrence of any adverse effects.

Supplementation with vitamin B₁₂ alone versus no intervention

The evidence is very uncertain about the effects of vitamin B₁₂ on height/length (2 trials, 70 children; very low-certainty evidence), compared to no intervention.

No trials reported data for the other critical outcomes.

Supplementation with vitamin B₁₂ and other micronutrients, compared to the same formulation without vitamin B₁₂

There may be little to no difference between groups in height/length (MD -0.22 cm, 95% CI -0.65 to 0.20; 2 trials, 608 children; low-certainty evidence).

There may be little to no difference between groups in cognitive function (e.g. attention) (SMD 0.00, 95% CI -0.13 to 0.13; 3 trials, 956 children; low-certainty evidence).

There may be little to no difference between groups in development outcomes (e.g. motor skill development) (SMD -0.05, 95% CI -0.18 to 0.08; 2 trials, 960 children; low-certainty evidence).

Children who received vitamin B₁₂ supplementation and other micronutrients may have little to no difference in vitamin B₁₂ deficiency (ARD 2 fewer per 1000, 95% CI 7 fewer to 32 more; 0.7% versus 0.8%; RR 0.77, 95% CI 0.12 to 5.16; 2 trials, 534 children; low-certainty evidence), or total vitamin B₁₂ concentrations (MD 34.32 pmol/L, 95% CI -31.13 to 99.77; 3 trials, 731 children; low-certainty evidence), compared to supplements without vitamin B₁₂.

There may be little to no difference between groups in anaemia (21.7% versus 25.0%; 1 trial, 35 children; low-certainty evidence).

There may be little to no difference between groups in any adverse effects (9.2% versus 5.9%; 1 trial, 574 children; low-certainty evidence).

Vitamin B12 supplementation for growth, development, and cognition in children (Review)

The certainty of evidence for critical outcomes ranged from high to very low; we downgraded for imprecision due to small sample sizes or few events, risk of bias in outcome measurement, randomisation methods, selective reporting bias, or inconsistency due to substantial statistical heterogeneity. Few studies reported growth, cognition, development, anaemia, or any adverse effects.

Authors' conclusions

Vitamin B₁₂ supplementation (alone or when supplemented with other micronutrients) in children may have little to no effect on growth (i.e. height/length), cognitive function (e.g. attention), development outcomes (e.g. motor skill development), or anaemia, compared to placebo, no intervention, or the same supplements without vitamin B₁₂. Vitamin B₁₂ supplementation in children reduces vitamin B₁₂ deficiency and probably improves total vitamin B₁₂ concentrations, compared to placebo. However, vitamin B₁₂ supplementation with other micronutrients may result in little to no difference in vitamin B₁₂ deficiency and total vitamin B₁₂ concentrations, compared to the same formulation without vitamin B₁₂.

Funding

This review had no dedicated funding.

Registration

Protocol available via DOI 10.1002/14651858.CD015264.

PLAIN LANGUAGE SUMMARY

What are the effects of vitamin B₁₂ supplementation for children less than 12 years of age?

Key messages

- Vitamin B₁₂ supplementation reduces the risk of vitamin B₁₂ deficiency, and probably improves vitamin B₁₂ levels in children less than 12 years of age, compared to placebo (a substance that looks the same as the intervention/treatment but does not contain any active ingredient) (that is, children who did not take vitamin B₁₂ supplements).
- Vitamin B₁₂ supplements, with other micronutrients (important vitamins and minerals the body needs in small amounts to function), may result in little to no difference in vitamin B₁₂ deficiency or total vitamin B₁₂ levels, compared to children who received the same supplements without vitamin B₁₂.
- The impact of vitamin B₁₂ supplements on other health outcomes in children is less clear.

What is vitamin B₁₂?

Vitamin B₁₂ is an important nutrient that helps make red blood cells and keep the body's cells and nerves healthy. Vitamin B₁₂ deficiency is an important public health problem, including in children and in resource-poor settings. Lower vitamin B₁₂ levels in children have been linked to anaemia (lower haemoglobin levels), and poorer growth, development, and cognition (a child's ability to gain knowledge through thought, understanding, and the senses).

What did we want to find out?

We wanted to find out if taking vitamin B₁₂ supplements would improve the nutrition and health of children less than 12 years of age.

What did we do?

We searched for studies that looked at the effects of vitamin B₁₂ compared to placebo (a substance that looks the same as the intervention/treatment but does not contain any active ingredient), no intervention, or the same supplements without vitamin B₁₂. We summarised the results of the studies and our confidence in the evidence, based on information such as how the study was done.

What did we find?

We included 16 studies involving 4083 children less than 12 years of age (2 days to 11 years). Nine studies, including 2391 children, had data that we could analyse. The studies were conducted in 10 countries around the world. In most studies, the outcomes were assessed immediately after the intervention from five weeks to 31 months, with one study assessing outcomes after nine years. Vitamin B₁₂ supplements were provided from 28 days to 31 months in amounts from 0.9 to 150 micrograms a day.

Supplementation with vitamin B₁₂ alone versus placebo

There may be little to no difference in height/length, cognition (for example, attention), or development (for example, motor skill development). Children who received vitamin B₁₂ supplements had less vitamin B₁₂ deficiency, and probably had higher vitamin B₁₂ levels,

compared to placebo. There may be little to no difference between groups in anaemia. No studies provided data on any unwanted (adverse) effects.

Vitamin B₁₂ alone versus no intervention

The evidence is very uncertain for height/length. No studies reported data we could analyse for other outcomes.

Vitamin B₁₂ and other micronutrients, compared to supplements with the same supplement without vitamin B₁₂

There may be little to no difference between groups for height/length, cognition (for example, attention), or development (for example, motor skill development). Children who received vitamin B₁₂ supplementation (with other nutrients) may have little to no difference in vitamin B₁₂ deficiency, vitamin B₁₂ levels, anaemia, or any unwanted effects.

What are the limitations of the evidence?

The small number and the size of studies were limitations of this review. Few studies reported on health outcomes, including growth, cognition, development, anaemia, or any adverse effects. The different ages of children, and differences in the amount of vitamin B₁₂ and when the outcomes were measured, are also limitations of this review.

How up to date is this evidence?

The evidence is up to date as of 23 September 2025.

SUMMARY OF FINDINGS

Summary of findings 1. Summary of findings table - Supplementation with vitamin B12 alone compared to placebo in children less than 12 years of age

Supplementation with vitamin B12 alone compared to placebo in children less than 12 years of age

Patient or population: children less than 12 years of age

Setting: schools, clinics/hospitals, or community settings in Europe, South-East Asia, or Western Pacific

Intervention: supplementation with vitamin B12 alone

Comparison: placebo

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with placebo	Risk with supplementation with vitamin B12 alone				
Growth: height/length (cm) (ALL) assessed with: stadiometer, or vertical scale follow-up: range 4 months to 12 months	The mean growth: height/length (cm) (ALL) ranged from 136.3 to 144.3 cm	The mean growth: height or length (cm) in the intervention group was 0.19 cm lower (2.44 lower to 2.06 higher)	-	143 (2 RCTs)	⊕⊕⊕⊕ Low ^a	There may be little to no difference between intervention groups in height/length for children receiving vitamin B12, compared to placebo.
Cognitive function, using validated instruments, including general cognitive ability or cognitive domains: attention (ALL) assessed with: NEPSY-II (higher scores = better attention; scale from: 1 to 19 (scaled scores)), or Color Trail Test (lower scores (time in seconds) = better attention; scale from: 0 to 240 (raw scores)) follow-up: range 1 years to 9 years ^b		Compared to the control group, the mean attention score in the intervention group was on average 0.09 standard deviations higher (0.1 lower to 0.28 higher)	-	438 (2 RCTs)	⊕⊕⊕⊕ Low ^a	There may be little to no difference between intervention groups in attention for children receiving vitamin B12, compared to placebo. Lower scores (time in seconds) for Color Trail Test reflect better attention; scores were reversed in the analysis so that higher values consistently reflect better attention across studies.
Development outcomes (e.g. motor skill development, language, and socio-emotional and adaptive behavior),	See comment			403 (1 RCT)	⊕⊕⊕⊕ Low ^c	There may be little to no difference between intervention groups in motor skill development for children receiving vit-



amin B12, compared to placebo (6 to 9 years).

assessed using validated instruments: motor skill development (ALL) assessed with: NEPSY-II (higher scores = better motor skill development; scale from: 1 to 19 (scaled scores)) follow-up: range 6 years to 9 years ^b						
Vitamin B12 status: vitamin B12 deficiency, as defined by trial authors (e.g. < 148, < 221 pmol/L) (ALL) assessed with: microbiologic assay, or radioisotope dilution (total vitamin B12 concentration <148 pmol/L, or 150 pmol/L) follow-up: range 4 months to 12 months	76 per 1000	21 per 1000 (8 to 53)	RR 0.27 (0.11 to 0.69)	642 (3 RCTs)	⊕⊕⊕⊕ High	Absolute risk difference: 56 fewer per 1000 (95% CI from 68 fewer to 24 fewer). Vitamin B12 supplementation reduces vitamin B12 deficiency, compared to placebo.
Vitamin B12 status: total vitamin B12 concentrations (pmol/L) (ALL) assessed with: microbiologic assay, or radioisotope dilution follow-up: range 5 weeks to 52 weeks	The mean vitamin B12 status: total vitamin B12 concentrations (pmol/L) (ALL) ranged from 162.36 to 475.56 pmol/L	The mean vitamin B12 status: total vitamin B12 concentrations (pmol/L) in the intervention group was 68.53 pmol/L higher (41.83 higher to 95.23 higher)	-	705 (7 RCTs)	⊕⊕⊕⊖ Moderated ^d	Vitamin B12 supplementation probably increases total vitamin B12 concentrations, compared to placebo. Sensitivity analysis using fixed-effect model: MD 58.82, 95% CI 46.02 to 71.63.
Anaemia and haemoglobin: anaemia, defined using age-specific cut-offs (ALL) assessed with: photoelectric calorimeter, or photometric method (Hb <110 g/L or 120 g/L) follow-up: range 4 months to 12 months	45 per 1000	66 per 1000 (18 to 236)	RR 1.48 (0.41 to 5.27)	144 (2 RCTs)	⊕⊕⊖⊖ Low ^e	Absolute risk difference: 21 more per 1000 (95% CI from 26 fewer to 191 more). There may be little to no difference between intervention groups in anaemia for children receiving vitamin B12, compared to placebo.
Any adverse effects (ALL) - not reported	-	-	-	-	-	None of the included trials reported data for this outcome

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_449278832325082205.

^a Downgraded two levels due to imprecision: total small sample size did not meet optimal information size threshold, and wide confidence intervals include the possibility of both appreciable benefit and appreciable harm.

^b NEPSY-II: NEuroPSYchological assessment, Second Edition; RCT: randomized controlled trial.

^c Downgraded two levels due to imprecision: total sample size did not meet optimal information size threshold, and effect estimate not precise.

^d Downgraded one level due to risk of bias: four out of seven studies contributing most to the effect estimate had serious concerns with bias arising from the randomisation process, bias due to deviations from intended interventions, and bias in selection of the reported result.

^e Downgraded two levels due to imprecision: total small sample size did not meet optimal information size threshold, few events, and wide confidence intervals include the possibility of both appreciable benefit and appreciable harm.

Summary of findings 2. Summary of findings table - Supplementation with vitamin B12 alone compared to no intervention in children less than 12 years of age

Supplementation with vitamin B12 alone compared to no intervention in children less than 12 years of age

Patient or population: children less than 12 years of age

Setting: obstetric clinic, or unspecified settings in Europe

Intervention: supplementation with vitamin B12 alone

Comparison: no intervention

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	N° of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with no intervention	Risk with supplementation with vitamin B12 alone				
Growth: height/length (cm) (ALL)	See comment			70 (2 RCTs)	⊕⊕⊕⊕ Very low ^{a,b}	There may be little to no difference between groups in height/length after 28 days (1 RCT), and children may have increased height/length after 10 months (1 RCT), compared to no intervention; however, the evidence is very uncertain. ^c

Cognitive function, using validated instruments, including general cognitive ability or cognitive domains: attention (ALL) - not reported	-	-	-	-	-	None of the included trials reported data for this outcome.
Development outcomes (e.g. motor skill development, language, and socio-emotional and adaptive behavior), assessed using validated instruments: motor skill development (ALL) - not reported	-	-	-	-	-	None of the included trials reported data for this outcome.
Vitamin B12 status: vitamin B12 deficiency, as defined by trial authors (e.g. < 148, < 221 pmol/L) (ALL) - not reported	-	-	-	-	-	None of the included trials reported data for this outcome.
Vitamin B12 status: total vitamin B12 concentrations (pmol/L) (ALL) - not reported	-	-	-	-	-	None of the included trials reported data for this outcome.
Anaemia and haemoglobin: anaemia, defined using age-specific cut-offs (ALL) - not reported	-	-	-	-	-	None of the included trials reported data for this outcome.
Any adverse effects (ALL) - not reported	-	-	-	-	-	None of the included trials reported data for this outcome.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **MD:** mean difference; **OR:** odds ratio; **RR:** risk ratio; **SMD:** standardised mean difference

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_449291733148853917.

^a Downgraded one level due to risk of bias: study contributing data had serious concerns with bias in randomization process, bias in measurement of the outcome, and bias in selection of the reported result.

^b Downgraded two levels due to imprecision: total small sample size did not meet optimal information size threshold, and effect estimate not precise.
^c RCT: randomised controlled trial.

Summary of findings 3. Summary of findings table - Supplementation with vitamin B12 and other micronutrients compared to supplements with the same formulation without vitamin B12 in children less than 12 years of age

Supplementation with vitamin B12 and other micronutrients compared to supplements with the same formulation without vitamin B12 in children less than 12 years of age

Patient or population: children less than 12 years of age
Setting: community settings in South-East Asia
Intervention: supplementation with vitamin B12 and other micronutrients
Comparison: supplements with the same formulation without vitamin B12

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with supplements with the same formulation without vitamin B12	Risk with supplementation with vitamin B12 and other micronutrients				
Growth: height/length (cm) (ALL) assessed with: portable length board follow-up: mean 12 months	The mean growth: height/length (cm) (ALL) ranged from 78.4 to 137.9 cm	The mean growth: height or length in the intervention group was 0.22 cm lower (0.65 lower to 0.2 higher)	-	608 (2 RCTs)	⊕⊕⊕⊕ Low ^a	There may be little to no difference between intervention groups in height/length for children receiving vitamin B12 and other micronutrients, compared to supplements with the same formulation without vitamin B12.
Cognitive function, using validated instruments, including general cognitive ability or cognitive domains: attention (ALL) assessed with: NEPSY-II (higher scores = better attention; scale from: 1 to 19 (scaled scores)), or Color Trail Test (lower scores (time in seconds) = better attention; scale from: 0 to 240 (raw scores))		Compared to the control group, the mean attention score was on average 0 standard deviations (0.13 lower to 0.13 higher)	-	956 (3 RCTs)	⊕⊕⊕⊕ Low ^a	There may be little to no difference between intervention groups in attention for children receiving vitamin B12 and other micronutrients, compared to supplements with the same formulation without vitamin B12. Lower scores (time in seconds) for Color Trail Test reflect better attention; scores were reversed in the analysis so that higher values consistently reflect better attention across studies.

follow-up: range 1 years to 9 years ^b						
Development outcomes (e.g. motor skill development, language, and socio-emotional and adaptive behavior), assessed using validated instruments: motor skill development (ALL) assessed with: BSID-III, or NEPSY-II (higher scores = better motor skill development) Scale from: 0 to 200 (raw scores) follow-up: range 2 years to 9 years ^b		Compared to the control group, the mean motor skill development score in the intervention group was on average 0.05 standard deviations lower (0.18 lower to 0.08 higher)	-	960 (2 RCTs)	⊕⊕⊕⊕ Low ^a	There may be little to no difference between intervention groups in motor skill development for children receiving vitamin B12 and other micronutrients, compared to supplements with the same formulation without vitamin B12.
Vitamin B12 status: vitamin B12 deficiency, as defined by trial authors (e.g. < 148, < 221 pmol/L) (ALL) assessed with: microbiologic assay (total vitamin B12 concentration < 148 pmol/L) follow-up: range 6 months to 12 months	8 per 1000	6 per 1000 (1 to 40)	RR 0.77 (0.12 to 5.16)	534 (2 RCTs)	⊕⊕⊕⊕ Low ^a	Absolute risk difference: 2 fewer per 1000 (95% CI from 7 fewer to 32 more). There may be little to no difference between intervention groups in vitamin B12 deficiency for children receiving vitamin B12 and other micronutrients, compared to supplements with the same formulation without vitamin B12.
Vitamin B12 status: total vitamin B12 concentrations (pmol/L) (ALL) assessed with: microbiologic assay follow-up: range 6 months to 12 months	The mean vitamin B12 status: total vitamin B12 concentrations (pmol/L) (ALL) ranged from 323.2 to 352.33 pmol/L	The mean vitamin B12 status: total vitamin B12 concentrations in the intervention group was 34.32 pmol/L higher (31.13 lower to 99.77 higher)	-	731 (3 RCTs)	⊕⊕⊕⊕ Low ^{c,d}	There may be little to no difference between intervention groups in total vitamin B12 concentrations for children receiving vitamin B12 and other micronutrients, compared to supplements with the same formulation without vitamin B12. Sensitivity analysis using fixed-effect model: MD 35.90, 95% CI 10.12 to 61.67.
Anaemia and haemoglobin: anaemia, defined using age-specific cut-offs (ALL) assessed with: photoelectric calorimeter, or photometric method (Hb < 120 g/L) follow-up: mean 12 months	See comment			35 (1 RCT)	⊕⊕⊕⊕ Low ^e	There may be little to no difference between intervention groups in anaemia for children receiving vitamin B12 and other micronutrients, compared to supplements with the same formulation without vitamin B12 (21.7% versus 25.0%).

Any adverse effects (ALL) follow-up: mean 6 months	See comment	574 (1 RCT)	⊕⊕⊕⊕ Low ^e	There may be little to no difference between intervention groups in any adverse effects for children receiving vitamin B12 and other micronutrients, compared to supplements with the same formulation without vitamin B12 (9.2% versus 5.9%).
---	-------------	----------------	--------------------------	--

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradeapro.org/presentations/#/isof/isof_question_revman_web_447803254987475211.

^a Downgraded two levels due to imprecision: total sample size did not meet the optimal information size threshold, and wide confidence intervals include the possibility of both appreciable benefit and appreciable harm.

^b BSID-III: Bayley Scale of Infant Development, Third Edition; NEPSY-II: NEuroPSYchological assessment, Second Edition; RCT: randomised controlled trial.

^c Downgraded one level due to inconsistency: substantial statistical heterogeneity ($I^2 = 77\%$) possibly due to a larger relative effect among healthy versus stunted (height-for-age z-score <-1) children involved in one study.

^d Downgraded one level due to imprecision: wide confidence intervals include the possibility of both appreciable benefit and appreciable harm.

^e Downgraded two levels due to imprecision: total small sample size did not meet optimal information size threshold, and effect estimate not precise.

BACKGROUND

Description of the condition

Vitamin B₁₂ is a water-soluble vitamin required for cell division, DNA synthesis and methylation, red blood cell formation, folate metabolism, and neurotransmitter synthesis [1, 2, 3]. Vitamin B₁₂ is needed to convert folate to its active form (tetrahydrofolate) to synthesise purines [2, 4, 5, 6] and convert homocysteine to methionine to produce S-adenosylmethionine, required in over 100 methylation reactions [7, 8, 9].

Vitamin B₁₂ is found predominantly in animal source foods, including meat, fish, poultry, eggs, and dairy [6]. Vitamin B₁₂ is also found in nutritional yeast, fortified foods, and cereals [8], shiitake mushrooms, Nori, and tempeh [10]. Vitamin B₁₂ requirements vary by age and life stage: 0.4 to 1.8 µg/day in children to 2.4 to 2.8 µg/day in adults [11, 12].

Vitamin B₁₂ deficiency

Vitamin B₁₂ deficiency is an important public health problem [8, 13, 14, 15], with the highest prevalence in the elderly [8, 16], pregnant women [17], and children [8], particularly in resource-limited settings (Latin America: 30% to 40% [14, 18, 19]; South Asia: 50% to 80% [20, 21]). In children, the prevalence of vitamin B₁₂ deficiency ranges from 5% to 64% (in Mexico 5% [22], Kenya 10% [23], Nepal 17% [24], Tanzania 26% [25], Guatemala 30% [26], and India 64% [27]), although there are limited population-based data.

Vitamin B₁₂ is synthesised by bacteria and obtained through animal-source foods [6, 13, 28]. Vitamin B₁₂ deficiency can result from low intake, bioavailability, or absorption (pernicious anaemia, gastrointestinal infections or conditions [8, 17, 28, 29]). Young children are at increased risk of vitamin B₁₂ deficiency, due to supply *in utero* and birth [30, 31, 32, 33], and low content in breast milk [8, 34], particularly with maternal vitamin B₁₂ deficiency and in exclusively breastfed infants [28, 35]. Vitamin B₁₂ deficiency may develop in children due to low content in complementary foods and intake of animal-source foods [28], and as requirements increase with age (0 to 6 months: 0.4 µg/day; 7 to 17 months: 0.5 µg/day; 1 to 3 years: 0.9 µg/day; 4 to 8 years: 1.2 µg/day; 9 to 13 years: 1.8 µg/day) [11, 12].

The classical manifestation of vitamin B₁₂ deficiency is haematological, as pernicious anaemia or megaloblastic anaemia [6, 8]. In addition to anaemia [36, 37, 38], in children, vitamin B₁₂ deficiency has been associated with impaired growth [39], cognition [40, 41], and development [41], which may be irreversible [42, 43]. This review focusses on the effects of vitamin B₁₂ supplementation among apparently healthy children, regardless of baseline vitamin B₁₂ status.

Vitamin B₁₂ biomarkers

Vitamin B₁₂ status is evaluated through circulating (i.e. total vitamin B₁₂, holotranscobalamin) or functional (i.e. methylmalonic acid, homocysteine) biomarkers. Total vitamin B₁₂ is the most common biomarker (e.g. vitamin B₁₂ < 148 and 221 pmol/L [44, 45], which includes holotranscobalamin (~30%) and holohaptocorrin (~70%) [46, 47]. Methylmalonic acid is a functional and specific vitamin

B₁₂ biomarker that is elevated in vitamin B₁₂ deficiency [44, 45]. In contrast, homocysteine is not specific to vitamin B₁₂ deficiency and is influenced by nutritional (e.g. folate) and non-nutritional (e.g. renal function) factors [8]. The use of both circulating and functional biomarkers is recommended [44, 45], although specific cut-offs have not been established for children.

Description of the intervention and how it might work

Vitamin B₁₂ supplementation (cyanocobalamin, methylcobalamin) is a common strategy to prevent and treat deficiency [48, 49], with standard doses ranging from 2.0 to 10.0 µg/day [48]. Daily oral vitamin B₁₂ supplementation is used in clinical practice for adults with vitamin B₁₂ deficiency [48, 50], although intermittent administration in adults, including pregnant women (6 to 14-week gestation; 18 to 28 years), has been reported [51], the effect of vitamin B₁₂ supplementation in children remains unclear. Intramuscular or intravenous administration of vitamin B₁₂ is more commonly used in patients with malabsorption disorders, while oral vitamin B₁₂ supplementation at higher doses (e.g. 1000 µg/day to 2000 µg/day) may need to be continued indefinitely [48, 50]. There is no tolerable upper level for vitamin B₁₂ [8, 11, 44].

In children, vitamin B₁₂ is administered via oral supplements (i.e. capsules, tablets, dispersible tablets), liquids, drops or syrups, micronutrient powders, ready-to-use therapeutic foods, or fortified foods.

This review focuses on oral vitamin B₁₂ supplementation, with any formulation, dose, frequency, or duration. It does not include intravenous/intramuscular injections, which are primarily for clinical populations at pharmaceutical doses.

Cobalamin is primarily absorbed via an active, receptor-mediated transport in the distal ileum. In contrast, passive diffusion of free vitamin B₁₂ can occur throughout the small intestine and to a limited extent in the large intestine [13, 50]. Dietary vitamin B₁₂ is tightly bound to proteins that must be degraded and bound to intrinsic factor for absorption (20% to 40%) via receptor-mediated endocytosis [6, 8, 52]. In supplements, approximately 50% of vitamin B₁₂ is absorbed with a 1.0 µg oral dose in the distal ileum, and absorption decreases with increasing dosages, as active transport is saturable [44]. Additionally, up to 1% of free cobalamin is passively absorbed in the terminal ileum [8, 48].

In observational studies in children, lower vitamin B₁₂ status [2, 8, 53] has been associated with anaemia [36, 37, 38, 53] and impaired brain development [9, 43], cognitive function [41, 54, 55], development [56], school performance [57, 58], and growth [39]. Vitamin B₁₂ supplementation in children (1 month to 17 years of age) with vitamin B₁₂ deficiency (< 148 pmol/L), compared to vitamin B₁₂ status at enrolment, improved vitamin B₁₂ status [59]. However, its effects on child health outcomes have not been established.

This review includes children without clinical vitamin B₁₂ deficiency. It includes presumably healthy children who may be at risk due to inadequate dietary intake, infection, or other factors, with an emphasis on oral preventive supplementation and

outcomes related to growth, development, and cognition, rather than treatment of vitamin B₁₂ deficiency.

Why it is important to do this review

Vitamin B₁₂ deficiency is an important health problem in children [8], and has been associated with anaemia, and impaired growth, cognition, and development [41, 57, 58]. Higher vitamin B₁₂ status [56, 57] and supplementation [39, 60] have been associated with improved health outcomes in some studies. The high prevalence of vitamin B₁₂ deficiency in children and its association with adverse health outcomes has stimulated interest in vitamin B₁₂ supplementation and fortification interventions [16]. However, screening or vitamin B₁₂ supplementation are not currently recommended in children.

No systematic reviews have been conducted to examine the effects of vitamin B₁₂ supplementation in children. This review complements previous Cochrane reviews, including vitamin B₁₂ during pregnancy [61], multiple micronutrient supplementation during pregnancy [62], and vitamin D [63], zinc [64, 65], iron [66, 67, 68], and polyunsaturated fatty acid supplementation [69] in children.

OBJECTIVES

To determine the effects of vitamin B₁₂ supplementation on growth, development, and cognition in children less than 12 years of age.

METHODS

We followed the Methodological Expectations for Cochrane Intervention Reviews when conducting the review [70], and PRISMA 2020 for the reporting [71].

Differences between protocol and review

Outcome measures: To improve the applicability of this review, we reclassified pre-specified important outcomes related to vitamin B₁₂ status (e.g. vitamin B₁₂ deficiency, total vitamin B₁₂ concentrations), and anaemia and haemoglobin, as critical outcomes as these are patient-centred and clinically meaningful indicators in children less than 12 years.

Risk of bias assessment in included studies: In addition to risk of bias assessments for critical outcomes in the long term (more than six months after the intervention), we assessed risk of bias for critical outcomes in the short term (immediately after the intervention), to provide a comprehensive assessment of internal validity.

Investigation of heterogeneity and subgroup analysis: We conducted subgroup analyses with critical outcomes that included at least three trials. These analyses were retained to explore biologically plausible effect differences across included studies. Subgroup analyses based on vitamin B₁₂ status at enrolment could not be undertaken due to inconsistent reporting across included studies; we summarised the available information in the Results and Discussion sections. In addition, we removed three subgroup analyses because all included studies fell into the same subgroup for the outcomes of interest, and comparisons were not feasible.

Equity-related assessment: We did not plan equity-related assessments *a priori*; however, we considered health inequities in this review using the PROGRESS-PLUS and INCLUDE Ethnicity Framework to improve transparency and interpretation in describing population characteristics and contextual factors that may influence the applicability of findings.

Criteria for considering studies for this review

Types of studies

We included the following types of studies:

1. Randomised controlled trials (RCTs), with randomisation at the individual or cluster level.
2. Quasi-randomised controlled trials (treatment allocated by a different method such as alternation, birthdate, or case record number).

We excluded cross-over trials or observational studies.

Types of participants

We included trials among children less than 12 years of age, regardless of geographic location or baseline vitamin B₁₂ status. Trials were included if the intervention was administered directly to the child. We excluded studies of interventions targeting pregnant or breastfeeding women, or participants with critical illnesses or severe comorbidities, including malabsorption or medical conditions that may affect vitamin B₁₂ absorption or metabolism.

If a study also included participants above 12 years, we contacted the authors to obtain data for participants less than 12 years of age.

Types of interventions

We included studies of oral supplementation of vitamin B₁₂, alone or combined with other vitamins/minerals.

Vitamin B₁₂ supplementation was delivered via tablet, capsule, dispersible tablet, syrup, drops, micronutrient powders, soft gels/pastes, or fortified foods, with no restrictions on dose, formulation, frequency, or duration. We excluded intravenous/intramuscular injections.

We included studies with co-interventions (e.g. nutrition education) if it was the same in both groups.

Comparators

We included studies where the comparators were placebo, no intervention, or other micronutrient supplements without vitamin B₁₂.

Main comparisons

1. Supplementation with vitamin B₁₂ alone versus placebo
2. Supplementation with vitamin B₁₂ alone versus no intervention
3. Supplementation with vitamin B₁₂ and other micronutrients, compared to supplements with the same formulation without vitamin B₁₂

Outcome measures

Critical outcomes

1. Growth: height/length (height or length-for-age, reported as height-for-age z-score (HAZ) or length-for-age z-score (LAZ)) (WHO growth standards)
2. Cognitive function, using validated instruments, including general cognitive ability or cognitive domains [72, 73, 74]
3. Development outcomes (e.g. motor skill development, language, and socio-emotional and adaptive behaviour), assessed using validated instruments [72]
4. Vitamin B₁₂ status: vitamin B₁₂ deficiency or insufficiency, as defined by trial authors (e.g. < 148, < 221 pmol/L) [44, 45]; total vitamin B₁₂ (pmol/L), methylmalonic acid (MMA) (μmol/L), holotranscobalamin (pmol/L), and homocysteine (μmol/L) concentrations
5. Anaemia and haemoglobin: anaemia, defined using age-specific cut-offs; haemoglobin (g/L) adjusted for altitude (and smoking if reported), when applicable
6. Any adverse effects

Important outcomes

1. Megaloblastic anaemia, as defined by trial authors
2. Body composition, as defined by trial authors
3. Health-related quality of life, including physical, personal, or emotional health problems, physical functioning, emotional well-being, social functioning, bodily pain, energy or fatigue, general health perceptions, as defined by trial authors, evaluated by validated instruments [75, 76]
4. Morbidity, as defined by trial authors (proportion with at least one reported illness)
5. All-cause mortality (death from any cause)

Timing of outcome assessment

We extracted any adverse events, all-cause mortality, and morbidity for any time during the study period after participants were randomised.

We extracted and analysed growth, cognitive function, development outcomes, vitamin B₁₂ status, haemoglobin, anaemia, body composition, and health-related quality of life group outcomes for short (immediate), medium (1 to 6 months), and long (> 6 months) term time points.

Search methods for identification of studies

The search strategies are included in [Supplementary material 1](#).

Electronic searches

Information specialists (MK, EE, KGK) searched the databases and trial registries listed below without date or language restrictions.

1. Cochrane Central Register of Controlled Trials (CENTRAL; 2025, Issue 12) in the Cochrane Library, which includes the Cochrane Developmental, Psychosocial and Learning Problems Specialised Register (searched 22 September 2025)
2. MEDLINE via PubMed (1946 to 22 September 2025)
3. Embase Ovid (1980 to 23 September 2025)

4. CINAHL EBSCO (Cumulative Index to Nursing and Allied Health Literature; 1982 to 23 September 2025)
5. Centre for Agriculture and Biosciences International (CABI): CAB Abstracts and Global Health Web of Science Clarivate (1973 to 22 September 2025)
6. PsycINFO EBSCO (1887 to 22 September 2025)
7. ERIC EBSCO (Educational Resources Information Center; 2004 to 22 September 2025)
8. Web of Science Core Collection: Science Citation Index Expanded, Social Sciences Citation Index, Conference Proceedings Citation Index-Science, Conference Proceedings Citation Index-Social Science & Humanities, and Emerging Sources Citation Index Clarivate (1900 to 22 September 2025)
9. Cochrane Database of Systematic Reviews (CDSR) in the Cochrane Library (searched 22 September 2025)
10. Proquest Dissertations & Theses Global (searched 22 September 2025)
11. Global Index Medicus: African Index Medicus, Latin America and the Caribbean Literature on Health Sciences, Index Medicus for Eastern Mediterranean Region, Index Medicus for South-East Asia Region, and Western Pacific Region Index Medicus (www.globalindexmedicus.net, searched 22 September 2025)
12. IBECs (Índice Bibliográfico Español en Ciencias de la Salud via the Virtual Health Library; www.bvsalud.org, searched 23 September 2025)
13. Scientific Electronic Library Online Clarivate (searched 22 September 2025)
14. WHO International Clinical Trials Registry Platform (ICTRP; apps.who.int/trialsearch, searched 22 September 2025)
15. EU Clinical Trials Register (www.clinicaltrialsregister.eu/ctr-search, searched 22 September 2025)
16. Epistemonikos (limited to systematic reviews; www.epistemonikos.org, searched 22 September 2025)
17. Scopus Elsevier (1788 to 23 September 2025)

Searching other resources

We searched the reference lists of relevant publications from the searches to identify additional studies. We attempted to obtain information on relevant unpublished, planned, and ongoing trials by contacting the authors of included trials and other organisations. We checked the included studies for retractions and errata in PubMed and via the Retraction Watch Database.

Data collection and analysis

Selection of studies

Review authors (CEGL, SV, DYL, LBS) independently screened record titles/abstracts identified in the search strategy ([Electronic searches](#)) using Covidence [77]. We retrieved, de-duplicated, and evaluated full-text records for abstracts for eligibility ([Criteria for considering studies for this review](#)). Discrepancies were resolved through discussion and consultation with the senior author (JLF). We contacted study authors for additional information or clarification. We summarised the decision process in a flow diagram [71].

Data extraction and management

We developed and piloted a data extraction form for this review. Review authors (CEGL, DYL, XPM, LBS) independently extracted

information including participants (age, sex, location, baseline vitamin B₁₂ status), intervention (type, formulation, dose, duration, comparisons), study methods (randomisation method, unit of randomisation, sample size), outcomes, outcome assessment, trial dates, sources of funding, adverse effects, declaration of interests, and information for risk of bias assessment. Any discrepancies were resolved through discussion and consultation with the senior author (JLF). We contacted study authors for missing data or clarification. We entered extracted data into Review Manager software [78].

Risk of bias assessment in included studies

Review authors (DYL, XPM, LBS) independently assessed the risk of bias for critical outcomes (i.e. growth, cognitive function, development, vitamin B₁₂ status, anaemia and haemoglobin, any adverse effects) in the short term (immediately after the intervention) and long term (more than six months after the intervention) using the Cochrane RoB 2 tool [79, 80]. This tool includes five domains: bias arising from the randomisation process, bias due to deviations from intended interventions, bias due to missing outcome data, bias in measurement of the outcome, and bias in selection of the reported result. We were interested in the effect of the assignment. We used the modified RoB 2 tool for cluster-randomised trials [80].

For each of the domains, review authors answered a set of signalling questions with a set of five possible responses (i.e. 'yes', 'probably yes', 'probably no', 'no', 'no information') and assigned a risk of bias level (i.e. 'low', 'some concerns', or 'high'). Any discrepancies were resolved through discussion and consultation with the senior author (JLF). We reached an overall low risk of bias when all domains were at low risk; some concerns of bias when the trial result was judged to raise some concerns in at least one domain, but was not at high risk of bias for any domain; and overall high risk of bias when the trial result was at high risk of bias in at least one domain or when it was judged to have some concerns for multiple domains ($\geq$ three domains). We used Excel to manage risk of bias assessments and report the timing of the outcome assessment. We summarised the risk of bias in traffic lights at the side of forest plots, and in the text. Detailed risk of bias assessments are available on reasonable request.

Measures of treatment effect

Dichotomous data

For dichotomous outcomes, we reported risk ratios (RR) and 95% confidence intervals (95% CIs) [81].

Continuous data

We reported mean differences (MD) and corresponding 95% CIs. If the assessment scale differed, we reported the standardised mean difference (SMD) and 95% CIs. We converted standard error (SE) and CI (group means or change estimates) to standard deviations (SD) [82] and ensured that scales were reported using the same direction [81].

Unit of analysis issues

Cluster-randomised trials

To adjust the standard errors of the cluster-randomised trial, we used an internal estimate of the intra-cluster correlation coefficient

(ICC) of 0.01 [83, 84] and 0.05 [85] from similar trials/populations, as recommended in the *Cochrane Handbook for Systematic Reviews of Interventions* [86]. We conducted sensitivity analyses to examine the effects of ICC values (e.g. assumptions or ICC value used) on findings. We combined findings from individually randomised and cluster-randomised trials if there was little heterogeneity in study designs and if it was unlikely that the randomisation unit altered the effect of the intervention.

Cross-over trials

Cross-over trials were not included in this review.

Multiple intervention arms

For trials with multiple arms, we extracted relevant intervention groups. For analyses, we created a single pairwise comparison by combining groups (if appropriate), or selected one pair of interventions.

Outcome assessment

If outcomes were reported at multiple time points, we considered the primary endpoint of the trial as the primary outcome. If multiple cognitive tests were assessed within the same domain, we calculated the SMD from the mean standardised scores.

Dealing with missing data

We attempted to contact trial authors to obtain missing data or clarifications ([Supplementary material 9](#)). Meta-analyses were intention-to-treat analyses, retaining participants in the randomised group, regardless of the intervention received. If study authors did not respond or were unable to supply missing data, we analysed the available data only [87].

We reported and assessed attrition in our risk of bias judgements. We conducted sensitivity analyses with/without trials with high attrition.

Reporting bias assessment

We planned to generate funnel plots if there were 10 or more studies for a particular outcome in meta-analyses and assess their symmetry visually. We planned to conduct sensitivity analyses (i.e. comparing fixed-effect and random-effects estimates, and regression-based method [88]) to investigate asymmetry, if suggested by visual assessment. However, we were unable to assess publication bias using funnel plots due to lack of data (< 10 studies). To reduce the risk of non-reporting bias, we searched trial protocols, and trial registries and databases, to evaluate potential discrepancies in methods or outcomes described.

Synthesis methods

We conducted meta-analyses in Review Manager [78], using a fixed-effect model. We used inverse-variance methods to minimise imprecision of the pooled effect estimate [87].

We used both clinical (participant characteristics, interventions, outcomes studied) and methodological (study design, outcome measurement tools, risk of bias) heterogeneity to inform our evaluation of heterogeneity. If there was substantial clinical (e.g. different populations, different doses of interventions) or methodological (e.g. different study design or outcome measure tools) heterogeneity to suggest that underlying treatment effects

may differ between trials, we used a random-effects model with inverse-variance [89]. We considered the summary estimate from random-effects meta-analyses the average of the range of possible treatment effects, and we discussed the clinical implications of treatment effects differing between trials. When random-effects meta-analyses were used, we presented the results as the average treatment effect with 95% CIs and the τ^2 and I^2 estimates.

To assess statistical heterogeneity, we used forest plots and estimated the χ^2 , I^2 , and τ^2 statistics. The χ^2 statistic assesses whether the differences in the observed results are compatible with chance. We interpreted the I^2 values according to Deeks et al. [87], where 30% to 60% may represent moderate heterogeneity, 50% to 90% may represent substantial heterogeneity, and 75% to 100% represents considerable heterogeneity. We used τ^2 as an estimate of the between-study variation in random-effects models [87]. We regarded heterogeneity as substantial if I^2 was greater than 30% and either τ^2 was greater than zero, or the P value of the χ^2 test was less than 0.10.

If there was substantial heterogeneity, we performed prespecified subgroup analyses to examine differences between these groups (Investigation of heterogeneity and subgroup analysis). For outcomes with substantial heterogeneity (based on clinical and methodological evaluation), we did not report a pooled estimate.

When meta-analyses were not conducted (e.g. due to incompletely reported outcome or effect estimates, different effect measures used across studies, or studies with high risk of bias), we described findings from individual trials, and considered synthesis and presentation of findings across studies using other methods, such as summarising effect estimates and combining P values [90].

Investigation of heterogeneity and subgroup analysis

We investigated heterogeneity using subgroup analyses, where data were available ($\geq$ three trials) for critical outcomes [91].

We conducted the following subgroup analyses to examine the potential effect of participant and intervention characteristics on the response to vitamin B₁₂ supplementation.

1. Age groups: less than 2 years, 2 to less than 5 years, 5 to less than 10 years, and 10 to less than 12 years
2. Sex: male, female, mixed, or not reported
3. Form of vitamin B₁₂ supplementation: cyanocobalamin versus methylcobalamin versus other forms
4. Cognitive assessment tools by age groups above (i.e. less than 2 years, 2 to less than 5 years, 5 to less than 10 years, and 10 to less than 12 years)

These subgroups were developed *a priori* based on variation in vitamin B₁₂ requirements by age group, potential to benefit based on vitamin B₁₂ status at baseline, variation in selected outcomes (e.g. growth) by sex, and anticipated variation in the effects of vitamin B₁₂ supplementation on status by frequency and duration of intervention.

We conducted a standard test for heterogeneity across subgroup results and reported an interaction test I^2 value [87].

Equity-related assessment

We examined to assess health equity using the PRO EDI Participant Characteristics Table, and we extracted participant characteristics based on the PROGRESS-PLUS and INCLUDE Ethnicity Framework [92].

Sensitivity analysis

We performed sensitivity analyses to explore the influence of the following factors (when applicable) on effect size.

1. Risk of bias of the included studies, by excluding studies assessed as having a high risk of bias (e.g. as determined by concealment of allocation, high attrition rates, or both)
2. Fixed-effect or random-effects analyses, to examine the potential impact of using a random-effects model instead of a fixed-effect approach on the effect estimate for outcomes with clinical heterogeneity
3. ICC value for cluster-randomised trials, to examine the impact of higher compared to lower ICC values for cluster-randomised trials on findings

Certainty of the evidence assessment

Review authors (DYL, XPM, LBS) used the GRADE approach to independently evaluate the certainty of the evidence for critical outcomes [93], considering five domains (risk of bias, consistency of effect, imprecision, indirectness, publication bias). We downgraded the certainty of the evidence, which started as high certainty, by one ('serious') or two ('very serious') levels depending on the presence of these factors. Discrepancies were resolved through discussion and consultation with the senior author (JLF).

We imported data from Review Manager and used the GRADE guideline development tool to create the summary of findings tables for our main comparisons (i.e. vitamin B₁₂ versus placebo; vitamin B₁₂ versus no intervention; and vitamin B₁₂ with other micronutrients, compared to the same formulation without vitamin B₁₂) [94]. In each table, we reported a summary of the intervention effect and a measure of certainty for each of the critical outcomes in the long term (i.e. more than six months).

Consumer involvement

Consumers were not involved in this review due to limited resources. In addition, the available timeline did not permit the adequate recruitment, training, and support of consumer contributors.

RESULTS

Description of studies

Results of the search

The study flow diagram is shown in Figure 1. The search identified 30,638 records (zero records were identified from other sources). After de-duplication, we removed 21,402 records and assessed 9236 records for eligibility; we excluded 9116 records as irrelevant after title and abstract screening. We screened 120 full-text articles for eligibility and excluded 69 (59 studies: $n = 48$ due to ineligible comparator, $n = 1$ due to ineligible intervention, and $n = 10$ due to ineligible route of administration). A total of 16 trials (41 records) met the eligibility criteria for inclusion in this review.

Figure 1. Study flow diagram

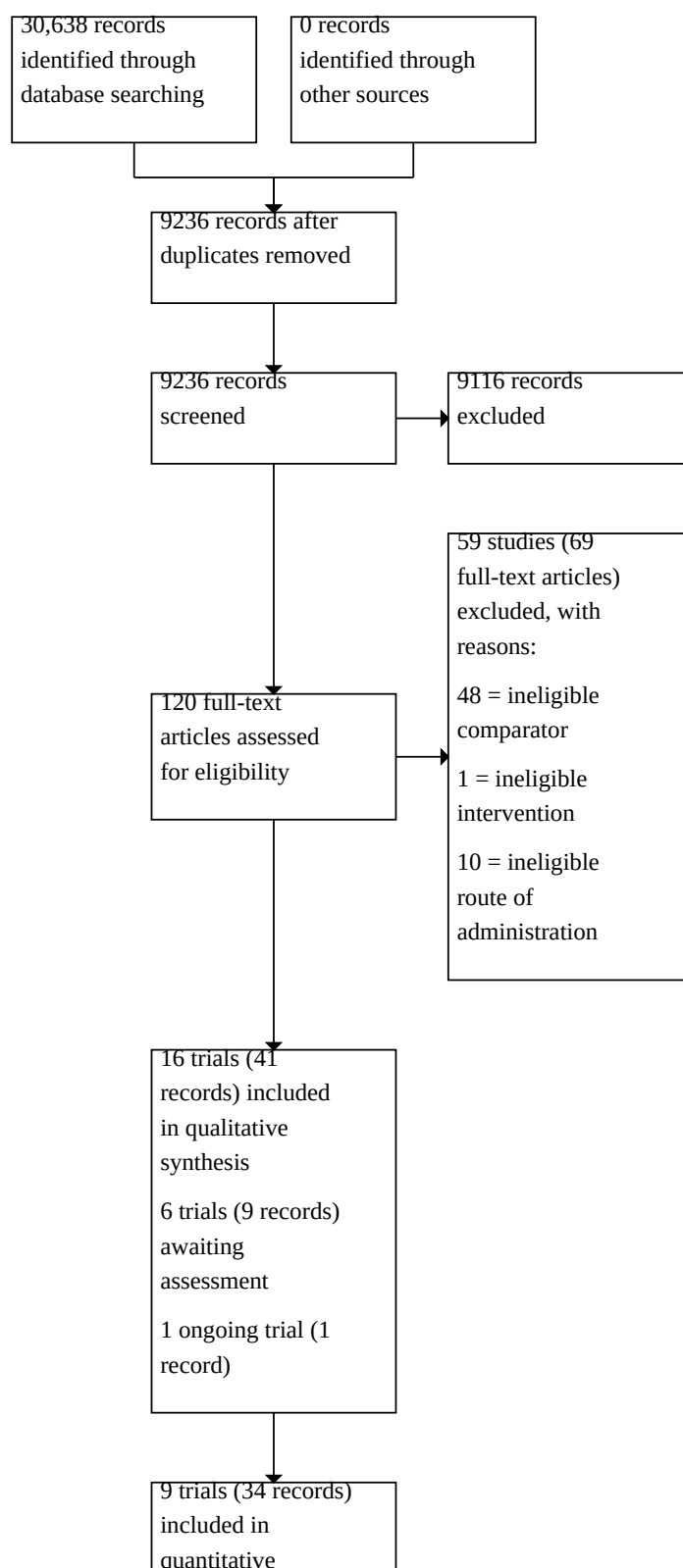


Figure 1. (Continued)

included in quantitative synthesis (meta-analysis)

The evidence for this review is up to date as of 23 September 2025.

Included studies

This review included 16 trials (Areekul 1979 [95]; Benjamin 1952 [96]; Chinnock 1952 [97]; Craigmile 1956 [98]; Deshmukh 2010 [99, 100, 101, 102]; Fukui 1959 [103]; Gutiérrez-Díaz 1959 [104]; Kudo 1962 [105, 106]; Murakami 1962 [107]; Scaglione 1955 [108]; Scrimshaw 1953 [109, 110]; Scrimshaw 1959 [111]; Shimazono 1963 [112]; Strand 2020 [113, 114, 115, 116, 117, 118, 119, 120, 121, 122]; Taneja 2013 [123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134]; Yajnik 2021 [135, 136, 137]). Nine trials contributed data to analyses (Areekul 1979; Craigmile 1956; Deshmukh 2010; Kudo 1962; Murakami 1962; Shimazono 1963; Strand 2020; Taneja 2013; Yajnik 2021). We contacted the authors of trials to request unpublished data, if applicable; the authors of two studies shared age-stratified data (Deshmukh 2010; Yajnik 2021). When we could identify contact information, we contacted the authors of all trials, including those awaiting classification, to request more information, and we kept the study categorised as 'Awaiting classification' if these attempts were unsuccessful.

Seven trials were eligible but did not contribute data to meta-analyses (Benjamin 1952; Chinnock 1952; Fukui 1959; Gutiérrez-Díaz 1959; Scaglione 1955; Scrimshaw 1953; Scrimshaw 1959).

Included trials are described in [Table 1](#) and [Supplementary material 2](#).

Settings

Trials were conducted in 10 countries in South-East Asia (Areekul 1979; Deshmukh 2010; Strand 2020; Taneja 2013; Yajnik 2021), Europe (Benjamin 1952; Craigmile 1956; Gutiérrez-Díaz 1959; Scaglione 1955), Western Pacific (Fukui 1959; Kudo 1962; Murakami 1962; Shimazono 1963), and the Americas (Chinnock 1952; Scrimshaw 1953; Scrimshaw 1959). They were conducted in high-income ($n = 9$) (Benjamin 1952; Chinnock 1952; Craigmile 1956; Fukui 1959; Gutiérrez-Díaz 1959; Kudo 1962; Murakami 1962; Scaglione 1955; Shimazono 1963), upper-middle ($n = 3$) (Areekul 1979; Scrimshaw 1953; Scrimshaw 1959), and lower-middle ($n = 4$) (Deshmukh 2010; Strand 2020; Taneja 2013; Yajnik 2021) income countries. Trials were conducted in communities ($n = 7$), schools ($n = 5$), and clinics/hospitals ($n = 3$). One trial did not report setting. See [Table 1](#).

Participants

Participants' ages ranged from two days to 11 years ($n = 4083$). Three trials did not report participant age. For trials including children ≥ 12 years, only age-stratified data for children < 12 years were included in analyses ([Table 1](#)).

The prevalence of vitamin B₁₂ deficiency at enrolment was 17% (< 150 pmol/L) (Taneja 2013) and 25% (< 148 pmol/L) (Deshmukh 2010; Yajnik 2021) in the three trials that reported it. Five trials reported total vitamin B₁₂ concentrations at enrolment but did not report vitamin B₁₂ deficiency (Areekul 1979; Kudo 1962; Murakami 1962; Shimazono 1963; Strand 2020); eight trials did not report vitamin B₁₂ status.

The prevalence of anaemia at enrolment ranged from 3% to 70% (Hb < 11.0 g/dL: 64% (Strand 2020) to 70% (Taneja 2013); Hb < 12.0 g/dL: 3% (Yajnik 2021) to 24% (Deshmukh 2010)). Two trials reported haemoglobin concentrations at enrolment (Murakami 1962; Shimazono 1963). Ten trials did not report haemoglobin or anaemia at enrolment.

Interventions

Daily dose of vitamin B₁₂

All included trials administered vitamin B₁₂ as a daily oral supplement, with doses ranging from 0.9 µg/day to 150 µg/day. Most trials used fixed doses of 2 µg/day, 5 µg/day, 10 µg/day, 20 µg/day, 25 µg/day, 30 µg/day, 50 µg/day, or 150 µg/day. One trial used variable dosing based on participant age, with doses of 0.9 µg/day or 1.8 µg/day (Taneja 2013).

Additional details on the dose, form, and compound are in [Table 1](#).

Duration of intervention

Duration of supplementation ranged from 28 days to 6 months ($n = 8$), 10 months ($n = 1$), and 12 to 31 months ($n = 7$) ([Table 1](#)).

Comparisons

Thirteen trials compared vitamin B₁₂ supplementation to placebo. Two trials compared vitamin B₁₂ alone to no intervention. Three trials compared vitamin B₁₂ supplementation with micronutrients to micronutrients with the same formulation without vitamin B₁₂. Two trials contributed data to more than one comparison ([Table 1](#)).

Outcomes

Timing of outcome assessment

In the short term (immediately after the intervention), 14 trials evaluated growth, cognitive function, development outcomes, vitamin B₁₂ status, anaemia, haemoglobin, body composition, or health-related quality of life, with follow-ups post-intervention ranging from five weeks to 31 months. No studies assessed outcomes in the medium term (one to six months post-intervention). In the long term (more than six months), two studies

assessed cognitive function and developmental outcomes, with follow-ups post-intervention extending from 12 to 24 months and up to six to nine years.

Outcomes and assessment are described in [Table 1](#), [Supplementary material 2](#), and [Supplementary material 10](#).

Sources of trial funding

Eleven trials reported funding sources. Five trials did not report funding sources (Areekul 1979; Fukui 1959; Gutiérrez-Díaz 1959; Murakami 1962; Scaglione 1955) ([Supplementary material 2](#); [Supplementary material 10](#)).

Declarations of interest

Four trials reported declarations of interest (Deshmukh 2010; Strand 2020; Taneja 2013; Yajnik 2021).

Excluded studies

We excluded a total of 59 trials (69 records) ([Supplementary material 3](#)), due to ineligible intervention ($n = 1$) (Grillenberger 2006 [138]), ineligible route of administration of vitamin B₁₂ ($n = 10$) (intramuscular (Bjørke-Monsen 2011 [139, 140, 141, 142]; Evdokimova 1971 [143]; Jobert 1975 [144]; Mena 2001 [145]; NCT00572741 [146]; Olivi 1971 [147]; Sun 2022 [148]; Torsvik 2015 [149, 150]; Worthington-White 1994 [151]); intravenous (Haiden 2006 [152]) injection), or ineligible comparator ($n = 48$).

Studies awaiting classification

Six trials (nine records) are awaiting classification ([Supplementary material 4](#)) (Bansal 2016 [153, 154, 155]; Chandelia 2012 [156]; Daute 1964 [157]; Mackay 1956 [158, 159]; ISRCTN67437725 [160]; Rusconi 1983 [161]. One trial registry (ISRCTN67437725) and five trials (Bansal 2016; Chandelia 2012; Daute 1964; Mackay 1956; Rusconi 1983) did not have sufficient information for eligibility.

Ongoing study

One ongoing randomised trial (one record) was identified (CTRI/2023/09/057950 2023 [162]) ([Supplementary material 5](#)).

Risk of bias in included studies

We applied the Cochrane RoB 2 tool to randomised trials. Risk of bias assessments for critical outcomes assessed in the short and long term, including all domain judgements and support for judgements, are located in [Supplementary material 6](#), [Supplementary material 11](#), and summarised in [Table 2](#) and [Table 3](#).

In the text below, the overall risk of bias assessment is summarised for nine trials that contributed quantitative outcome data to the critical outcomes of the review. Risk of bias assessments are also reported for outcomes synthesised narratively, due to inclusion of a single study (i.e. height/length (Scaglione 1955), methylmalonic acid (Strand 2020), holotranscobalamin (Areekul 1979), and any adverse effects (Strand 2020)).

Five of the included trials had a high risk of bias overall: one for growth (i.e. height/length; Scaglione 1955), four for vitamin B₁₂ status (i.e. total vitamin B₁₂ concentrations and/or holotranscobalamin) (Areekul 1979; Kudo 1962; Murakami 1962; Shimazono 1963), and two for haemoglobin concentrations (Murakami 1962; Shimazono 1963).

Two of the included trials had 'some concerns' regarding the overall risk of bias: one for vitamin B₁₂ status (i.e. homocysteine concentrations (Yajnik 2021)) and one for any adverse effects (Strand 2020).

Four of the included trials had a low risk of bias overall for growth (i.e. height/length or HAZ: Deshmukh 2010; Strand 2020; Taneja 2013; Yajnik 2021); three for cognitive function (Deshmukh 2010; Strand 2020; Taneja 2013); two for development outcomes (Strand 2020; Taneja 2013); four for vitamin B₁₂ status (i.e. vitamin B₁₂ deficiency, total vitamin B₁₂ concentrations, methylmalonic acid, and/or homocysteine: Deshmukh 2010; Strand 2020; Taneja 2013; Yajnik 2021); and four for anaemia and haemoglobin (Deshmukh 2010; Strand 2020; Taneja 2013; Yajnik 2021).

Synthesis of results

The effects of vitamin B₁₂ supplementation on critical outcomes are summarised in [Summary of findings 1](#), [Summary of findings 2](#), and [Summary of findings 3](#). The analyses can be found in [Supplementary material 7](#).

In this review, 16 trials including 4083 children were included; nine trials (2391 children) contributed data to meta-analyses. Two trials contributed data to more than one comparison (Deshmukh 2010; Taneja 2013).

In Deshmukh 2010, we adjusted the reported sample size and event rate to account for the design effect of the cluster-randomised trial and reported the effective sample size. The results presented below are based on low intracluster correlation coefficients (ICC). In addition, we report the results of sensitivity analyses using a high ICC. These results can be found in [Table 4](#) and [Table 5](#).

(1) Supplementation with vitamin B₁₂ alone versus placebo (13 trials)

In this review, 13 trials evaluated the effects of vitamin B₁₂ compared to placebo (Areekul 1979; Benjamin 1952; Chinnock 1952; Craigmile 1956; Deshmukh 2010; Fukui 1959; Kudo 1962; Murakami 1962; Scrimshaw 1953; Scrimshaw 1959; Shimazono 1963; Taneja 2013; Yajnik 2021); eight trials contributed data to meta-analyses (Areekul 1979; Craigmile 1956; Deshmukh 2010; Kudo 1962; Murakami 1962; Shimazono 1963; Taneja 2013; Yajnik 2021).

Critical outcomes

See [Summary of findings 1](#); [Supplementary material 12](#).

1. Growth

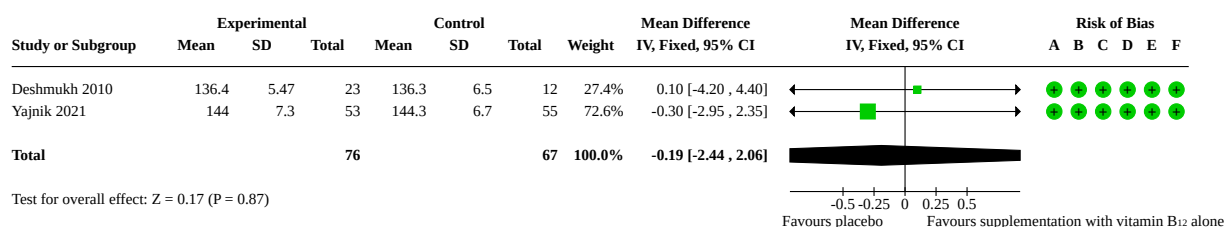
Height/length (cm)

Nine trials reported height/length in children (Benjamin 1952; Chinnock 1952; Craigmile 1956; Deshmukh 2010; Fukui 1959; Scrimshaw 1953; Scrimshaw 1959; Shimazono 1963; Yajnik 2021). Two trials reported data for meta-analyses (Deshmukh 2010; Yajnik 2021).

There may be little to no difference in height/length between groups (MD -0.19 cm, 95% CI -2.44 to 2.06; 2 trials, effective sample size 143; low-certainty evidence; heterogeneity: $\chi^2 = 0.02$, $df = 1$ ($P = 0.88$); $I^2 = 0\%$; Analysis 1.1; [Figure 2](#)). Both trials contributing

data to this outcome were conducted in India, with 25% of children reported as vitamin B₁₂ deficient (< 148 pmol/L) at enrolment.

Figure 2. Forest plot for Supplementation with vitamin B12 alone versus placebo, Outcome 1.1 Growth: Height or length (cm) (ALL)



Risk of bias legend

- (A) Bias arising from the randomization process
- (B) Bias due to deviations from intended interventions
- (C) Bias due to missing outcome data
- (D) Bias in measurement of the outcome
- (E) Bias in selection of the reported result
- (F) Overall bias

Findings were similar with sensitivity analyses using a high ICC for a cluster-randomised trial (Deshmukh 2010) (MD -0.25 cm, 95% CI -2.70 to 2.21; 2 trials, effective sample size 123; heterogeneity: $\text{Chi}^2 = 0.01$, $df = 1$ ($P = 0.91$); $I^2 = 0\%$; Table 4).

In individual trials, there was little to no difference between groups for height/length at different time points (Benjamin 1952: 8 weeks (vitamin B₁₂ 0.76 cm versus placebo 0.78 cm); Chinnock 1952: 8 weeks (vitamin B₁₂ 6.68 cm versus placebo 8.69 cm); Craigmile 1956: 6 months (vitamin B₁₂ 1.17 cm versus placebo 1.17 cm); Fukui 1959: 6 months (vitamin B₁₂ 3.14 cm versus placebo 2.96 cm); Scrimshaw 1953: 11 to 28 months (vitamin B₁₂ 0.43 cm versus placebo 0.39 cm) and 28 to 31 months (vitamin B₁₂ 0.48 cm versus placebo 0.40 cm); Scrimshaw 1959: 18 months (vitamin B₁₂ 0.43 cm versus placebo 0.48 cm); Shimazono 1963: 12 months (vitamin B₁₂ 5.53 cm versus placebo 5.34 cm)).

Height-for-age z-score (HAZ) or length-for-age z-score (LAZ) (WHO growth standards)

In ancillary analyses of one trial (Taneja 2013) [134], there may be little to no difference between groups in HAZ (MD 0.05, 95% CI -0.12 to 0.23; 403 children; low-certainty evidence).

2. Cognitive function, using validated instruments, including general cognitive ability or cognitive domains

Two trials reported cognitive outcomes (Deshmukh 2010; Taneja 2013) in children aged 6 to 9 years (Taneja 2013 [129]) and 11 years (Deshmukh 2010). Studies reporting cognitive function were conducted in India, with baseline vitamin B₁₂ deficiency prevalence ranging from 17% (< 150 pmol/L) to 25% (< 148 pmol/L).

Cognitive domains

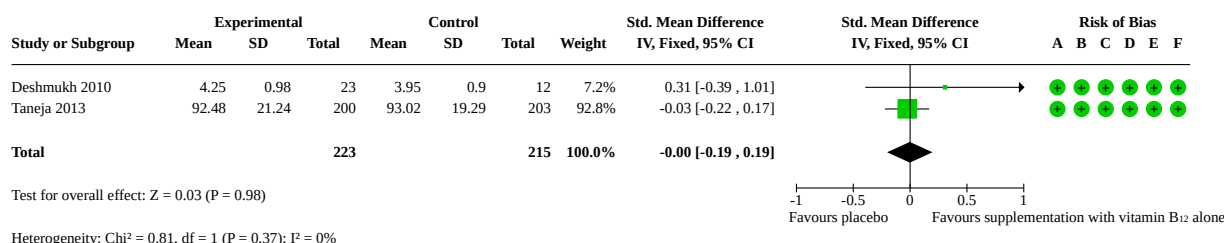
There may be little to no difference in cognitive domains (perceptual reasoning, verbal comprehension, visuo-spatial function, working memory, attention, processing speed) between groups.

- **Perceptual reasoning:** In ancillary analyses of one trial (Taneja 2013 [129]), there may be little to no difference in perceptual reasoning (WISC-IV^{INDIA}; Wechsler Intelligence Scale for Children, Fourth Edition (India)) between groups (mean (SD): vitamin B₁₂ 94.77 (17.03) versus placebo 95.83 (16.22); 403 children; low-certainty evidence).
- **Verbal comprehension:** Similarly, in ancillary analyses of Taneja 2013 [129], there may be little to no difference in verbal comprehension (CVS) between groups (mean (SD): vitamin B₁₂ 98.26 (12.78) versus placebo 100.20 (SD 13.76); 403 children; low-certainty evidence).
- **Visuo-spatial function:** Two trials contributed data on visuo-spatial function (Taneja 2013 [129] using the NEPSY-II, and Deshmukh 2010 using Malin's Intelligence Scale for Children). There may be little to no difference in visuo-spatial function between groups (SMD -0.13, 95% CI -0.32 to 0.06; 2 trials, effective sample size 438; low-certainty evidence; heterogeneity: $\text{Chi}^2 = 0.61$, $df = 1$ ($P = 0.43$); $I^2 = 0\%$; Analysis 1.2). Findings were similar in sensitivity analyses using a high ICC for a cluster-randomised trial (Deshmukh 2010) (SMD -0.12, 95% CI -0.31 to 0.08; 2 trials, effective sample size 418; heterogeneity: $\text{Chi}^2 = 0.24$, $df = 1$ ($P = 0.62$); $I^2 = 0\%$; Table 4).
- **Working memory:** Two trials reported data on working memory (Taneja 2013 [129] using WISC-IV^{INDIA} and Deshmukh 2010 using Malin's Intelligence Scale for Children). There may be little to no difference between groups in working memory (SMD -0.00, 95% CI -0.19 to 0.19; 2 trials, effective sample size 438; low-certainty evidence; heterogeneity: $\text{Chi}^2 = 0.81$, $df = 1$ ($P = 0.37$); $I^2 = 0\%$; Analysis 1.3; Figure 3). Findings were similar in sensitivity analyses using a high ICC for a cluster-randomised trial (Deshmukh 2010) (SMD -0.02, 95% CI -0.21 to 0.18; 2 trials, effective sample size 418; heterogeneity: $\text{Chi}^2 = 0.33$, $df = 1$ ($P = 0.57$); $I^2 = 0\%$; Table 4).
- **Attention:** Two trials (Deshmukh 2010; Taneja 2013 [129]) contributed data on attention (NEPSY-II, attention and executive function (Taneja 2013); Color Trail Test, decision-making and attention domain (Deshmukh 2010)). There may be little to no difference between groups (SMD 0.09, 95% CI -0.10 to

0.28; 2 trials, effective sample size 438; low-certainty evidence; heterogeneity: $\text{Chi}^2 = 1.08$, $\text{df} = 1$ ($P = 0.30$); $I^2 = 7\%$; Analysis 1.4). Findings were similar in sensitivity analyses using a high ICC for a cluster-randomised trial (Deshmukh 2010) (SMD 0.11, 95% CI -0.08 to 0.30; 2 trials, effective sample size 418; heterogeneity: $\text{Chi}^2 = 0.45$, $\text{df} = 1$ ($P = 0.50$); $I^2 = 0\%$; Table 4).

- **Processing speed:** In one trial (Taneja 2013 [129]), there may be little to no difference in processing speed (WISC-IV^{INDIA}) between groups (403 children) (mean (SD): vitamin B₁₂ 97.12 (15.94) versus placebo 98.10 (15.18); 403 children; low-certainty evidence).

Figure 3. Forest plot for Supplementation with vitamin B12 alone versus placebo, Outcome 1.3 Cognitive function: Working memory (ALL)



Intelligence quotient (IQ)

One trial, Deshmukh 2010, conducted in India (25% of participants were vitamin B₁₂ deficient (< 148 pmol/L) at enrolment) reported data for IQ (i.e. Raven's Color Progressive Matrices; color progressive matrices score). There may be little to no difference between groups (mean (SD): vitamin B₁₂ 26.90 (7.85) versus placebo 30.00 (6.00); effective sample size: 35 children; low-certainty evidence).

3. Development outcomes (e.g. motor skill development, language, and socio-emotional and adaptive behaviour), assessed using validated instruments

Ancillary analyses of one trial (Taneja 2013 [128, 129]) conducted in India (17% of participants were vitamin B₁₂ deficient (< 150 pmol/L) at baseline) reported data on development outcomes in children at 12 to 36 months and 6 to 9 years of age. There may be little to no difference in development outcomes (fine and gross motor skill development, socio-emotional and adaptive behaviour, language) between groups.

- **Motor skill development:** There may be little to no difference between groups in motor skill development domains (i.e. NEPSY-II: sensorimotor domain, mean (SD): vitamin B₁₂ 9.11 (2.71) versus placebo 9.30 (2.80); 403 children; low-certainty evidence) or fine motor domains (ASQ-3: vitamin B₁₂ 47.10 (13.30) versus placebo 47.60 (13.20); 214 children; low-certainty evidence); however children receiving vitamin B₁₂

supplementation may have an improved gross motor domain (ASQ-3: vitamin B₁₂ 46.80 (13.60) versus placebo 42.80 (15.20); 214 children; low-certainty evidence).

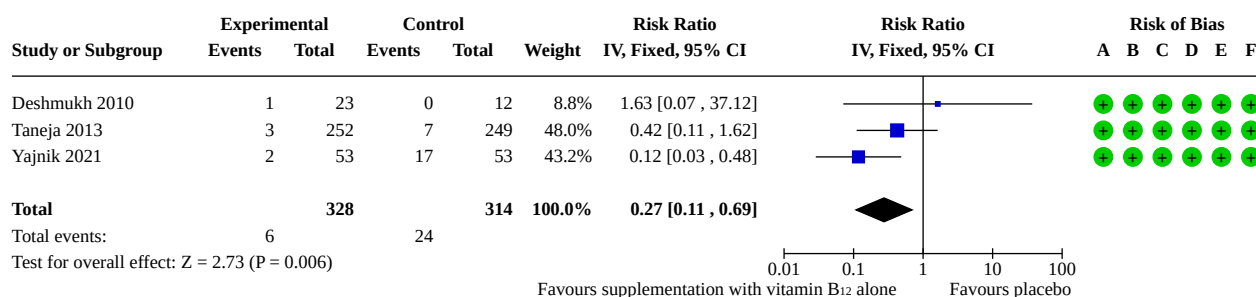
- **Socio-emotional and adaptive behaviour:** There may be little to no difference in socio-emotional and adaptive behaviour (NEPSY-II: social perception domain) between groups (mean (SD): vitamin B₁₂ 10.01 (2.44) versus placebo 9.98 (2.38); 403 children; low-certainty evidence).
- **Language:** Similarly, there may be little to no difference in language (NEPSY-II) between groups (mean (SD): vitamin B₁₂ 7.54 (2.62) versus placebo 7.62 (2.62); 403 children; low-certainty evidence).

4. Vitamin B12 status

Vitamin B12 deficiency, as defined by trial authors (e.g. < 148, < 221 pmol/L)

Three trials reported data on vitamin B₁₂ deficiency (< 148 pmol/L: Deshmukh 2010; Yajnik 2021; < 150 pmol/L: Taneja 2013). Included trials were conducted in India, with baseline vitamin B₁₂ deficiency prevalence ranging from 17% (< 150 pmol/L) to 25% (< 148 pmol/L).

Vitamin B₁₂ supplementation reduces vitamin B₁₂ deficiency, compared to placebo (1.8% versus 7.6%; RR 0.27, 95% CI 0.11 to 0.69; 3 trials, effective sample size 642; high-certainty evidence; heterogeneity: $\text{Chi}^2 = 3.02$, $\text{df} = 2$ ($P = 0.22$); $I^2 = 34\%$; Analysis 1.5; Figure 4).

Figure 4. Forest plot for Supplementation with vitamin B12 alone versus placebo, Outcome 1. 5 Vitamin B₁₂ status: Vitamin B₁₂ deficiency, as defined by trial authors (e.g. < 148, < 221 pmol/L) (ALL)

Heterogeneity: Chi² = 3.02, df = 2 (P = 0.22); I² = 34%

Risk of bias legend

- (A) Bias arising from the randomization process
- (B) Bias due to deviations from intended interventions
- (C) Bias due to missing outcome data
- (D) Bias in measurement of the outcome
- (E) Bias in selection of the reported result
- (F) Overall bias

There was little to no evidence of a difference between subgroups for age for this outcome (test for subgroup differences: Chi² = 2.42, df = 1 (P = 0.12); I² = 58.7%; Analysis 1.6).

Findings were similar in sensitivity analyses using a high ICC for a cluster-randomised trial (Deshmukh 2010) (RR 0.23, 95% CI 0.09 to 0.61; 3 trials, effective sample size 622; heterogeneity: Chi² = 1.66, df = 1 (P = 0.20); I² = 40%; Table 4).

Total vitamin B₁₂ concentrations (pmol/L)

Seven trials reported total vitamin B₁₂ concentrations (Areekul 1979; Deshmukh 2010; Kudo 1962; Murakami 1962; Shimazono 1963; Taneja 2013; Yajnik 2021); trials were conducted in Thailand, India, and Japan. Baseline vitamin B₁₂ deficiency was reported in trials conducted in India (Deshmukh 2010; Taneja 2013; Yajnik 2021), with prevalence ranging from 17% (< 150 pmol/L) to 25% (< 148 pmol/L). Trials conducted in Thailand (Areekul 1979) and Japan (Kudo 1962; Murakami 1962; Shimazono 1963) reported total vitamin B₁₂ concentrations but did not report vitamin B₁₂ deficiency (see Supplementary material 2).

In analyses of seven trials, vitamin B₁₂ supplementation probably increases total vitamin B₁₂ concentrations, compared to placebo (MD 68.53 pmol/L, 95% CI 41.83 to 95.23; 7 trials, effective sample size 705; moderate-certainty evidence; random-effects, heterogeneity: Tau² = 648.03; Chi² = 16.08, df = 6, P = 0.01; I² = 63%; Analysis 1.7).

There was little to no evidence of a difference between subgroups by age (Chi² = 0.59, df = 1, P = 0.44; I² = 0%; Analysis 1.8) or form of vitamin B₁₂ supplementation (Chi² = 0.72, df = 1, P = 0.40; I² = 0%; Analysis 1.10); however, tests for subgroup differences suggested variation by sex (Chi² = 13.21, df = 2, P = 0.001; I² = 84.9%; Analysis 1.9). These findings should be interpreted with caution given the heterogeneity, small number of trials, and distribution across subgroups.

Findings from sensitivity analyses were similar after excluding four trials with a high overall risk of bias (Areekul 1979; Kudo 1962; Murakami 1962; Shimazono 1963) (MD 76.62 pmol/L, 95% CI 37.64 to 115.60; 3 trials, effective sample size 273; random-effects, heterogeneity: Tau² = 349.52; Chi² = 2.80, df = 2, P = 0.25; I² = 29%) (see Table 6). Findings were also similar in sensitivity analyses using a fixed-effect model (MD 58.82 pmol/L, 95% CI 46.02 to 71.63; 7 trials, effective sample size 705; heterogeneity: Chi² = 16.08, df = 6, P = 0.01; I² = 63%; Table 6). Findings were similar when using a high ICC for a cluster-randomised trial (Deshmukh 2010) (MD 70.45 pmol/L, 95% CI 43.07 to 97.84; 7 trials, effective sample size 685; random-effects, heterogeneity: Tau² = 649.03; Chi² = 15.67, df = 6, P = 0.02; I² = 62%; Table 4).

Methylmalonic acid (MMA) concentrations (μmol/L)

No studies reported on MMA concentrations.

Holotranscobalamin concentrations (pmol/L)

In one trial, conducted in Thailand (Areekul 1979), the evidence is very uncertain for holotranscobalamin concentrations between groups (8 children) (mean (SD): vitamin B₁₂ 162.22 (31.85) versus placebo 234.07 (185.19) pmol/L; very low-certainty evidence).

Homocysteine concentrations (μmol/L)

Three trials contributed data on homocysteine (Deshmukh 2010; Taneja 2013; Yajnik 2021). Included trials were conducted in India, with vitamin B₁₂ deficiency prevalence at enrolment ranging from 17% (< 150 pmol/L) to 25% (< 148 pmol/L).

Vitamin B₁₂ supplementation may lower homocysteine, compared to placebo (MD -3.97 μmol/L, 95% CI -6.16 to -1.78; 3 trials, effective sample size 254; low-certainty evidence; random-effects, heterogeneity: Tau² = 2.47; Chi² = 6.72, df = 2; P = 0.03; I² = 70%; Analysis 1.11); however, due to the heterogeneity in analyses, the findings should be interpreted with caution.

There was little to no difference between subgroups by age for this outcome (test for subgroup differences: $\text{Chi}^2 = 2.01$, $\text{df} = 1$ ($P = 0.16$), $I^2 = 50.2\%$; Analysis 1.12).

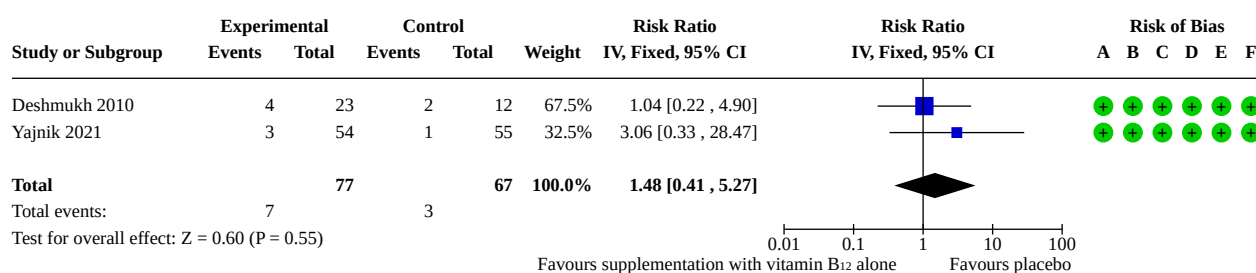
Findings were similar in sensitivity analyses using a high ICC for a cluster-randomised trial (Deshmukh 2010) (MD -3.85 $\mu\text{mol/L}$, 95% CI -6.17 to -1.53; 3 trials, effective sample size 244; random-effects, heterogeneity: $\text{Tau}^2 = 2.51$; $\text{Chi}^2 = 6.24$, $\text{df} = 2$ ($P = 0.04$); $I^2 = 68\%$; Table 4).

5. Anaemia and haemoglobin

Anaemia, defined using age-specific cut-offs

In two trials conducted in India (25% of participants were vitamin B₁₂ deficient (< 148 pmol/L) at enrolment) (Deshmukh 2010; Yajnik 2021), vitamin B₁₂ supplementation may result in little to no difference in anaemia, compared to placebo (9.1% versus 4.5%; RR 1.48, 95% CI 0.41 to 5.27; 2 trials, effective sample size 144; low-certainty evidence; heterogeneity: $\text{Chi}^2 = 0.60$, $\text{df} = 1$ ($P = 0.44$); $I^2 = 0\%$; Analysis 1.13; Figure 5).

Figure 5. Forest plot for Supplementation with vitamin B12 alone versus placebo, Outcome 1.13 Anaemia and haemoglobin: anaemia, defined using age-specific cut-offs (ALL)



Risk of bias legend

- (A) Bias arising from the randomization process
- (B) Bias due to deviations from intended interventions
- (C) Bias due to missing outcome data
- (D) Bias in measurement of the outcome
- (E) Bias in selection of the reported result
- (F) Overall bias

Findings were similar in sensitivity analyses using a high ICC for a cluster-randomised trial (Deshmukh 2010) (RR 1.71, 95% CI 0.36 to 8.04; 2 trials, effective sample size 124; heterogeneity: $\text{Chi}^2 = 0.50$, $\text{df} = 1$ ($P = 0.48$); $I^2 = 0\%$; Table 4).

Haemoglobin (g/L) adjusted for altitude (and smoking if reported), when applicable

Five trials reported haemoglobin concentrations (Murakami 1962; Scrimshaw 1953; Shimazono 1963; Taneja 2013; Yajnik 2021). Included trials were conducted in Japan, India, and El Salvador and Guatemala. Baseline vitamin B₁₂ deficiency was reported in trials conducted in India (Taneja 2013; Yajnik 2021), with prevalence ranging from 17% (< 150 pmol/L) to 25% (< 148 pmol/L). Trials conducted in Japan (Murakami 1962; Shimazono 1963), but not El Salvador and Guatemala (Scrimshaw 1953), reported baseline total vitamin B₁₂ concentrations (see Supplementary material 2). Four trials contributed data to analyses (Murakami 1962; Shimazono 1963; Taneja 2013; Yajnik 2021).

There may be little to no difference between groups for haemoglobin (MD -0.43 g/L, 95% CI -1.72 to 0.87; 4 trials, 607 children; very low-certainty evidence; heterogeneity: $\text{Chi}^2 = 2.86$, $\text{df} = 3$; $P = 0.41$; $I^2 = 0\%$; Analysis 1.14); however, the evidence is very uncertain.

There was little to no difference in effect estimates across prespecified subgroups by age (test for subgroup differences: $\text{Chi}^2 = 1.06$, $\text{df} = 1$ ($P = 0.30$), $I^2 = 5.7\%$; Analysis 1.15) or form of vitamin

B₁₂ supplementation (test for subgroup differences: $\text{Chi}^2 = 0.45$, $\text{df} = 1$ ($P = 0.50$), $I^2 = 0\%$; Analysis 1.16).

In one trial (Scrimshaw 1953 [163]), there was little to no difference in haemoglobin between groups after two years (vitamin B₁₂ 145 $\pm$ 6 versus placebo 141 $\pm$ 9 g/L; 220 children).

6. Any adverse effects

No trials reported data on any adverse effects.

Important outcomes

See Supplementary material 13.

(2) Supplementation with vitamin B₁₂ alone versus no intervention (two trials)

In this review, two trials evaluated the effects of vitamin B₁₂ supplementation compared to no intervention (Gutiérrez-Díaz 1959; Scaglione 1955); no studies contributed data to quantitative analysis.

Critical outcomes

See Summary of findings 2; Supplementary material 14.

1. Growth

Height/length (cm)

Two trials, conducted in Italy and Spain, reported data for height/length at two days of age (Scaglione 1955), or age was not reported (Gutiérrez-Díaz 1959). The included trials did not report total vitamin B₁₂ concentrations or vitamin B₁₂ deficiency at enrolment.

In individual trials, there may be little to no difference between groups in height/length after 28 days (Scaglione 1955; mean (SD): vitamin B₁₂ 48.8 (0.9) cm versus no intervention 48.3 (0.9) cm, $P = 0.21$; 30 children); children who received vitamin B₁₂ may have increased height/length after 10 months versus no intervention (Gutiérrez-Díaz 1959; mean: vitamin B₁₂ 6.54 cm versus no intervention 5.96 cm; 40 children) (both very low-certainty evidence); however, the evidence is very uncertain. See Analysis 2.1 in [Supplementary material 7](#).

Height-for-age z-score (HAZ) or length-for-age z-score (LAZ) (WHO growth standards)

No trials reported data on HAZ.

2. Cognitive function, using validated instruments, including general cognitive ability or cognitive domains

No trials reported data on cognitive function.

3. Development outcomes (e.g. motor skill development, language, and socio-emotional and adaptive behaviour), assessed using validated instruments

No trials reported developmental outcomes.

4. Vitamin B₁₂ status

No trials reported data on vitamin B₁₂ status.

5. Anaemia and haemoglobin

No trials reported data for anaemia or haemoglobin.

6. Any adverse effects

No trials reported adverse effects.

Important outcomes

See Analysis 2.2 in [Supplementary material 15](#).

(3) Supplementation with vitamin B₁₂ and other micronutrients, compared to supplements with the same formulation without vitamin B₁₂ (three trials)

Three trials evaluated vitamin B₁₂ and other micronutrients compared to the same supplements without vitamin B₁₂ (Deshmukh 2010; Strand 2020; Taneja 2013). Two of the included trials were conducted in India (Deshmukh 2010; Taneja 2013), with baseline vitamin B₁₂ deficiency prevalence ranging from 17% (< 150 pmol/L) to 25% (< 148 pmol/L), whereas Strand 2020, which was conducted in Nepal, reported total vitamin B₁₂ concentrations but did not report vitamin B₁₂ deficiency at enrolment (see [Supplementary material 2](#)).

Critical outcomes

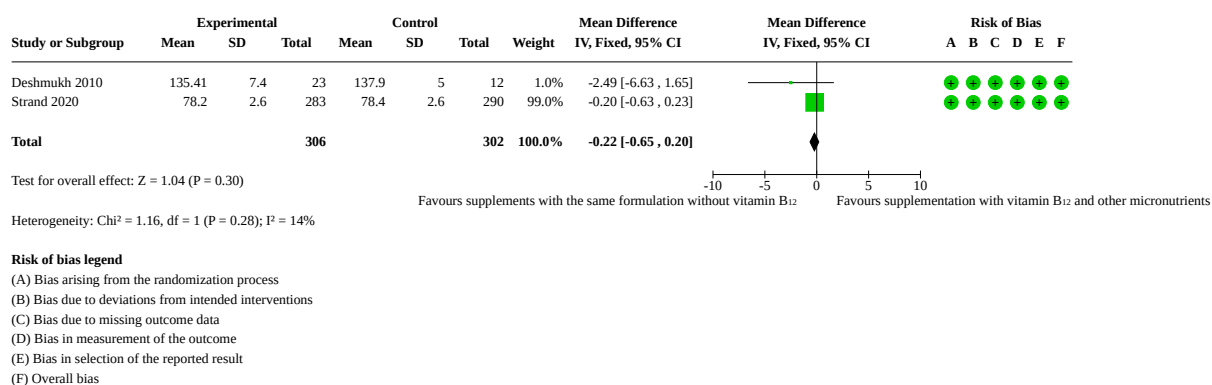
See [Summary of findings 3](#); [Supplementary material 16](#).

1. Growth

Height/length (cm)

In analyses of two trials (Deshmukh 2010; Strand 2020), there may be little to no difference between groups in height/length (MD -0.22 cm, 95% CI -0.65 to 0.20; 2 trials, effective sample size 608; low-certainty evidence; heterogeneity: $\text{Chi}^2 = 1.16$, $\text{df} = 1$ ($P = 0.28$); $I^2 = 14\%$; Analysis 2.1; [Figure 6](#)).

Figure 6. Forest plot for Supplementation with vitamin B₁₂ and other micronutrients, compared to supplements with the same formulation without vitamin B₁₂, Outcome 2.1 Growth: Height or length (cm) (ALL)



Findings were similar in sensitivity analyses using a high ICC for a cluster-randomised trial (Deshmukh 2010) (MD -0.21 cm, 95% CI -0.64 to 0.21; 2 trials, effective sample size 588; heterogeneity: $\text{Chi}^2 = 0.50$, $\text{df} = 1$ ($P = 0.48$); $I^2 = 0\%$; [Table 5](#)).

Height-for-age z-score (HAZ) or length-for-age z-score (LAZ) (WHO growth standards)

In two trials (Strand 2020; Taneja 2013), there may be little to no difference between intervention groups for HAZ (MD 0.02, 95% CI -0.24 to 0.27; 2 trials, 961 children; very low-certainty evidence; random-effects, heterogeneity: $\text{Tau}^2 = 0.03$; $\text{Chi}^2 = 5.22$, $\text{df} = 1$

($P = 0.02$); $I^2 = 81\%$; Analysis 2.2); however, the evidence is very uncertain.

2. Cognitive function, using validated instruments, including general cognitive ability or cognitive domains

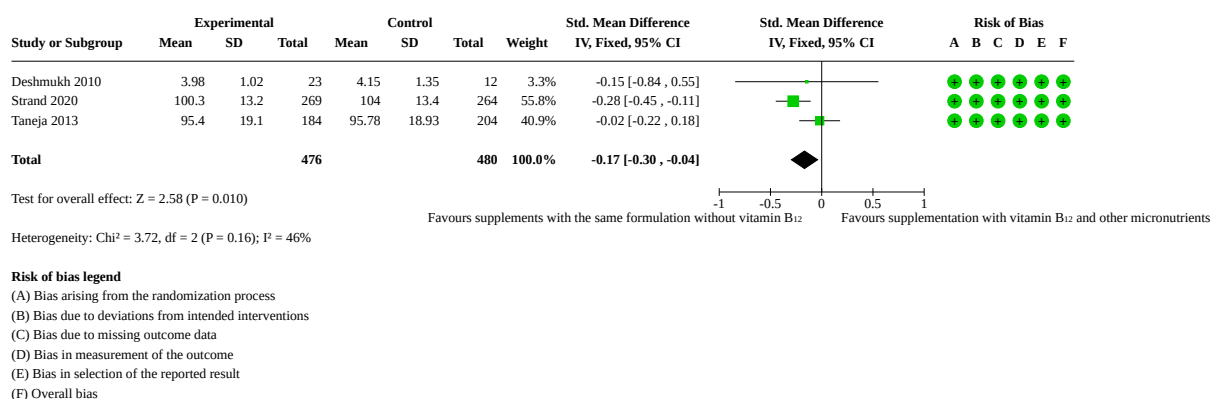
Three trials reported data on cognitive outcomes (Deshmukh 2010; Strand 2020; Taneja 2013), at 42 to 48 months (Strand 2020: 122), and 6 to 9 (Taneja 2013 [129]) and 11 years of age (Deshmukh 2010).

Cognitive domains

There may be little to no difference in most cognitive domains (including perceptual reasoning, verbal comprehension, visuo-spatial function, attention, processing speed) between groups.

- **Perceptual reasoning:** In one trial (Taneja 2013), there may be little to no difference in perceptual reasoning (WISC-IV^{INDIA}) between groups (mean: vitamin B₁₂ + micronutrients 97.18 (16.26) versus micronutrients alone 97.60 (16.51); 388 children; low-certainty evidence).
- **Verbal comprehension:** In analyses of two trials, there may be little or no difference in verbal comprehension (WPPSI-IV (Strand 2020); CVS (Taneja 2013)) between groups (SMD -0.06, 95% CI -0.19 to 0.07; 2 trials, 921 children; low-certainty evidence; heterogeneity: $\text{Chi}^2 = 0.00$, $\text{df} = 1$ ($P = 0.97$); $I^2 = 0\%$; Analysis 2.3).
- **Visuo-spatial function:** In three trials, there may be little to no difference in visuo-spatial function (NEPSY-II (Strand 2020; Taneja 2013); Malin's Intelligence Scale for Children (Deshmukh 2010)) between groups (SMD 0.00, 95% CI -0.12 to 0.13; 3 trials, effective sample size 956 children; low-certainty evidence; heterogeneity: $\text{Chi}^2 = 0.41$, $\text{df} = 2$ ($P = 0.81$); $I^2 = 0\%$; Analysis 2.4). For this outcome, there was little to no evidence of a difference between subgroups by age (test for subgroup differences: $\text{Chi}^2 = 0.36$, $\text{df} = 1$ ($P = 0.55$), $I^2 = 0\%$; Analysis 2.5) or cognitive assessment tools by age groups (test for subgroup differences: $\text{Chi}^2 = 0.36$, $\text{df} = 1$ ($P = 0.55$), $I^2 = 0\%$; Analysis 2.6). Findings were similar in sensitivity analyses using a high ICC for a cluster-randomised trial (Deshmukh 2010) (SMD 0.01, 95% CI -0.12 to 0.13; 3 trials, effective sample size 936; heterogeneity: $\text{Chi}^2 = 0.34$, $\text{df} = 2$ ($P = 0.84$); $I^2 = 0\%$; Table 5).
- **Attention:** In three trials (Deshmukh 2010; Strand 2020; Taneja 2013), there may be little to no difference in attention (NEPSY-II (Strand 2020; Taneja 2013); Color Trail Test (Deshmukh 2010)) (SMD 0.00, 95% CI -0.13 to 0.13; 3 trials, effective sample size 956; low-certainty evidence; heterogeneity: $\text{Chi}^2 = 0.63$, $\text{df} = 2$ ($P = 0.73$); $I^2 = 0\%$; Analysis 2.10). There was little to no evidence of a difference between subgroups by age (test for subgroup differences: $\text{Chi}^2 = 0.27$, $\text{df} = 1$ ($P = 0.61$), $I^2 = 0\%$; Analysis 2.11) or cognitive assessment tools by age groups (test for subgroup differences: $\text{Chi}^2 = 0.06$, $\text{df} = 1$ ($P = 0.81$), $I^2 = 0\%$; analysis 2.12) for this outcome. Findings were similar in sensitivity analyses using a high ICC for a cluster-randomised trial (Deshmukh 2010) (SMD 0.01, 95% CI -0.12 to -0.13; 3 trials, effective sample size 936; heterogeneity: $\text{Chi}^2 = 0.34$, $\text{df} = 2$ ($P = 0.84$); $I^2 = 0\%$; Table 5).
- **Processing speed:** In one trial (Taneja 2013), there may be little to no difference between groups for processing speed (WISC-IV^{INDIA}) (mean: vitamin B₁₂ + micronutrients 98.28 (15.06) versus micronutrients alone 98.29 (14.36); 388 children; low-certainty evidence).

Figure 7. Forest plot for Supplementation with vitamin B12 and other micronutrients, compared to supplements with the same formulation without vitamin B12, Outcome 2.7 Cognitive function: Working memory (ALL)



Intelligence quotient

In analyses of two trials, there may be little to no difference in intelligence quotient (Raven's Color Progressive matrices, colour

progressive matrices score (Deshmukh 2010); WPPSI-IV (Strand 2020)) between groups (SMD -0.07, 95% CI -0.23 to 0.10; 2 trials, effective sample size 568; low-certainty evidence; heterogeneity:

$\text{Chi}^2 = 0.03$, $\text{df} = 1$ ($P = 0.87$); $I^2 = 0\%$; Analysis 2.13). Findings were similar with sensitivity analyses using a high ICC for a cluster-randomised trial (Deshmukh 2010) (SMD -0.07, 95% CI -0.24 to 0.10; 2 trials, effective sample size 548; heterogeneity: $\text{Chi}^2 = 0.01$, $\text{df} = 1$ ($P = 0.92$); $I^2 = 0\%$; Table 5).

3. Development outcomes (e.g. motor skill development, language, and socio-emotional and adaptive behaviour), assessed using validated instruments

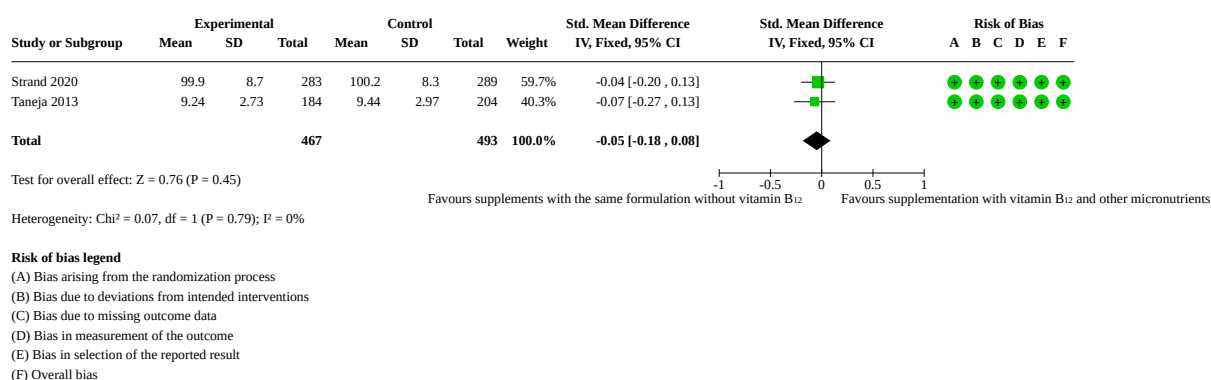
Two trials (Strand 2020; Taneja 2013) reported data on development outcomes (12 to 36 months (Taneja 2013 [128]); 30 to 35 months (Strand 2020 [122])). There may be little to no difference in development outcomes (fine and gross motor skill development, socio-emotional and adaptive behaviour, language) between groups.

- Motor skill development:** There may be little to no difference in motor skill development (NEPSY-II (Taneja 2013); Bayley Scale of Infant Development, Third Edition (BSID-III), (Strand 2020)) (SMD -0.05, 95% CI -0.18 to 0.08; 2 trials, 960 children; low-certainty evidence; heterogeneity: $\text{Chi}^2 = 0.07$, $\text{df} = 1$ ($P = 0.79$); $I^2 = 0\%$; Analysis 2.14; Figure 8), fine motor domains (ASQ-3 (Taneja 2013); BSID-III (Strand 2020)) (SMD 0.04, 95% CI -0.30 to 0.37;

2 trials, 780 children; low-certainty evidence; random-effects, heterogeneity: $\text{Tau}^2 = 0.05$; $\text{Chi}^2 = 4.47$, $\text{df} = 1$ ($P = 0.03$); $I^2 = 78\%$; Analysis 2.15), or gross motor domains (ASQ-3 (Taneja 2013); BSID-III (Strand 2020)) (SMD 0.11, 95% CI -0.03 to 0.25; 2 trials, 780 children; low-certainty evidence; heterogeneity: $\text{Chi}^2 = 1.57$, $\text{df} = 1$ ($P = 0.21$); $I^2 = 36\%$; Analysis 2.16).

- Socio-emotional and adaptive behaviour:** There may be little to no difference in socio-emotional and adaptive behaviour (NEPSY-II (Taneja 2013); BSID-III (Strand 2020)) between groups (SMD 0.07, 95% CI -0.06 to 0.19; 2 trials, 960 children; low-certainty evidence; heterogeneity: $\text{Chi}^2 = 0.08$, $\text{df} = 1$ ($P = 0.78$); $I^2 = 0\%$; Analysis 2.17).
- Language:** There may be little to no difference in language domains (NEPSY-II (Taneja 2013); BSID-III (Strand 2020)) between groups (SMD 0.07, 95% CI -0.06 to 0.20; 2 trials, 960 children; low-certainty evidence; heterogeneity: $\text{Chi}^2 = 0.57$, $\text{df} = 1$ ($P = 0.45$); $I^2 = 0\%$; Analysis 2.18). In one trial (Strand 2020), there may be little to no difference in receptive language domains (MD -0.13, 95% CI -0.53 to 0.27; 1 trial, 572 children; low-certainty evidence) or expressive language domains (MD -0.01, 95% CI -0.43 to 0.41; 1 trial, 572 children; low-certainty evidence) (BSID-III).

Figure 8. Forest plot for Supplementation with vitamin B12 and other micronutrients, compared to supplements with the same formulation without vitamin B12, Outcome 2.14 Developmental outcome: Motor skill development (ALL)



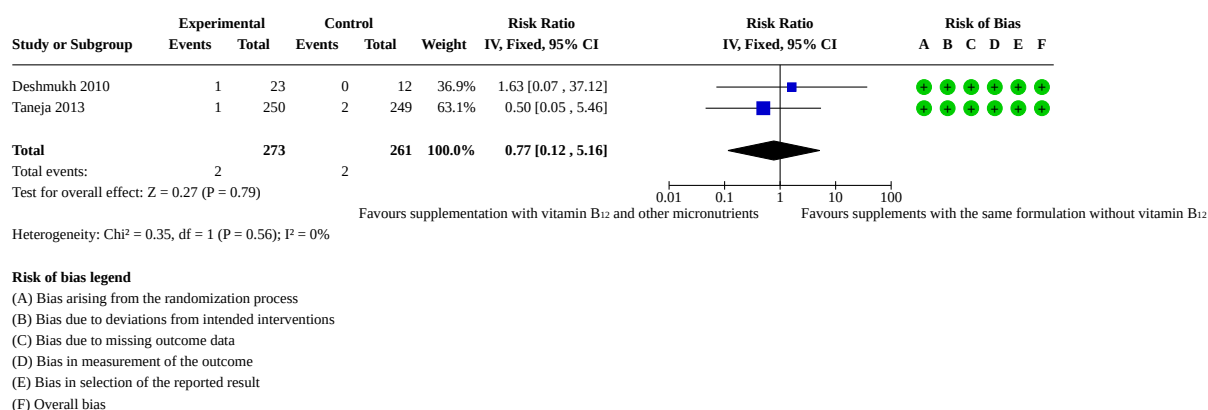
4. Vitamin B12 status

Vitamin B12 deficiency, as defined by trial authors (e.g. < 148, < 221 pmol/L)

Two trials reported data on vitamin B12 deficiency (< 148 pmol/L: Deshmukh 2010; < 150 pmol/L: Taneja 2013). Supplementation

with vitamin B12 and other micronutrients may result in little to no difference in vitamin B12 deficiency, compared to supplements without vitamin B12 (0.7% versus 0.8%; RR 0.77, 95% CI 0.12 to 5.16; 2 trials, effective sample size 534; low-certainty evidence; heterogeneity: $\text{Chi}^2 = 0.35$, $\text{df} = 1$ ($P = 0.56$); $I^2 = 0\%$; Analysis 2.19; Figure 9).

Figure 9. Forest plot for Supplementation with vitamin B12 and other micronutrients, compared to supplements with the same formulation without vitamin B12, Outcome 2.19 Vitamin B₁₂ status: Vitamin B₁₂ deficiency, as defined by trial authors (e.g. < 148, < 221 pmol/L) (ALL)



Findings were similar in sensitivity analyses using a high ICC for a cluster-randomised trial (Deshmukh 2010) (RR 0.78, 95% CI 0.12 to 5.15; 2 trials, effective sample size 514; heterogeneity: Chi² = 0.36, df = 1 (P = 0.55); I² = 0%; Table 5).

Total vitamin B₁₂ concentrations (pmol/L)

Three trials reported total vitamin B₁₂ concentrations (Deshmukh 2010; Strand 2020; Taneja 2013). Vitamin B₁₂ supplementation with other micronutrients may result in little to no difference in total vitamin B₁₂ concentrations, compared to supplements without vitamin B₁₂ (MD 34.32 pmol/L, 95% CI -31.13 to 99.77; 3 trials, effective sample size 731; low-certainty evidence; random-effects, heterogeneity: Tau² = 2545.69; Chi² = 8.81, df = 2 (P = 0.01); I² = 77%; Analysis 2.20). However, due to the heterogeneity in study results, the findings should be interpreted with caution.

There was little to no evidence of a difference between subgroups by age (test for subgroup differences: Chi² = 0.01, df = 1 (P = 0.91), I² = 0%; Analysis 2.21) or cognitive assessment tools by age groups (test for subgroup differences: Chi² = 0.01, df = 1 (P = 0.91), I² = 0%; Analysis 2.22) for this outcome.

Findings differed in sensitivity analyses using a fixed-effect model (MD 35.90, 95% CI 10.12 to 61.67; 3 trials, effective sample size 731; heterogeneity: Chi² = 8.81, df = 2 (P = 0.01); I² = 77%; Table 7). Findings were similar in sensitivity analyses using a high ICC for a cluster-randomised trial (Deshmukh 2010) (MD 42.21 pmol/L, 95% CI -18.50 to 102.91; 3 trials, effective sample size 711; random-effects, heterogeneity: Tau² = 1811.90; Chi² = 5.54, df = 2 (P = 0.06); I² = 64%; Table 5).

Methylmalonic acid (MMA) concentrations (μmol/L)

In one trial (Strand 2020), vitamin B₁₂ with other micronutrients may reduce mean MMA, compared to supplements without vitamin B₁₂ (mean: vitamin B₁₂ + micronutrients 0.37 (SD 0.26) versus micronutrients alone 0.53 (SD 0.45) μmol/L; 567 children; low-certainty evidence).

Holotranscobalamin concentrations (pmol/L)

No studies reported on holotranscobalamin concentrations.

Homocysteine concentrations (μmol/L)

Three trials reported homocysteine concentrations (Deshmukh 2010; Strand 2020; Taneja 2013). Vitamin B₁₂ supplementation with other micronutrients may reduce homocysteine, compared to the same formulation without vitamin B₁₂ (MD -1.35 μmol/L, 95% CI -2.68 to -0.01; 3 trials, effective sample size 716; low-certainty evidence; random-effects, heterogeneity: Tau² = 0.88; Chi² = 8.86, df = 2 (P = 0.01); I² = 77%; Analysis 2.23). Due to the heterogeneity in analyses, the findings should be interpreted with caution.

There was little to no difference between subgroups by age (test for subgroup differences: Chi² = 0.31, df = 1 (P = 0.57), I² = 0%; Analysis 2.24) or cognitive assessment tools by age groups (test for subgroup differences: Chi² = 0.31, df = 1 (P = 0.57), I² = 0%; Analysis 2.25) for this outcome.

In sensitivity analyses using a high ICC for a cluster-randomised trial (Deshmukh 2010), the findings differed (MD -1.25 μmol/L, 95% CI -2.60 to 0.09; 3 trials, effective sample size 706; random-effects, heterogeneity: Tau² = 0.85; Chi² = 8.45, df = 2 (P = 0.01); I² = 76%; Table 5).

5. Anaemia and haemoglobin

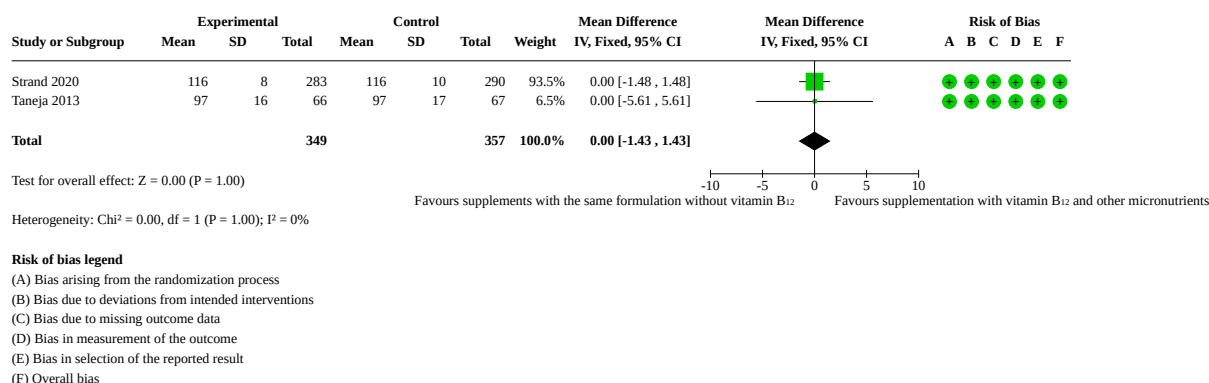
Anaemia, defined using age-specific cut-offs

In one trial (Deshmukh 2010), there may be little to no difference in anaemia between groups (vitamin B₁₂ + micronutrients 21.7% versus micronutrients alone 25.0%; 35 children; low-certainty evidence).

Haemoglobin (g/L) adjusted for altitude (and smoking if reported), when applicable

In two trials (Strand 2020; Taneja 2013), there may be little to no difference in haemoglobin concentrations between intervention groups (MD 0.00 g/L, 95% CI -1.43 to 1.43; 2 trials, 706 children; low-certainty evidence; heterogeneity: Chi² = 0.00, df = 1 (P = 1.00); Analysis 2.26; I² = 0%; Figure 10).

Figure 10. Forest plot for Supplementation with vitamin B12 and other micronutrients, compared to supplements with the same formulation without vitamin B12, Outcome 2.26 Anaemia and haemoglobin: haemoglobin (g/L) adjusted for altitude (and smoking if reported), when applicable (ALL)



6. Any adverse effects

One trial, Strand 2020, reported data on any adverse effects. There may be little to no difference between groups (vitamin B₁₂ + micronutrients 9.2% versus micronutrients alone 5.9%; 574 children; low-certainty evidence).

Important outcomes

See [Supplementary material 17](#).

Equity assessment

In this review, the included studies enrolled a wide age range, with children aged 2 days to 11 years. Most studies included both sexes, although some trials included boys alone, limiting equity-sensitive interpretation by sex. Most participants were apparently healthy, often recruited through schools, clinics or community settings, which may reflect access to healthcare or education services.

Reporting on race/ethnicity, socioeconomic status, education, and disability was limited. Most trials were conducted in high-income countries ($n = 9$), and the lack of participant-level equity data, including socioeconomic status, constrains interpretation within and across settings. In addition, most included studies did not report the prevalence of vitamin B₁₂ deficiency or insufficiency at enrolment among children. Studies that reported vitamin B₁₂ concentrations at baseline typically included mixed populations and did not provide enough detail to differentiate subgroups by baseline vitamin B₁₂ status.

See [Supplementary material 18](#).

Reporting biases

The assessment of reporting bias for the included studies is reported in [Risk of bias in included studies](#).

DISCUSSION

Summary of main results

This review was conducted to examine the effects of vitamin B₁₂ supplementation on health outcomes in children, including growth, development, and cognition. Sixteen trials involving 4083 children were included, comparing vitamin B₁₂ supplementation to

placebo ($n = 13$) or no intervention ($n = 2$), or vitamin B₁₂ with other micronutrients compared to micronutrients alone ($n = 3$). Nine trials contributed data to meta-analyses, and two trials contributed data to more than one comparison.

Trials were conducted in South-East Asia ($n = 5$; India, Nepal, Thailand); Western Pacific ($n = 4$; Japan); Europe ($n = 4$; Italy, Spain, United Kingdom); and the Americas ($n = 3$; El Salvador, Guatemala, United States). Most trials were conducted in high-income countries ($n = 9$; UK, US, Japan, Italy, Spain), with three trials in upper-middle income countries ($n = 3$; El Salvador, Guatemala, Thailand), and four trials in lower-middle income countries ($n = 4$; India, Nepal). Participant age range (2 days to 11 years), supplementation duration (28 days to 31 months), and prevalence of anaemia (3% to 70%) and vitamin B₁₂ deficiency (17% to 25%) ranged considerably, although most included trials did not report vitamin B₁₂ status at enrolment. Child growth (height/length), vitamin B₁₂ status (total vitamin B₁₂ concentrations), and haemoglobin were the commonly reported outcomes. Three trials evaluated cognitive or development outcomes.

Supplementation with vitamin B₁₂ alone versus placebo (13 trials)

There may be little to no difference in growth (height/length), cognitive function (e.g. attention), or development outcomes (e.g. motor skill development) (all low-certainty evidence). In children, trials ($n = 3$) conducted in India, where vitamin B₁₂ deficiency prevalence at enrolment ranged from 17% (< 150 pmol/L) to 25% (< 148 pmol/L), suggest that vitamin B₁₂ supplementation reduces the risk of vitamin B₁₂ deficiency by 56 fewer per 1000 children (high-certainty evidence) and probably increases total vitamin B₁₂ concentrations by 68.5 pmol/L (moderate-certainty evidence), compared to placebo. Evidence from one trial conducted in Thailand contributed to meta-analyses of total vitamin B₁₂ concentrations but did not report vitamin B₁₂ deficiency at baseline. There may be little to no difference between groups for anaemia (low-certainty evidence). No trials reported on the occurrence of any adverse effects. See [Summary of findings 1](#).

Supplementation with vitamin B₁₂ alone versus no intervention (two trials)

Two trials conducted in Italy and Spain (total vitamin B₁₂ concentrations or vitamin B₁₂ deficiency at enrolment not reported) contributed to narrative synthesis in this comparison (i.e. growth (height/length); very low-certainty evidence). No trial reported data for other critical outcomes, including any adverse effects. See [Summary of findings 2](#).

Supplementation with vitamin B₁₂ and other micronutrients, compared to supplements with the same formulation without vitamin B₁₂ (three trials)

Two of the trials included in this comparison were conducted in India (Deshmukh 2010; Taneja 2013), with baseline vitamin B₁₂ deficiency prevalence ranging from 17% (< 150 pmol/L) to 25% (< 148 pmol/L), whereas Strand 2020, conducted in Nepal, reported total vitamin B₁₂ concentrations but did not report vitamin B₁₂ deficiency at enrolment. There may be little to no difference between groups in growth (i.e. height/length), cognitive function (e.g. attention), development outcomes (e.g. motor skill development), vitamin B₁₂ deficiency, total vitamin B₁₂ concentrations, anaemia, or any adverse effects (all low-certainty evidence). See [Summary of findings 3](#).

Limitations of the evidence included in the review

Based on trial setting, studies conducted in India [27, 164, 165], Nepal [166], Guatemala [26, 167], or El Salvador [168] may have enrolled participants with a high prevalence of vitamin B₁₂ deficiency. In contrast, studies from the United States, United Kingdom, or Europe probably included predominantly vitamin B₁₂ replete participants. However, due to the lack of participant-level data and variability in recruitment settings, this constrained the interpretation of findings. Future trials should report baseline nutritional status to improve the interpretability of pooled estimates.

In this review, although we aimed to assess growth, development, and cognition, child growth (height/length), vitamin B₁₂ status (total vitamin B₁₂ concentrations), and haemoglobin concentrations were the most commonly reported outcomes; hence, we designated these as critical outcomes. The wide age range of children, and the timing of outcome assessment, constrained comparability of findings across trials. It is noteworthy that variations in the bioavailability of vitamin B₁₂ based on its form, combined with the lack of detailed reporting in included trials, could influence the effectiveness of supplementation and the comparability of results across studies. The limited number of trials reporting on *a priori* outcomes constrained synthesis and interpretation of findings and subgroup analyses. For instance, subgroup analyses included fewer than 10 trials, and covariates were unequally distributed across subgroups, which may increase the risk of type I error. These analyses were retained to explore biologically plausible effect differences but should be interpreted with caution.

In addition, in this review, cognitive function was analysed at the domain level by consolidating similar sub-tests from different cognitive tools (e.g. BSID-III, WISC-IV, or NEPSY-II) to reduce heterogeneity and improve precision, leveraging their shared

constructs [169]. However, the diversity of instruments used across trials limits comparability with other reviews that may have focused on specific cognitive tools or standardised effect sizes [170]. While this approach allows for meaningful meta-analysis, instrument variability constrains cross-study comparability and warrants cautious interpretation.

Risk of bias

Allocation concealment and random sequence generation methods were adequately reported in four trials and five trials lacked sufficient information for these methods. The main sources of bias were: deviations from intended interventions, outcome measurement, and selection of reported results, due to inadequate blinding information, adherence data, and inappropriate methods for intervention effects.

The risk of bias for outcome measurement was due to insufficient details on ascertainment methods. For laboratory outcomes, non-blinding of assessors was less likely to affect results. However, for other outcomes (e.g. any adverse effects), the lack of blinding raised some concerns. Seven trials also did not provide enough information on pre-specified analysis plans, limiting the ability to evaluate risk of bias for outcomes or analysis.

Certainty of the evidence

The certainty of evidence for critical outcomes was of moderate to very low certainty (See [Summary of findings 1](#); [Summary of findings 2](#); [Summary of findings 3](#)).

Supplementation with vitamin B₁₂ versus placebo

The certainty of evidence for this comparison ranged from high to low-certainty. The overall certainty of evidence for growth (i.e. height/length), cognitive function (e.g. attention), development outcomes (e.g. motor skill development), and anaemia was low, due to imprecision (small sample sizes, few events, imprecise estimates) [171]. We downgraded vitamin B₁₂ status (i.e. total vitamin B₁₂; moderate-certainty) by one level due to serious concerns regarding randomisation, deviations from intended interventions, and selective reporting. We did not further downgrade vitamin B₁₂ deficiency (high-certainty).

Supplementation with vitamin B₁₂ versus no intervention

No trials provided data that could be included in meta-analyses. We downgraded the evidence (very low-certainty for growth) by three levels due to serious concerns about randomisation, outcome measurement, selective reporting, and imprecision.

Supplementation with vitamin B₁₂ and other micronutrients, compared to supplements with the same formulation without vitamin B₁₂

The certainty of evidence for this comparison was low. We downgraded growth (height/length), cognition, development outcomes, vitamin B₁₂ deficiency, anaemia, and any adverse effects (all low-certainty) by two levels due to imprecision. We downgraded vitamin B₁₂ status (total vitamin B₁₂; low-certainty) by two levels due to inconsistency and imprecision.

Limitations of the review processes

We were aware of the potential for introducing bias throughout the review process. In this review, we attempted to minimise bias in several ways. This review was conducted using Cochrane methodology, to minimise bias throughout. An information specialist independently conducted searches of trial registries and databases, with no restrictions on date or language. Review authors independently evaluated eligibility, extracted data, and evaluated risk of bias and quality of evidence. Discrepancies were resolved through discussion and consultation with the senior author.

There are several important limitations in this review. Most outcomes were informed by few trials (e.g. one trial), rare events were not reported (e.g. all-cause mortality), or estimates were imprecise despite large sample sizes. We conducted subgroup analyses ($\geq$ three trials) to explore biological plausibility and transparently present all available evidence on the effect of vitamin B₁₂ supplementation, given the clinical relevance in children less than 12 years of age. However, the small number of studies limits the reliability of subgroup comparisons, and the results should be interpreted with caution.

We observed differences in effect estimates between the fixed-effect and random-effects meta-analyses (i.e. for total vitamin B₁₂ concentrations), which may indicate the presence of small-study effects ([Summary of findings 3](#)).

Some included studies did not report results following the intention-to-treat principle, and data were missing for participants lost to follow-up or excluded post-randomisation. This limited the completeness of outcome data and may have introduced attrition bias or impacted the precision of effect estimates. However, we downgraded the certainty of evidence for risk of bias if missingness was likely to have affected the outcome or for imprecision due to small sample sizes or wide confidence intervals.

It was possible that we did not include unpublished or unregistered studies. However, to ensure inclusion of important studies, we checked the reference lists of included studies and systematic reviews.

In addition, assessing the certainty of the evidence is a challenging process. Subjective judgements, beliefs, and attitudes could introduce bias. Although there were no restrictions on language in searches, most reports were in English and relied largely on published reports.

Agreements and disagreements with other studies or reviews

Systematic reviews of randomised trials

To date, no systematic reviews have been conducted to evaluate the effects of vitamin B₁₂ supplementation in children on health outcomes, to which findings from this review are directly comparable.

In a recent Cochrane review of the effects of vitamin B₁₂ supplementation during pregnancy on maternal and child health outcomes [61], prenatal vitamin B₁₂ supplementation improved vitamin B₁₂ concentrations in children. However, the effects of maternal vitamin B₁₂ supplementation on longer-term child health

outcomes, including anaemia, growth, cognition, developmental outcomes, or adverse effects, are uncertain. These findings are consistent with the current review. However, its focus on maternal vitamin B₁₂ supplementation constrains the comparability of findings.

Reviews of observational studies

In a review of observational (cross-sectional, case-control, cohort) studies on vitamin B₁₂ intake/status and cognition in children, higher child vitamin B₁₂ status was associated with improved cognitive performance, school performance, developmental indexes, and problem-solving skills [41]. In a systematic review (including cross-sectional and case-control studies), lower vitamin B₁₂ concentrations were associated with delays in infant motor development [172]. In a systematic review and meta-analysis of micronutrients and cognition in children (6 to 11 years), vitamin B₁₂ intake (animal-source foods) was associated with improved academic performance ($n = 1$ study) [173]. In contrast, in a systematic review of vitamin B₁₂ biomarkers/intake and child health, higher vitamin B₁₂ intake was associated with increased serum vitamin B₁₂ concentrations, but increased vitamin B₁₂ intake was not associated with cognitive outcomes in children ($n = 2$ studies) [174].

Other reviews

The effects of multiple micronutrients (including vitamin B₁₂) on child health outcomes have been evaluated in other systematic reviews [175, 176], meta-analyses [177], and other Cochrane reviews [178, 179].

Additional trials

We identified one ongoing trial in this review (CTRI/2023/09/057950) on the effect of vitamin B₁₂ + micronutrients versus micronutrients alone in children (1 to 60 months) (CTRI/2023/09/057950 2023).

AUTHORS' CONCLUSIONS

Implications for practice

Vitamin B₁₂ supplementation (alone or when supplemented with other micronutrients) in children may make little to no difference to growth (i.e. height/length), cognitive function (e.g. attention), development outcomes (e.g. motor skill development), or anaemia, compared to placebo, no intervention, or the same supplements without vitamin B₁₂; the evidence is of low certainty. Vitamin B₁₂ supplementation alone reduces the risk of vitamin B₁₂ deficiency in children and probably improves total vitamin B₁₂ concentrations, compared to placebo. Vitamin B₁₂ supplementation with other micronutrients, compared to micronutrients alone, may result in little to no difference in the risk of vitamin B₁₂ deficiency and may result in little to no difference in total vitamin B₁₂ concentrations. However, the evidence on the effects of vitamin B₁₂ supplementation on other child health outcomes is limited.

The heterogeneity in study design (including instruments, participant age, duration) constrained the comparability of findings in this review.

Equity-related implications for practice

Most of the trials in this review were conducted in high-income countries ($n = 9$), with minimal information on ethnicity. Findings may not fully capture the magnitude of effect of vitamin B₁₂ supplementation in high-risk groups (e.g. children in low-resource settings with a high prevalence of vitamin B₁₂ deficiency), and care should be taken in applying the findings in this review broadly without considering setting-specific or individual-specific contextual risk factors.

Implications for research

Vitamin B₁₂ supplementation alone improves vitamin B₁₂ deficiency and probably improves vitamin B₁₂ levels in children, compared to placebo. However, the effects of vitamin B₁₂ supplementation on longer-term child health outcomes in children have not been established.

This review emphasises the importance of larger trials conducted for longer durations on the effects of vitamin B₁₂ supplementation on child health outcomes. Future research is needed on:

- prospective evaluation of vitamin B₁₂ biomarkers;
- inclusion of additional health outcomes, using standardised instruments and time points (including cognition and development);
- larger trials to examine child health outcomes, including growth, cognition, development, and any adverse effects;
- evidence to inform recommendations for screening and vitamin B₁₂ supplementation as part of standard of care; and
- the effect of baseline vitamin B₁₂ status, vitamin B₁₂ dose, and vitamin B₁₂ formulation on child health outcomes.

Equity-related implications for research

Studies that include populations with diverse ethnicities and conducted in a range of socioeconomic settings are needed.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD015264.pub2](https://doi.org/10.1002/14651858.CD015264.pub2).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of studies awaiting classification

Supplementary material 5 Characteristics of ongoing studies

Supplementary material 6 Risk of bias

Supplementary material 7 Analyses

Supplementary material 8 Data package

Supplementary material 9 Author correspondence

Supplementary material 10 Key characteristics of included studies: outcomes, timing of outcome assessment, and sources of trial funding

Supplementary material 11 Summary of the risk of bias assessment by domain

Supplementary material 12 Summary of findings table 1 - (Comparison 1: Additional critical outcomes) Supplementation with vitamin B₁₂ alone compared to placebo in children less than 12 years of age

Supplementary material 13 Synthesis of results 1: supplementation with vitamin B₁₂ alone versus placebo; important outcomes

Supplementary material 14 Summary of findings table 2 - (Comparison 2: Additional critical outcomes) Supplementation with vitamin B₁₂ compared to no intervention in children less than 12 years of age

Supplementary material 15 Synthesis of results 2: supplementation with vitamin B₁₂ alone versus no intervention; important outcomes

Supplementary material 16 Summary of findings table 3 - (Comparison 3: Additional critical outcomes) Supplementation with vitamin B₁₂ and other micronutrients, compared to supplements with the same formulation without vitamin B₁₂ in children less than 12 years of age

Supplementary material 17 Synthesis of results 3: supplementation with vitamin B₁₂ and other micronutrients, compared to supplements with the same formulation without vitamin B₁₂; important outcomes

Supplementary material 18 PRO EDI Participant Characteristics Table

ADDITIONAL INFORMATION

Acknowledgements

We thank the editorial office staff of Cochrane Developmental, Psychosocial and Learning Problems (CDPLP) for their support in the preparation of the protocol.

The title registration for the protocol was initially developed during the WHO/Cochrane/Cornell University Summer Institute for Systematic Reviews in Nutrition for Global Policy Making, hosted at the Division of Nutritional Sciences, Cornell University, Ithaca, New York, USA. The WHO has partially supported this course since 2014.

We thank Matt Kibbee, Erin Eldemire, and Kate Ghezzi-Kopel, information specialists at Cornell University, for their guidance in developing the search strategy for this review.

Editorial and peer-reviewer contributions

The following people conducted the editorial process for this article:

- Sign-off Editor (final editorial decision): Celeste E. Naude, Centre for Evidence-based Health Care, Division of Epidemiology

and Biostatistics, Department of Global Health, Stellenbosch University, South Africa;

- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Joanne Duffield, Cochrane Editorial Service;
- Editorial Assistant (conducted editorial policy checks, collated peer-reviewer comments and supported the editorial team): Andrew Savage, Cochrane Editorial Service;
- Copy Editor (copy editing and production): Jenny Bellorini, Cochrane Central Production Service;
- Peer-reviewers (provided comments and recommended an editorial decision): Tor Strand, Innlandet Hospital Trust, Norway (clinical/content review); Brian Duncan (patient and public review); Jo-Ana Chase, Evidence Production and Methods Directorate (methods review); Douglas M Salzwedel, Information Specialist, Cochrane Hypertension (search review).

Contributions of authors

CEGL wrote the initial draft of the protocol with JLF, developed the initial search strategy and methods section, conducted study selection and data extraction, and developed the results section.

DYL wrote the initial draft of the review with JLF, and conducted study selection, data extraction, risk of bias and GRADE assessment, and developed the results and discussion sections.

SV conducted study selection, and revised the background and results sections.

XEPM developed the background section, conducted data extraction, risk of bias and GRADE assessment, and developed the results section.

LBS conducted study selection, data extraction, risk of bias and GRADE assessment, and revised the results section.

RK contributed to the development of the methods section of the protocol and review, and provided clinical expertise on paediatric health outcomes and interpretation of findings.

JLF (senior author) coordinated the review; developed the protocol and review; arbitrated disagreements in study selection, data extraction, risk of bias assessment, and GRADE ratings; provided expert guidance on statistical and epidemiological methods; and has responsibility for the final content.

CEGL, DYL, SV, XEPM, LBS, RK, and JLF revised the review.

All review authors contributed to and approved the final version of the review. JLF has responsibility for final content.

Declarations of interest

Dr Doreen Y Larvie declares no known conflicts of interest.

Cristina E Güitrón Leal reports a Fellowship from Consejo Nacional de Ciencia y Tecnología, Mexico, to support her PhD training at Cornell. This fellowship is not related to this review or its subject. CGL declares no known conflicts of interest.

History

Protocol first published: Issue 11, 2022

Dr Sudha Venkatramanan declares no known conflicts of interest.

Ximena E Palma Molina declares no known conflicts of interest.

Lili B Steel declares no known conflicts of interest.

Dr Rebecca Kuriyan is a Clinical Nutritionist at St John's Medical College and Hospital, Bangalore, India. She declares no known conflicts of interest.

Dr Julia L Finkelstein is Director of the Cornell Cochrane Center and serves as a Handling Editor and Sign-off Editor for Cochrane (since 2022); JLF did not have any role in the editorial process for this review. JLF is a Principal Investigator on research grants (not related to this review) to examine the burden and aetiology of anaemia in women of reproductive age and adolescents (US Centers for Disease Control and Prevention; CDC), biomarkers of anaemia and nutritional status in women of reproductive age (National Institutes of Health), and to conduct randomised trials with micronutrient interventions to improve the health of women and adolescents (United States Department of Agriculture, CDC); all were institutional research grants awarded to Cornell University. JLF declares no known conflicts of interest.

None of the authors are investigators in any trials included in this review.

Sources of support

Internal sources

- No internal support received, Other

Not applicable

External sources

- No external support received, Other

Not applicable

Registration and protocol

Protocol (2022) DOI: 10.1002/14651858.CD015264

Data, code and other materials

As part of the published Cochrane review, the following are made available for download for users of the Cochrane Library: full search strategies for each database; full citations of each unique report for all studies included, ongoing or awaiting classification, or excluded at the full-text screen, in the final review; study data, including study information, study arms, and study results or test data; consensus risk of bias assessments; and analysis data, including overall estimates and settings, subgroup estimates, and individual data rows. Appropriate permissions have been obtained for such use. Analyses and data management were conducted within Cochrane's authoring tool, Review Manager, using the inbuilt computation methods. Template data extraction forms are available from the authors on reasonable request.

Study data, consensus risk of bias assessments, and analysis data are provided in [Supplementary material 8](#).

REFERENCES

1. Reynolds E. Vitamin B12, folic acid, and the nervous system. *Lancet. Neurology* 2006;**5**(11):949-60. [DOI: [10.1016/S1474-4422\(06\)70598-1](https://doi.org/10.1016/S1474-4422(06)70598-1)] [PMID: 17052662]
2. Stover PJ. Physiology of folate and vitamin B12 in health and disease. *Nutrition Reviews* 2004;**62**(6 Pt 2):S3-12; discussion S13. [DOI: [10.1111/j.1753-4887.2004.tb00070.x](https://doi.org/10.1111/j.1753-4887.2004.tb00070.x)] [PMID: 15298442]
3. Winter-Vann AM, Kamen BA, Bergo MO, Young SG, Melnyk S, James SJ, et al. Targeting Ras signaling through inhibition of carboxyl methylation: an unexpected property of methotrexate. *Proceedings of the National Academy of Sciences of the United States of America* 2003;**100**(11):6529-34. [DOI: [10.1073/pnas.1135239100](https://doi.org/10.1073/pnas.1135239100)] [PMCID: PMC164480] [PMID: 12750467]
4. Banerjee RV, Matthews RG. Cobalamin-dependent methionine synthase. *FASEB Journal* 1990;**4**(5):1450-9. [DOI: [10.1096/fasebj.4.5.2407589](https://doi.org/10.1096/fasebj.4.5.2407589)] [PMID: 2407589]
5. Shane B, Stokstad EL. Vitamin B12-folate interrelationships. *Annual Review of Nutrition* 1985;**5**:115-41. [DOI: [10.1146/annurev.nu.05.070185.000555](https://doi.org/10.1146/annurev.nu.05.070185.000555)] [PMID: 3927946]
6. Stabler SP. Vitamin B12. In: Marriott BP, Birt DF, Stallings V, Yates AA, editor(s). *Present Knowledge in Nutrition*. 11 edition. Vol. **1: Basic Nutrition and Metabolism**. London, UK: Elsevier, 2020:257-71. [ISBN: 978-0-323-66162-1]
7. Froese DS, Fowler B, Baumgartner MR. Vitamin B₁₂, folate, and the methionine remethylation cycle-biochemistry, pathways, and regulation. *Journal of Inherited Metabolic Disease* 2019;**42**(4):673-85. [DOI: [10.1002/jimd.12009](https://doi.org/10.1002/jimd.12009)] [PMID: 30693532]
8. Green R, Allen LH, Björke-Monsen AL, Brito A, Guéant JL, Miller JW, et al. Vitamin B₁₂ deficiency. *Nature Reviews. Disease Primers* 2017;**3**:17040. [DOI: [10.1038/nrdp.2017.40](https://doi.org/10.1038/nrdp.2017.40)] [PMID: 28660890]
9. Pourié G, Guéant JL, Quadros EV. Behavioral profile of vitamin B₁₂ deficiency: a reflection of impaired brain development, neuronal stress and altered neuroplasticity. *Vitamins and Hormones* 2022;**119**:377-404. [DOI: [10.1016/bs.vh.2022.02.002](https://doi.org/10.1016/bs.vh.2022.02.002)] [PMID: 35337627]
10. Watanabe F, Yabuta Y, Bito T, Teng F. Vitamin B₁₂-containing plant food sources for vegetarians. *Nutrients* 2014;**6**(5):1861-73. [CENTRAL: PMC4042564] [DOI: [10.3390/nu6051861](https://doi.org/10.3390/nu6051861)] [PMID: 24803097]
11. Institute of Medicine (US) Standing Committee on the Scientific Evaluation of Dietary Reference Intakes and its Panel on Folate, Other B Vitamins, and Choline. *Dietary Reference Intakes for Thiamin, Riboflavin, Niacin, Vitamin B6, Folate, Vitamin B12, Pantothenic Acid, Biotin, and Choline*. Washington (DC): National Academies Press, 1998. [BOOKSHELF ID: NBK114310] [DOI: [10.17226/6015](https://doi.org/10.17226/6015)] [PMID: 23193625]
12. Institute of Medicine. *Dietary Reference Intakes: The Essential Guide to Nutrient Requirements*. Washington (DC): National Academies Press, 2006. [DOI: [10.17226/11537](https://doi.org/10.17226/11537)]
13. Allen LH. How common is vitamin B-12 deficiency? *American Journal of Clinical Nutrition* 2009;**89**(2):693S-6S. [DOI: [10.3945/ajcn.2008.26947A](https://doi.org/10.3945/ajcn.2008.26947A)] [PMID: 19116323]
14. Brito A, Mujica-Coopman MF, López de Romaña D, Cori H, Allen LH. Folate and vitamin B12 status in Latin America and the Caribbean: an update. *Food and Nutrition Bulletin* 2015;**36**(Suppl 2):S109-18. [DOI: [10.1177/0379572115585772](https://doi.org/10.1177/0379572115585772)] [PMID: 26125196]
15. McLean E, de Benoist B, Allen LH. Review of the magnitude of folate and vitamin B12 deficiencies worldwide. *Food and Nutrition Bulletin* 2008;**29**(Suppl 2):S38-51. [DOI: [10.1177/156482650802925107](https://doi.org/10.1177/156482650802925107)] [PMID: 18709880]
16. Allen LH, Rosenberg IH, Oakley GP, Omenn GS. Considering the case for vitamin B12 fortification of flour. *Food and Nutrition Bulletin* 2010;**31**(1 Suppl):S36-46. [DOI: [10.1177/156482651003115104](https://doi.org/10.1177/156482651003115104)] [PMID: 20629351]
17. Finkelstein JL, Layden AJ, Stover PJ. Vitamin B-12 and perinatal health. *Advances in Nutrition* 2015;**6**(5):552-63. [DOI: [10.3945/an.115.008201](https://doi.org/10.3945/an.115.008201)] [PMCID: PMC4561829] [PMID: 26374177]
18. Anaya-Loyola MA, Brito A, Vergara-Castañeda H, Sosa C, Rosado JL, Allen LH. Low serum B12 concentrations are associated with low B12 dietary intake but not with helicobacter pylori infection or abnormal gastric function in rural Mexican women. *Nutrients* 2019;**11**(12):2922. [DOI: [10.3390/nu11122922](https://doi.org/10.3390/nu11122922)] [PMCID: PMC6950710] [PMID: 31810343]
19. Arazo-Rusindo MC, Zúñiga RN, Cortés-Segovia P, Benavides-Valenzuela S, Pérez-Bravo F, Castillo-Valenzuela O, et al. Nutritional status and serum levels of micronutrients in an elderly group who participate in the program for complementary food in older people (PACAM) from the metropolitan region, Santiago de Chile. *Nutrients* 2021;**14**(1):3. [DOI: [10.3390/nu14010003](https://doi.org/10.3390/nu14010003)] [PMCID: PMC8746835] [PMID: 35010878]
20. Finkelstein JL, Fothergill A, Johnson CB, Guetterman HM, Bose B, Jabbar S, et al. Anemia and vitamin B-12 and folate status in women of reproductive age in southern India: estimating population-based risk of neural tube defects. *Current Developments in Nutrition* 2021;**5**(5):nzab069. [DOI: [10.1093/cdn/nzab069](https://doi.org/10.1093/cdn/nzab069)] [PMCID: PMC8128722] [PMID: 34027296]
21. Gonmei Z, Toteja GS. Micronutrient status of Indian population. *Indian Journal of Medical Research* 2018;**148**(5):511-21. [DOI: [10.4103/ijmr.IJMR_1768_18](https://doi.org/10.4103/ijmr.IJMR_1768_18)] [PMCID: PMC6366258] [PMID: 30666978]
22. De la Cruz-Góngora V, Martínez-Tapia B, Shamah-Levy T, Villalpando S. Nutritional status of iron, vitamin B12, vitamin A and anemia in Mexican children: results from the Ensanut 2018-19. *Salud Pública de México* 2021;**63**(3):359-70. [DOI: [10.21149/12158](https://doi.org/10.21149/12158)] [PMID: 34098609]
23. Williams AM, Stewart CP, Shahab-Ferdows S, Hampel D, Kiprotich M, Achando B, et al. Infant serum and maternal milk vitamin B-12 are positively correlated in Kenyan infant-mother

dyads at 1-6 months postpartum, irrespective of infant feeding practice. *Journal of Nutrition* 2018;**148**(1):86-93. [DOI: [10.1093/jn/nxx009](https://doi.org/10.1093/jn/nxx009)] [PMCID: PMC5955065] [PMID: 29378045]

24. Ulak M, Chandyo RK, Thorne-Lyman AL, Henjum S, Ueland PM, Middtun Ø, et al. Vitamin status among breastfed infants in Bhaktapur, Nepal. *Nutrients* 2016;**8**(3):149. [DOI: [10.3390/nu8030149](https://doi.org/10.3390/nu8030149)] [PMCID: PMC4808878] [PMID: 27005657]

25. Bellows AL, Smith ER, Muhihi A, Briegleb C, Noor RA, Mshamu S, et al. Micronutrient deficiencies among breastfeeding infants in Tanzania. *Nutrients* 2017;**9**(11):1258. [DOI: [10.3390/nu9111258](https://doi.org/10.3390/nu9111258)] [PMCID: PMC5707730] [PMID: 29149073]

26. Jones KM, Ramirez-Zea M, Zuleta C, Allen LH. Prevalent vitamin B-12 deficiency in twelve-month-old Guatemalan infants is predicted by maternal B-12 deficiency and infant diet. *Journal of Nutrition* 2007;**137**(5):1307-13. [DOI: [10.1093/jn/137.5.1307](https://doi.org/10.1093/jn/137.5.1307)] [PMID: 17449597]

27. Kadiyala A, Palani A, Rajendraprasath S, Venkatramanan P. Prevalence of vitamin B12 deficiency among exclusively breast fed term infants in South India. *Journal of Tropical Pediatrics* 2021;**67**(1):fmaa114. [DOI: [10.1093/tropej/fmaa114](https://doi.org/10.1093/tropej/fmaa114)] [PMID: 33346823]

28. Allen LH. Causes of vitamin B12 and folate deficiency. *Food and Nutrition Bulletin* 2008;**29**(Suppl 2):S20-34; discussion S35-7. [DOI: [10.1177/15648265080292S105](https://doi.org/10.1177/15648265080292S105)] [PMID: 18709879]

29. Layden AJ, Täse K, Finkelstein JL. Neglected tropical diseases and vitamin B12: a review of the current evidence. *Transactions of The Royal Society of Tropical Medicine and Hygiene* 2018;**112**(10):423-35. [DOI: [10.1093/trstmh/try078](https://doi.org/10.1093/trstmh/try078)] [PMCID: PMC6457089] [PMID: 30165408]

30. Bergen NE, Schalekamp-Timmermans S, Jaddoe VW, Hofman A, Lindemans J, Russcher H, et al. Maternal and neonatal markers of the homocysteine pathway and fetal growth: the Generation R study. *Paediatric and Perinatal Epidemiology* 2016;**30**(4):386-96. [DOI: [10.1111/ppe.12297](https://doi.org/10.1111/ppe.12297)] [PMID: 27271101]

31. Finkelstein JL, Guillet R, Pressman EK, Fothergill A, Guetterman HM, Kent TR, et al. Vitamin B₁₂ status in pregnant adolescents and their infants. *Nutrients* 2019;**11**(2):397. [DOI: [10.3390/nu11020397](https://doi.org/10.3390/nu11020397)] [PMCID: PMC6413223] [PMID: 30781902]

32. Finkelstein JL, Fothergill A, Krisner JT, Thomas T, Kurpad AV, Dwarkanath P. Maternal vitamin B12 deficiency and perinatal outcomes in southern India. *PLOS One* 2021;**16**(4):e0248145. [DOI: [10.1371/journal.pone.0248145](https://doi.org/10.1371/journal.pone.0248145)] [PMCID: PMC8023483] [PMID: 33822790]

33. Greibe E, Lildballe DL, Streym S, Vestergaard P, Rejnmark L, Mosekilde L, et al. Cobalamin and haptocorrin in human milk and cobalamin-related variables in mother and child: a 9-mo longitudinal study. *American Journal of Clinical Nutrition* 2013;**98**(2):389-95. [DOI: [10.3945/ajcn.113.058479](https://doi.org/10.3945/ajcn.113.058479)] [PMID: 23783295]

34. Hay G, Clausen T, Whitelaw A, Trygg K, Johnston C, Henriksen T, et al. Maternal folate and cobalamin status predicts

vitamin status in newborns and 6-month-old infants. *Journal of Nutrition* 2010;**140**(3):557-64. [DOI: [10.3945/jn.109.117424](https://doi.org/10.3945/jn.109.117424)] [PMID: 20071650]

35. Anaya-Loyola MA, Brito A, Brown KH, Allen LH. Breast milk provides inadequate amounts of vitamin B12 for predominantly breastfed Guatemalan infants. *International Journal for Vitamin and Nutrition Research* 2020;**90**(5-6):395-402. [DOI: [10.1024/0300-9831/a000583](https://doi.org/10.1024/0300-9831/a000583)] [PMID: 30987548]

36. Kose E, Besci O, Gudeloglu E, Suncak S, Oymak Y, Ozen S, et al. Transcobalamin II deficiency in twins with a novel variant in the TCN2 gene: case report and review of literature. *Journal of Pediatric Endocrinology and Metabolism* 2020;**33**(11):1487-99. [DOI: [10.1515/jpem-2020-0096](https://doi.org/10.1515/jpem-2020-0096)] [PMID: 32841161]

37. Mullikin D, Pillai N, Sanchez R, O'Donnell-Luria AH, Kritzer A, Tal L, et al. Megaloblastic anemia progressing to severe thrombotic microangiopathy in patients with disordered vitamin B₁₂ metabolism: case reports and literature review. *Journal of Pediatrics* 2018;**202**:315-9.e2. [DOI: [10.1016/j.jpeds.2018.06.054](https://doi.org/10.1016/j.jpeds.2018.06.054)] [PMID: 30057141]

38. Qasrawi R, Abu Al-Halawa D. Cluster analysis and classification model of nutritional anemia associated risk factors among Palestinian schoolchildren, 2014. *Frontiers in Nutrition* 2022;**9**:838937. [DOI: [10.3389/fnut.2022.838937](https://doi.org/10.3389/fnut.2022.838937)] [PMCID: PMC9127973] [PMID: 35619964]

39. Strand TA, Taneja S, Kumar T, Manger MS, Refsum H, Yajnik CS, et al. Vitamin B-12, folic acid, and growth in 6- to 30-month-old children: a randomized controlled trial. *Pediatrics* 2015;**135**(4):e918-26. [DOI: [10.1542/peds.2014-1848](https://doi.org/10.1542/peds.2014-1848)] [PMID: 25802345]

40. Dror DK, Allen LH. Effect of vitamin B12 deficiency on neurodevelopment in infants: current knowledge and possible mechanisms. *Nutrition Reviews* 2008;**66**(5):250-5. [DOI: [10.1111/j.1753-4887.2008.00031.x](https://doi.org/10.1111/j.1753-4887.2008.00031.x)] [PMID: 18454811]

41. Venkatramanan S, Armata IE, Strupp BJ, Finkelstein JL. Vitamin B-12 and cognition in children. *Advances in Nutrition* 2016;**7**(5):879-88. [DOI: [10.3945/an.115.012021](https://doi.org/10.3945/an.115.012021)] [PMCID: PMC5015033] [PMID: 27633104]

42. Black MM. Effects of vitamin B12 and folate deficiency on brain development in children. *Food and Nutrition Bulletin* 2008;**29**(Suppl 2):S126-31. [DOI: [10.1177/15648265080292S117](https://doi.org/10.1177/15648265080292S117)] [PMCID: PMC3137939] [PMID: 18709887]

43. Pepper MR, Black MM. B12 in fetal development. *Seminars in Cell & Developmental Biology* 2011;**22**(6):619-23. [DOI: [10.1016/j.semcdb.2011.05.005](https://doi.org/10.1016/j.semcdb.2011.05.005)] [PMID: 21664980]

44. Allen LH, Miller JW, de Groot L, Rosenberg IH, Smith AD, Refsum H, et al. Biomarkers of nutrition for development (BOND): vitamin B-12 review. *Journal of Nutrition* 2018;**148**(Suppl 4):1995S-2027S. [DOI: [10.1093/jn/nxy201](https://doi.org/10.1093/jn/nxy201)] [PMCID: PMC6297555] [PMID: 30500928]

45. Yetley EA, Pfeiffer CM, Phinney KW, Bailey RL, Blackmore S, Bock JL, et al. Biomarkers of vitamin B-12 status in NHANES: a roundtable summary. *American Journal of Clinical Nutrition*

2011;**94**(1):313S-21S. [DOI: [10.3945/ajcn.111.013243](https://doi.org/10.3945/ajcn.111.013243)] [PMCID: PMC3127527] [PMID: 21593512]

46. Quadros EV. Advances in the understanding of cobalamin assimilation and metabolism. *British Journal of Haematology* 2010;**148**(2):195-204. [DOI: [10.1111/j.1365-2141.2009.07937.x](https://doi.org/10.1111/j.1365-2141.2009.07937.x)] [PMCID: PMC2809139] [PMID: 19832808]

47. Quadros EV, Sequeira JM. Cellular uptake of cobalamin: transcobalamin and the TCBLR/CD320 receptor. *Biochimie* 2013;**95**(5):1008-18. [DOI: [10.1016/j.biochi.2013.02.004](https://doi.org/10.1016/j.biochi.2013.02.004)] [PMCID: PMC3902480] [PMID: 23415653]

48. Stabler SP. Clinical practice. Vitamin B12 deficiency. *New England Journal of Medicine* 2013;**368**(2):149-60. [DOI: [10.1056/NEJMcpl113996](https://doi.org/10.1056/NEJMcpl113996)] [PMID: 23301732]

49. Obeid R, Fedosov SN, Nexo E. Cobalamin coenzyme forms are not likely to be superior to cyano- and hydroxyl-cobalamin in prevention or treatment of cobalamin deficiency. *Molecular Nutrition & Food Research* 2015;**59**(7):1364-72. [DOI: [10.1002/mnfr.201500019](https://doi.org/10.1002/mnfr.201500019)] [PMCID: PMC4692085] [PMID: 25820384]

50. Carmel R. How I treat cobalamin (vitamin B₁₂) deficiency. *Blood* 2008;**112**(6):2214-21. [DOI: [10.1182/blood-2008-03-040253](https://doi.org/10.1182/blood-2008-03-040253)] [PMCID: PMC2532799] [PMID: 18606874]

51. Bhowmik B, Siddiquee T, Mdala I, Quamrun Nesa L, Jahan Shelly S, Hassan Z, et al. Vitamin D3 and B12 supplementation in pregnancy. *Diabetes Research and Clinical Practice* 2021;**174**:108728. [DOI: [10.1016/j.diabres.2021.108728](https://doi.org/10.1016/j.diabres.2021.108728)] [PMID: 33662489]

52. Rashid S, Meier V, Patrick H. Review of vitamin B12 deficiency in pregnancy: a diagnosis not to miss as veganism and vegetarianism become more prevalent. *European Journal of Haematology* 2021;**106**(4):450-5. [DOI: [10.1111/ejh.13571](https://doi.org/10.1111/ejh.13571)] [PMID: 33341967]

53. Green R, Miller JW. Vitamin B12 deficiency. *Vitamins and Hormones* 2022;**119**:405-39. [DOI: [10.1016/bs.vh.2022.02.003](https://doi.org/10.1016/bs.vh.2022.02.003)] [PMID: 35337628]

54. Ars CL, Nijs IM, Marroun HE, Muetzel R, Schmidt M, Steenweg-de Graaff J, et al. Prenatal folate, homocysteine and vitamin B₁₂ levels and child brain volumes, cognitive development and psychological functioning: the Generation R Study. *British Journal of Nutrition* 2019;**122**(Suppl 1):S1-9. [DOI: [10.1017/S0007114515002081](https://doi.org/10.1017/S0007114515002081)] [PMID: 31638501]

55. D'souza N, Behere RV, Patni B, Deshpande M, Bhat D, Bhalerao A, et al. Pre-conceptional maternal vitamin B12 supplementation improves offspring neurodevelopment at 2 years of age: PRIYA trial. *Frontiers in Pediatrics* 2021;**9**:755977. [DOI: [10.3389/fped.2021.755977](https://doi.org/10.3389/fped.2021.755977)] [PMCID: PMC8697851] [PMID: 34956975]

56. Strand TA, Taneja S, Ueland PM, Refsum H, Bahl R, Schneede J, et al. Cobalamin and folate status predicts mental development scores in North Indian children 12-18 mo of age. *American Journal of Clinical Nutrition* 2013;**97**(2):310-7. [DOI: [10.3945/ajcn.111.032268](https://doi.org/10.3945/ajcn.111.032268)] [PMID: 23283502]

57. Duong MC, Mora-Plazas M, Marín C, Villamor E. Vitamin B-12 deficiency in children is associated with grade repetition and school absenteeism, independent of folate, iron, zinc, or vitamin A status biomarkers. *Journal of Nutrition* 2015;**145**(7):1541-8. [DOI: [10.3945/jn.115.211391](https://doi.org/10.3945/jn.115.211391)] [PMID: 25972530]

58. Louwman MW, van Dusseldorp M, van de Vijver FJ, Thomas CM, Schneede J, Ueland PM, et al. Signs of impaired cognitive function in adolescents with marginal cobalamin status. *American Journal of Clinical Nutrition* 2000;**72**(3):762-9. [DOI: [10.1093/ajcn/72.3.762](https://doi.org/10.1093/ajcn/72.3.762)] [PMID: 10966896]

59. Bahadir A, Reis PG, Erduran E. Oral vitamin B12 treatment is effective for children with nutritional vitamin B12 deficiency. *Journal of Paediatrics and Child Health* 2014;**50**(9):721-5. [PMID: 10.1111/jpc.12652] [PMID: 24944005]

60. Kvestad I, Taneja S, Kumar T, Hysing M, Refsum H, Yajnik CS, et al. Vitamin B12 and folic acid improve gross motor and problem-solving skills in young North Indian children: a randomized placebo-controlled trial. *PLOS One* 2015;**10**(6):e0129915. [DOI: [10.1371/journal.pone.0129915](https://doi.org/10.1371/journal.pone.0129915)] [PMCID: PMC4476750] [PMID: 26098427]

61. Finkelstein JL, Fothergill A, Venkatramanan S, Layden AJ, Williams JL, Crider KS, et al. Vitamin B12 supplementation during pregnancy for maternal and child health outcomes. *Cochrane Database of Systematic Reviews* 2024, Issue 1. Art. No: CD013823. [DOI: [10.1002/14651858.CD013823.pub2](https://doi.org/10.1002/14651858.CD013823.pub2)]

62. Keats EC, Haider BA, Tam E, Bhutta ZA. Multiple-micronutrient supplementation for women during pregnancy. *Cochrane Database of Systematic Reviews* 2019, Issue 3. Art. No: CD004905. [DOI: [10.1002/14651858.CD004905.pub6](https://doi.org/10.1002/14651858.CD004905.pub6)] [PMCID: PMC6418471] [PMID: 30873598]

63. Huey SL, Acharya N, Silver A, Shen R, Yu EA, Peña-Rosas JP, et al. Effects of oral vitamin D supplementation on linear growth and other health outcomes among children under five years of age. *Cochrane Database of Systematic Reviews* 2020, Issue 12. Art. No: CD012875. [DOI: [10.1002/14651858.CD012875.pub2](https://doi.org/10.1002/14651858.CD012875.pub2)] [PMCID: PMC8121044] [PMID: 33305842]

64. Gogia S, Sachdev HS. Zinc supplementation for mental and motor development in children. *Cochrane Database of Systematic Reviews* 2012, Issue 12. Art. No: CD007991. [DOI: [10.1002/14651858.CD007991.pub2](https://doi.org/10.1002/14651858.CD007991.pub2)] [PMID: 23235652]

65. Lassi ZS, Kurji J, Oliveira CS, Moin A, Bhutta ZA. Zinc supplementation for the promotion of growth and prevention of infections in infants less than six months of age. *Cochrane Database of Systematic Reviews* 2020, Issue 4. Art. No: CD010205. [DOI: [10.1002/14651858.CD010205.pub2](https://doi.org/10.1002/14651858.CD010205.pub2)] [PMCID: PMC7140593] [PMID: 32266964]

66. De-Regil LM, Jefferds ME, Sylvetsky AC, Dowswell T. Intermittent iron supplementation for improving nutrition and development in children under 12 years of age. *Cochrane Database of Systematic Reviews* 2011, Issue 12. Art. No: CD009085. [DOI: [10.1002/14651858.CD009085.pub2](https://doi.org/10.1002/14651858.CD009085.pub2)] [PMCID: PMC4547491] [PMID: 22161444]

67. Finkelstein JL, Herman HS, Guetterman HM, Peña-Rosas JP, Mehta S. Daily iron supplementation for prevention or treatment of iron deficiency anaemia in infants, children, and adolescents. *Cochrane Database of Systematic Reviews* 2018, Issue 12. Art. No: CD013227. [DOI: [10.1002/14651858.CD013227](https://doi.org/10.1002/14651858.CD013227)]
68. Rajagopalan K, Hackl L, Peña-Rosas JP, Garcia-Casal MN, Rizal M, Mehta NH, et al. Daily iron supplementation for prevention or treatment of iron deficiency anaemia in infants, children, and adolescents. Review under development. To be published in the Cochrane Database of Systematic Reviews: <https://www.cochrane.library.com>.
69. Delgado-Noguera MF, Calvache JA, Bonfill Cosp X, Kotanidou EP, Galli-Tsinopoulou A. Supplementation with long chain polyunsaturated fatty acids (LCPUFA) to breastfeeding mothers for improving child growth and development. *Cochrane Database of Systematic Reviews* 2015, Issue 7. Art. No: CD007901. [DOI: [10.1002/14651858.CD007901.pub3](https://doi.org/10.1002/14651858.CD007901.pub3)] [PMID: 26171898]
70. Higgins JP, Lasserson T, Thomas J, Flemmyng E, Churchill R. Methodological expectations of Cochrane intervention reviews. Cochrane: London, Version August 2023. Available from <https://community.cochrane.org/mecir-manual>.
71. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews [Declaración PRISMA 2020: una guía actualizada para la publicación de revisiones sistemáticas]. *Revista Espanola de Cardiologia (English Edition)* 2021;**74**(9):790-9. [DOI: [10.1016/j.rec.2021.07.010](https://doi.org/10.1016/j.rec.2021.07.010)] [PMID: 34446261]
72. Weiss LG, Oakland T, Aylward GP, editor(s). Bayley-III clinical use and interpretation. San Diego (CA): Academic Press, 2010.
73. Wechsler D. Manual for Weschler Intelligence Scale for Children. 3 edition. San Antonio (TX): Psychological Corporation, 1994.
74. Brooks BL, Sherman EMS, Strauss E. NEPSY-II: a developmental neuropsychological assessment, Second Edition. *Child Neuropsychology* 2009;**16**(1):80-101. [DOI: [10.1080/09297040903146966](https://doi.org/10.1080/09297040903146966)]
75. Ware JE Jr, Sherbourne CD. The MOS 36-item short-form health survey (SF-36). I. Conceptual framework and item selection. *Medical Care* 1992;**30**(6):473-83. [PMID: 1593914]
76. EuroQol Group. EuroQol-a new facility for the measurement of health-related quality of life. *Health Policy* 1990;**16**(3):199-208. [DOI: [10.1016/0168-8510\(90\)90421-9](https://doi.org/10.1016/0168-8510(90)90421-9)] [PMID: 10109801]
77. Covidence. Melbourne, Australia: Veritas Health Innovation, accessed 13 April 2022. Available at covidence.org.
78. Review Manager (RevMan). Version 8.1.0. The Cochrane Collaboration, 2024. Available at <https://revman.cochrane.org>.
79. Higgins JP, Savović J, Page MJ, Elbers RG, Sterne JA. Chapter 8: Assessing risk of bias in a randomized trial. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.3 (updated February 2022). Cochrane, 2022. Available from <https://www.cochrane.org/handbook>.
80. Sterne JA, Savović J, Page MJ, Elbers RG, Blencowe NS, Boutron I, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ* 2019;**366**:l4898. [DOI: [10.1136/bmj.l4898](https://doi.org/10.1136/bmj.l4898)] [PMID: 31462531]
81. Higgins JP, Li T, Deeks JJ, editor(s). Chapter 6: Choosing effect measures and computing estimates effect. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.3 (updated February 2022). Cochrane, 2022. Available from <https://www.cochrane.org/handbook>.
82. Higgins JP, Green S, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 5.1.0 (updated March 2011). The Cochrane Collaboration, 2011. Available from <https://www.cochrane.org/authors/handbooks-and-manuals/handbook/archive/v5.1.0>.
83. Berry DC, McMurray R, Schwartz TA, Skelly A, Sanchez M, Neal M, et al. Rationale, design, methodology and sample characteristics for the family partners for health study: a cluster randomized controlled study. *BMC Public Health* 2012;**12**:250. [DOI: [10.1186/1471-2458-12-250](https://doi.org/10.1186/1471-2458-12-250)]
84. Nanney MS, Shanafelt A, Wang Q, Leduc R, Dodds E, Hearst M, et al. Project BreakFAST: rationale, design, and recruitment and enrollment methods of a randomized controlled trial to evaluate an intervention to improve School Breakfast Program participation in rural high schools. *Contemporary Clinical Trial Communications* 2016;**3**:12-22. [DOI: [10.1016/j.conctc.2015.12.009](https://doi.org/10.1016/j.conctc.2015.12.009)]
85. Cliffer IR, Yussuf MH, Millogo O, Mwanyika-Sando M, Barry Y, Yusufu IS, et al. Scaling-up high-impact micronutrient supplementation interventions to improve adolescents' nutrition and health in Burkina Faso and Tanzania: protocol for a cluster-randomised controlled trial. *BMJ Open* 2023;**13**(2):e063686. [DOI: [10.1136/bmjopen-2022-063686](https://doi.org/10.1136/bmjopen-2022-063686)]
86. Higgins JP, Eldridge S, Li T, editor(s). Chapter 23: Including variants on randomized trials. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.3 (updated February 2022). Cochrane, 2022. Available from <https://www.cochrane.org/handbook>.
87. Deeks JJ, Higgins JP, Altman DG, editor(s). Chapter 10: Analysing data and undertaking meta-analyses. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.3 (updated February 2022). Cochrane, 2022. Available from <https://www.cochrane.org/handbook>.
88. Moreno SG, Sutton AJ, Turner EH, Abrams KR, Cooper NJ, Palmer TM, et al. Novel methods to deal with publication biases: secondary analysis of antidepressant trials in the FDA trial registry database and related journal publications. *BMJ* 2009;**339**:b2981. [DOI: [10.1136/bmj.b2981](https://doi.org/10.1136/bmj.b2981)] [PMID: PMC2723217]

- 89.** DerSimonian R, Laird N. Meta-analysis in clinical trials. *Controlled Clinical Trials* 1986;**7**(3):177-88. [DOI: [10.1016/0197-2456\(86\)90046-2](https://doi.org/10.1016/0197-2456(86)90046-2)] [PMID: 3802833]
- 90.** McKenzie JE, Brennan SE. Chapter 12: Synthesizing and presenting findings using other methods. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.3 (updated February 2022). Cochrane, 2022. Available from <https://www.cochrane.org/handbook>.
- 91.** Leal CEG, Molina XEP, Venkatramanan S, Williams JL, Kuriyan R, Crider KS, et al. Vitamin B12 supplementation for growth, development, and cognition in children. *Cochrane Database of Systematic Reviews* 2022, Issue 11. Art. No: CD015264. [DOI: [10.1002/14651858.CD015264](https://doi.org/10.1002/14651858.CD015264)] [PMCID: PMC9651173]
- 92.** PRO EDI participant characteristics table. Available in <https://www.trialforge.org/trial-diversity/pro-edi-improving-how-equity-diversity-and-inclusion-is-handled-in-evidence-synthesis> (accessed 4 November 2024).
- 93.** Balshem H, Helfand M, Schünemann HJ, Oxman AD, Kunz R, Brozek J, et al. GRADE guidelines: 3. Rating the quality of evidence. *Journal of Clinical Epidemiology* 2011;**64**(4):401-6. [DOI: [10.1016/j.jclinepi.2010.07.015](https://doi.org/10.1016/j.jclinepi.2010.07.015)] [PMID: 21208779]
- 94.** GRADEpro. Hamilton (ON): McMaster University (developed by Evidence Prime), accessed 24 October 2022. Available at gradepro.org.
- 95.** Areekul S, Sabcharoen A, Cheeramakara C, Srisukawat K. Effect of vitamin B12 supplementation on serum vitamin B12 and vitamin B12 binding protein. *Southeast Asian Journal of Tropical Medicine and Public Health* 1979;**10**(1):158-9.
- 96.** Benjamin B, Pirrie GD. Vitamin B12 by mouth. *Medical Officer* 1952;**87**:137-40.
- 97.** Chinnock RF, Rosenberg HW. Results of administration of vitamin B12 to newborn infants. *Journal of Pediatrics* 1952;**40**(2):182-5. [DOI: [10.1016/S0022-3476\(52\)80180-5](https://doi.org/10.1016/S0022-3476(52)80180-5)]
- 98.** Craigmile DA, Preston JE, Berry WTC. Similarity of growth and health in day nursery children with and without vitamin B12 supplements. *Monthly Bulletin of the Ministry of Health and the Emergency Public Health Laboratory Service* 1956;**15**:78-9.
- 99.** Deshmukh US, Joglekar CV, Lubree HG, Ramdas LV, Bhat DS, Naik SS, et al. Effect of physiological doses of oral vitamin B12 on plasma homocysteine: a randomized, placebo-controlled, double-blind trial in India. *European Journal of Clinical Nutrition* 2010;**64**(5):495-502. [DOI: [10.1038/ejcn.2010.15](https://doi.org/10.1038/ejcn.2010.15)]
- 100.** Larvie DY. Request for study information to include in cochrane systematic review [personal communication]. Email to: US Deshmukh 2 January 2024.
- 101.** Yadav DK, Shrestha S, Lillycrop KA, Joglekar CV, Pan H, Holbrook JD, et al. Vitamin B(12) supplementation influences methylation of genes associated with type 2 diabetes and its intermediate traits. *Epigenomics* 2018;**10**(1):71-90. [DOI: [10.2217/epi-2017-0102](https://doi.org/10.2217/epi-2017-0102)]
- 102.** ISRCTN59289820. Can developmental programming of adiposity and insulin resistance in rural Indian children be reversed during adolescent growth by vitamin B12 supplementation? A pilot study. <https://trialsearch.who.int/Trial2.aspx?TrialID=ISRCTN59289820> (first posted 7 September 2007).
- 103.** Fukui T, Fukui H, Sasaki T. Experiment of enriching school lunch with L-lysine and vitamin B12 for observation of their effects on physical development of children. *Tokushima Journal of Experimental Medicine* 1959;**6**:89-95.
- 104.** Gutiérrez-Díaz A. The influence on growth of children of magnesium molybdate, vitamin B12, the vitamin T complex and a combination of them [La influencia en el desarrollo infantil del molibdato de magnesio, la vitamina B12, el complejo vitamínico T y su asociación]. *Acta Pediátrica Española* 1959;**17**:125-53.
- 105.** Kudo Z. An aggravation of folic acid deficiency by an administration of vitamin B12 upon children with nutritional dystrophy. *Tohoku Journal of Experimental Medicine* 1962;**76**:23-33. [DOI: [10.1620/tjem.76.23](https://doi.org/10.1620/tjem.76.23)]
- 106.** Abe T. Aggravation of folic acid deficiency by vitamin B12 administration on children with nutritional dystrophy; from view-point of nuclear shift of neutrophils. *Tohoku Journal of Experimental Medicine* 1962;**76**:34-52. [DOI: [10.1620/tjem.76.34](https://doi.org/10.1620/tjem.76.34)]
- 107.** Murakami T. Studies on the improvement of nutritional intake of school children in the mountainous area. 2. Experiments on vitamin B12 supplementation. *Shikoku Acta Medica* 1962;**18**:273-91.
- 108.** Scaglione G, Cocciante G. Effect of certain growth factors, vitamin T Goetsch, vitamin B12 and methylandrostenediol, on premature infants [Osservazioni sull'azione di alcuni fattori di accrescimento (vitamin T Goetsch, vitamina B12, metilandrostenidiolo) negli immaturi]. *Pediatrics* 1955;**68**:1038-54.
- 109.** Scrimshaw NS, Guzman MA. Effect of dietary supplements and of the administration of vitamin B12 and aureomycin on the growth of children of school age [Efecto de los suplementos dietéticos y de la administración de vitamina B12 y aureomicina sobre el crecimiento de los niños de edad]. *Boletín de la Oficina Sanitaria Panamericana* 1953;**34**:551-62.
- 110.** Aguirre F, Scrimshaw NS, Munoz JA, Cabezas A. The effect of supplements of animal and vegetable protein, vitamin B12, and aureomycin on hematological values in Central American school children. *American Journal of Clinical Nutrition* 1955;**3**:225-9. [DOI: [10.1093/ajcn/3.3.225](https://doi.org/10.1093/ajcn/3.3.225)]
- 111.** Scrimshaw NS, Munoz JA, Tandon OB, Guzman MA. Growth and development of Central American children. 2. The effect of oral administration of vitamin B12 to rural children of preschool and school age. *American Journal of Clinical Nutrition* 1959;**7**:180-4. [DOI: [10.1093/ajcn/7.2.180](https://doi.org/10.1093/ajcn/7.2.180)]
- 112.** Shimazono N, Hayashi M, Nakagawa I, Sahashi Y, Ueda K, Arakawa M, et al. The effect of vitamin B12 administration on

Japanese children. *Journal of Vitaminology* 1963;**9**:79-88. [DOI: [10.5925/jnsv1954.9.79](https://doi.org/10.5925/jnsv1954.9.79)]

113. Strand TA, Ulak M, Hysing M, Ranjitkar S, Kvestad I, Shrestha M, et al. Effects of vitamin B12 supplementation on neurodevelopment and growth in Nepalese infants: a randomized controlled trial. *PLOS Medicine* 2020;**17**(12):e1003430. [DOI: [10.1371/journal.pmed.1003430](https://doi.org/10.1371/journal.pmed.1003430)]

114. Ask TF, Ranjitkar S, Ulak M, Chandyo RK, Hysing M, Strand TA, et al. The association between heart Rate variability and neurocognitive and socio-emotional development in Nepalese infants. *Frontiers in Neuroscience* 2019;**13**:Article 411. [DOI: [10.3389/fnins.2019.00411](https://doi.org/10.3389/fnins.2019.00411)]

115. Ulak M, Kvestad I, Chandyo RK, Schwinger C, Basnet S, Shrestha M, et al. The effect of vitamin B(12) supplementation on leukocyte telomere length in mildly stunted Nepalese children: a secondary outcome of a randomized controlled trial. *Journal of Nutrition* 2024;**154**(8):2546-50. [DOI: [10.1016/j.tjnut.2023.10.015](https://doi.org/10.1016/j.tjnut.2023.10.015)]

116. Hysing M, Strand TA, Chandyo RK, Ulak M, Ranjitkar S, Schwinger C, et al. The effect of vitamin B12-supplementation on actigraphy measured sleep pattern; a randomized control trial. *Clinical Nutrition* 2022;**41**(2):307-12. [DOI: [10.1016/j.clnu.2021.11.040](https://doi.org/10.1016/j.clnu.2021.11.040)]

117. NCT02272842. Vitamin B12, neurodevelopment and growth in Nepal [The effect of vitamin B12 supplementation in Nepali infants on growth and development]. <https://clinicaltrials.gov/study/NCT02272842> (first posted 23 October 2014).

118. Ranjitkar S, Kvestad I, Strand TA, Ulak M, Shrestha M, Chandyo RK, et al. Acceptability and reliability of the Bayley Scales of Infant and Toddler Development-III among children in Bhaktapur, Nepal. *Frontiers in Psychology* 2018;**9**:1265. [DOI: [10.3389/fpsyg.2018.01265](https://doi.org/10.3389/fpsyg.2018.01265)]

119. Chandyo MU, Kvestad I, Chandyo RK, Ranjitkar S, Hysing M, Schwinger C, et al. The effect of infant vitamin B12 supplementation on neurodevelopment: a follow-up of a randomized placebo-controlled trial in Nepal. *Tropical Medicine and International Health* 2021;**26**(Suppl 1):176-7. [DOI: [10.1111/tmi.13632](https://doi.org/10.1111/tmi.13632)]

120. Strand TA, Ulak M, Chandyo RK, Kvestad I, Hysing M, Shrestha M, et al. The effect of vitamin B(12) supplementation in Nepalese infants on growth and development: study protocol for a randomized controlled trial. *Trials* 2017;**18**(1):187. [DOI: [10.1186/s13063-017-1937-0](https://doi.org/10.1186/s13063-017-1937-0)]

121. Strand T, Chandyo R, McCann A, Ulak M, Hysing M, Ranjitkar S, et al. The effect of daily B12 supplementation on the incidence of common childhood infections in Nepal: secondary outcomes from a double blind randomized controlled trial. *Current Developments in Nutrition* 2023;**7**(Suppl 1):100955. [DOI: [10.1016/j.cdnut.2023.100955](https://doi.org/10.1016/j.cdnut.2023.100955)]

122. Ulak M, Kvestad I, Chandyo RK, Ranjitkar S, Hysing M, Schwinger C, et al. The effect of infant vitamin B12 supplementation on neurodevelopment: a follow-up of a

randomized placebo-controlled trial in Nepal. *British Journal of Nutrition* 2022;**129**(1):1-18. [DOI: [10.1017/s0007114522000071](https://doi.org/10.1017/s0007114522000071)]

123. Taneja S, Strand TA, Kumar T, Mahesh M, Mohan S, Manger MS, et al. Folic acid and vitamin B-12 supplementation and common infections in 6-30-mo-old children in India: a randomized placebo-controlled trial. *American Journal of Clinical Nutrition* 2013;**98**(3):731-7. [DOI: [10.3945/ajcn.113.059592](https://doi.org/10.3945/ajcn.113.059592)]

124. Manapurath R, Chowdhury R, Taneja S, Bhandari N, Strand T. Effect of folate and/or vitamin B12 supplementation between 6 to 30 months of age on markers of cardiovascular illness at school age: a secondary analysis. *Current Developments in Nutrition* 2022;**6**(Suppl 1):928. [DOI: [10.1093/cdn/nzac067.048](https://doi.org/10.1093/cdn/nzac067.048)]

125. Chowdhury R, Taneja S, Bhandari N, Kvestad I, Strand TA, Bhan MK. Vitamin-D status and neurodevelopment and growth in young north Indian children: a secondary data analysis. *Nutrition Journal* 2017;**16**(1):59. [DOI: [10.1186/s12937-017-0285-y](https://doi.org/10.1186/s12937-017-0285-y)]

126. Chowdhury R, Taneja S, Bhandari N, Sinha B, Upadhyay RP, Bhan MK, et al. Vitamin-D deficiency predicts infections in young north Indian children: a secondary data analysis. *PLOS One* 2017;**12**(3):e0170509. [DOI: [10.1371/journal.pone.0170509](https://doi.org/10.1371/journal.pone.0170509)]

127. Chowdhury R, Taneja S, Bhandari N, Strand TA, Bhan MK. Vitamin D deficiency and mild to moderate anemia in young North Indian children: a secondary data analysis. *Nutrition* 2019;**57**:63-8. [DOI: [10.1016/j.nut.2018.05.034](https://doi.org/10.1016/j.nut.2018.05.034)]

128. Kvestad I, Taneja S, Kumar T, Hysing M, Refsum H, Yajnik CS, et al. Vitamin B12 and folic acid improve gross motor and problem-solving skills in young North Indian children: a randomized placebo-controlled trial. *PLOS One* 2015;**10**(6):e0129915. [DOI: [10.1371/journal.pone.0129915](https://doi.org/10.1371/journal.pone.0129915)]

129. Kvestad I, Taneja S, Upadhyay RP, Hysing M, Bhandari N, Strand TA. Vitamin B(12), folate, and cognition in 6- to 9-year-olds: a randomized controlled trial. *Pediatrics* 2020;**145**(3):e20192316. [DOI: [10.1542/peds.2019-2316](https://doi.org/10.1542/peds.2019-2316)]

130. NCT00717730. Folic acid and vitamin B12 in young Indian children [Routine administration of folic acid and vitamin B12 to prevent childhood infections in young Indian children]. <https://clinicaltrials.gov/ct2/show/NCT00717730> (first posted 17 July 2008).

131. Strand TA, Taneja S, Kumar T, Manger MS, Refsum H, Yajnik CS, et al. Vitamin B-12, folic acid, and growth in 6- to 30-month-old children: a randomized controlled trial. *Pediatrics* 2015;**135**(4):e918-26. [DOI: [10.1542/peds.2014-1848](https://doi.org/10.1542/peds.2014-1848)]

132. Kumar T, Taneja S, Sachdev HPS, Refsum H, Yajnik CS, Bhandari N, et al. Supplementation of vitamin B12 or folic acid on hemoglobin concentration in children 6-36 months of age: a randomized placebo controlled trial. *Clinical Nutrition* 2017;**36**(4):986-91. [DOI: [10.1016/j.clnu.2016.07.002](https://doi.org/10.1016/j.clnu.2016.07.002)]

133. Winje BA, Kvestad I, Krishnamachari S, Manji K, Taneja S, Bellinger DC, et al. Does early vitamin B(12) supplementation

improve neurodevelopment and cognitive function in childhood and into school age: a study protocol for extended follow-ups from randomised controlled trials in India and Tanzania. *BMJ Open* 2018;**8**(2):e018962. [DOI: [10.1136/bmjopen-2017-018962](https://doi.org/10.1136/bmjopen-2017-018962)]

134. Taneja S, Chowdhury R, Kvestad I, Bhandari N, Strand TA. Vitamin B12 and/or folic acid supplementation on linear growth; a 6 years follow-up study of a randomised controlled trial in early childhood in North India. *British Journal of Nutrition* 2022;**129**:1-22. [DOI: [10.1017/s0007114522002343](https://doi.org/10.1017/s0007114522002343)]

135. Yajnik C, Kasture S, Kantikar V, Lubree H, Bhat D, Raut D, et al. Efficacy of B(12) fortified nutrient bar and yogurt in improving plasma B12 concentrations-results from 2 double-blind randomized placebo controlled trials. *Food and Nutrition Bulletin* 2021;**42**(4):480-9. [DOI: [10.1177/03795721211025448](https://doi.org/10.1177/03795721211025448)]

136. CTRI/2012/07/002799. Effect of vitamin B12 fortified nutrient bar supplementation in Indian children [Efficacy of vitamin B12 fortified food supplement in Indian children]. <https://trialsearch.who.int/Trial2.aspx?TrialID=CTRI/2012/07/002799> (first posted 16 July 2012).

137. Larvie DY. Request for study information to include in cochrane systematic review [personal communication]. Email to: C Yajnik 2 January 2024.

138. Grillenberger M. Impact of animal source foods on growth, morbidity and iron bioavailability in Kenyan school children [PhD thesis]. Netherlands: Wageningen University and Research, 2006.

139. Bjørke-Monsen AL, Torsvik IK, Markestad T, Ueland PM. Cobalamin supplementation improves motor development and gastrointestinal reflux in moderately deficient infants. *Clinical Biochemistry* 2011;**44**(7):524-5. [DOI: <https://dx.doi.org/10.1016/j.clinbiochem.2011.03.050>]

140. Bjørke-Monsen AL, Torsvik I, Stran H, Markestad T, Ueland PM. Effect of cobalamin supplementation in infants: a randomized, controlled trial. *Pediatrics* 2008;**121**(Suppl 2):S109-10.

141. NCT00479479. Cobalamin supplementation during infancy; effect on B-vitamin status, growth and psychomotor development. <https://clinicaltrials.gov/study/NCT00479479> (first posted 28 May 2007).

142. NCT05005897. Early infant micronutrition and development (RART) [Vitamin B12 status in infancy and the effect of providing vitamin B12 to infants with signs of suboptimal vitamin B12 status - a registry-based, randomized controlled trial]. <https://www.clinicaltrials.gov/study/NCT05005897> (first posted 16 August 2021).

143. Evdokimova AI, Aksenova OV. Vitamin B 12 treatment of children with hemolytic anemia. *Voprosy Okhrany Materinstva i Detstva* 1971;**16**(3):88.

144. Jobert J, Bachelot C, Jerome JM, Sele M, Bost M. Trial treatment of early anemia in premature infants [Essai de traitement de l'anémie précoce du prématuré]. *Pediatric* 1975;**30**(8):809-20.

145. Mena P, Pittaluga E, Blanco A, Castillo C, Toboada H. Vitamin B12 does not change the course of the anemia of preterm infants. *Revista Chilena Pediatría* 2001;**72**(1):34-9.

146. NCT00572741. Treating oxidative stress and the metabolic pathology of autism. <https://www.cdek.liu.edu/trial/NCT00572741/>.

147. Olivi O, Mugnaini P, Di Battista C. Enhancement of steroid anabolism by the association of cobamamide [Potenziamento dell'attività anabolizzante steroidea ottenuto mediante associazione cobamamidica]. *Minerva Pediatrica* 1971;**23**(30):914-21.

148. Sun L, Zhu L, Li C, Shen H, Zhai H, Liu D. Effect of vitamin B12 combined with mouse nerve growth factor in children with cerebral palsy. *Northwest Pharmaceutical Journal* 2022;**37**(6):116-9.

149. Torsvik IK, Ueland PM, Markestad T, Midttun O, Monsen AB. Motor development related to duration of exclusive breastfeeding, B vitamin status and B12 supplementation in infants with a birth weight between 2000-3000 g, results from a randomized intervention trials. *BMC Pediatrics* 2015;**15**(1):218. [DOI: <https://dx.doi.org/10.1186/s12887-015-0533-2>]

150. NCT01201005. B vitamin status in premature and small for gestational age (SGA) infants [B vitamin status in premature and small for gestational age infants]. <https://classic.clinicaltrials.gov/ct2/show/NCT01201005> (first posted 14 September 2010).

151. Worthingtonwhite DA, Behnke M, Gross S. Premature infants require additional folate and vitamin-B-12 to reduce the severity of the anemia of prematurity. *American Journal of Clinical Nutrition* 1994;**60**(6):930-5. [DOI: [10.1093/ajcn/60.6.930](https://doi.org/10.1093/ajcn/60.6.930)]

152. Haiden N, Klebermass K, Cardona F, Schwindt J, Berger A, Kohlhauser-Vollmuth C, et al. A randomized, controlled trial of the effects of adding vitamin B12 and folate to erythropoietin for the treatment of anemia of prematurity. *Pediatrics* 2006;**118**(1):180-8. [DOI: [10.1542/peds.2005-2475](https://doi.org/10.1542/peds.2005-2475)]

153. Bansal PG, Toteja GS, Bhatia N, Vikram NK, Siddhu A. Impact of weekly iron folic acid supplementation with and without vitamin B12 on anaemic adolescent girls: a randomised clinical trial. *European Journal of Clinical Nutrition* 2016;**70**(6):730-7. [DOI: [10.1038/ejcn.2015.215](https://doi.org/10.1038/ejcn.2015.215)]

154. NCT01490944. Impact of vitamin B12 supplementation with iron and folic acid on adolescent girls. <https://clinicaltrials.gov/show/NCT01490944> (first posted 13 December 2011).

155. CTRI/2011/12/002217. Clinical trial to assess impact of iron, folic acid and Vitamin B12 supplementation on anaemic adolescent girls [To assess the impact of vitamin B12 supplementation with iron and folic acid on anaemic adolescent girls]. <https://trialsearch.who.int/Trial2.aspx?TrialID=CTRI/2011/12/002217> (first posted 8 December 2011).

156. Chandelia S, Chandra J, Narayan S, Aneja S, Chawla HM, Sharma S, et al. Addition of cobalamin to iron and folic acid improves hemoglobin rise in nutritional anemia. *Indian*

Journal of Pediatrics 2012;**79**(12):1592-6. [DOI: [10.1007/s12098-012-0725-9](https://doi.org/10.1007/s12098-012-0725-9)]

157. Daute KH, Raab D, Kiessig R. Effect of vitamin B12 on the behaviour of schoolchildren with poor concentration [Über die Wirkung von Vitamin B12 auf das Verhalten konzentrationsgestörter Schulkinder]. *Deutsche Gesundheitswesen* 1964;**19**:580-4.

158. Mackay IFS, Patrick SJ, Stafford D, Cleveland FS. The influence of vitamin B12 and aureomycin upon the growth of protein-deficient children. *Journal of Nutrition* 1956;**59**:155-70. [DOI: [10.1093/jn/59.1.155](https://doi.org/10.1093/jn/59.1.155)]

159. Patrick SJ. Some observations on the metabolism of vitamin B12 by Jamaican children. *Journal of Nutrition* 1955;**55**:129-35. [DOI: [10.1093/jn/55.1.129](https://doi.org/10.1093/jn/55.1.129)]

160. ISRCTN67437725. Growth and body composition in undernourished children: effect of vitamin B12 supplementation [Growth and body composition in acute severe malnutrition (SAM) in Indian children: effect of vitamin B12 supplementation in a double-blind randomised controlled pilot study]. <https://www.isrctn.com/ISRCTN67437725> (first posted 15 November 2010).

161. Rusconi R, Seidita F. Controlled study of 3 different agents (ferritin, folic acid and coenzyme B12) in treatment of iron-deficiency anemia in children. *Clinica Europea* 1983;**22**(3):543-57.

162. CTRI/2023/09/057950. Effect of vitamin B12 supplementation on anthropometric indices in children with severe acute malnutrition (SAM) aged 1 month to 60 month in a nutritional rehabilitation center - a randomized control trial [A randomized control double blind parallel group without placebo clinical trial to look the antropometric outcome of vitamin B12 supplementation in children with severe acute malnutrition (SAM) aged b1 month to 60 months in Nutritional Rehabilitation Center]. <https://trialsearch.who.int/Trial2.aspx?TrialID=CTRI/2023/09/057950> (first posted 22 September 2023).

163. Aguirre F, Schrimshaw NS. The effect of supplements of animal and vegetable protein, vitamin B₁₂, and aureomycin on hematological values in Central American school children. *American Journal of Clinical Nutrition* 1955;**3**(3):225-9.

164. Shalini T, Pullakhandam R, Ghosh S, Kularni B, Rajkumar H, Sachdev HS, et al. Prevalence of vitamin B12 and folate deficiencies in Indian children and adolescents. *Nutrients* 2023;**15**(13):3026. [DOI: [10.3390/nu15133026](https://doi.org/10.3390/nu15133026)]

165. Umasanker S, Bhakat R, Mehta S, Rathaur VK, Verma PK, Bhat NK, et al. Vitamin B12 deficiency in children from Northern India: time to reconsider nutritional handicaps. *Journal of Family Medicine and Primary Care* 2020;**9**(9):4985-91. [DOI: [10.4103/jfmpc.jfmpc_712_20](https://doi.org/10.4103/jfmpc.jfmpc_712_20)]

166. Ng'eno B, Perrine CG, Whitehead RD, Subedi GR, Mebrahtu S, Dahal P, et al. High prevalence of vitamin B12 deficiency and no folate deficiency in young children in Nepal. *Nutrients* 2017;**9**(11):1-13. [DOI: [10.3390/nu9111126](https://doi.org/10.3390/nu9111126)]

167. Wong E, Molina-Cruz R, Rose C, Bailey L, Kauwell GPA, Rosenthal J. Prevalence and disparities in folate and vitamin B12 deficiency among children aged 6–59 months in Guatemala. *Maternal and Child Health Journal* 2022;**26**(1):156-67.

168. Galicia L, Grajeda R, de Romaña DL. Nutrition situation in Latin America and the Caribbean: current scenario, past trends, and data gaps. *Revista Panamericana de Salud Pública* 2016;**40**(2):104-13.

169. Kuo SS, Eack SM. Meta-analysis of cognitive performance in neurodevelopmental disorders during adulthood: comparisons between autism spectrum disorder and schizophrenia on the Wechsler Adult Intelligence Scales. *Frontiers in Psychiatry* 2020;**11**:187. [DOI: [10.3389/fpsy.2020.00187](https://doi.org/10.3389/fpsy.2020.00187)]

170. Griffith LE, van den Heuvel E, Fortier I, Sohel N, Hofer SM, Payette H, et al. Statistical approaches to harmonize data on cognitive measures in systematic reviews are rarely reported. *Journal of Clinical Epidemiology* 2015;**68**(2):154-62. [DOI: [10.1016/j.jclinepi.2014.09.003](https://doi.org/10.1016/j.jclinepi.2014.09.003)] [PMCID: PMC4685455] [PMID: 25497980]

171. Ryan R, Hill S. How to GRADE the quality of the evidence. Cochrane Consumers and Communication Group Version 3.0 December, 2016. Available at cccg.cochrane.org/author-resources.

172. dos Santos, da Silva KG, Souza AP, da Silva, Souza APD, da Silva ABJ, et al. Relationship between vitamin B12 levels and motor development: a systematic review. *Clinical Nutrition ESPEN* 2024;**63**:148-56.

173. Meli AM, Ali A, Mhd Jalil AM, Mohd Yusof H, Tan MMC. Effects of physical activity and micronutrients on cognitive performance in children aged 6 to 11 years: a systematic review and meta-analysis of randomized controlled trials. *Medicina (Kaunas)* 2021;**58**(1):57. [DOI: [10.3390/medicina58010057](https://doi.org/10.3390/medicina58010057)]

174. Iglesia I, Dhonukshe-Rutten RA, Bel-Serrat S, Doets EL, Cavelaars AE, van't Veer P, et al. Association between vitamin B12 intake and EURRECA's prioritized biomarkers of vitamin B12 in young populations: a systematic review. *Public Health Nutrition* 2013;**16**(10):1843-60. [DOI: [10.1017/s1368980012003953](https://doi.org/10.1017/s1368980012003953)]

175. Benton D. Micro-nutrient supplementation and the intelligence of children. *Neuroscience & Behavioural Reviews* 2001;**25**(4):297-309. [DOI: [10.1016/s0149-7634\(01\)00015-x](https://doi.org/10.1016/s0149-7634(01)00015-x)]

176. Ramakrishnan U, Goldenberg T, Allen LH. Do multiple micronutrient interventions improve child health, growth, and development? *Journal of Nutrition* 2011;**141**(11):2066-75. [PMID: 21956959]

177. Eilander A, Gera T, Sachdev HS, Transler C, van der Knaap HC, Kok FJ, et al. Multiple micronutrient supplementation for improving cognitive performance in children: systematic review of randomized controlled trials. *American Journal of Clinical Nutrition* 2010;**91**(1):115-30. [DOI: [10.3945/ajcn.2009.28376](https://doi.org/10.3945/ajcn.2009.28376)]

178. Das JK, Salam RA, Kumar R, Bhutta ZA. Micronutrient fortification of food and its impact on woman and child health: a systematic review. *BMC Systematic Reviews* 2013;**2**:67. [DOI: [10.1186/2046-4053-2-67](https://doi.org/10.1186/2046-4053-2-67)]

179. Das JK, Salam RA, Mahmood SB, Moin A, Kumar R, Mukhtar K, et al. Food fortification with multiple micronutrients: impact on health outcomes in general population. *Cochrane Database of Systematic Reviews* 2019, Issue 12. Art. No: CD011400. [DOI: [10.1002/14651858.CD011400.pub2](https://doi.org/10.1002/14651858.CD011400.pub2)]

ADDITIONAL TABLES

Table 1. Overview of included studies and synthesis table illustrating key study characteristics sorted by comparison group^{a,b}

Study	Study design	Setting (World Bank region, WHO region, country, facility/setting type)	Study population (age range, number randomised)	Intervention				Outcome assessment	
				Duration	Form	Dose	Vitamin B ₁₂ compound	Method of measurement/assessment tools (synthesis method)	Time points/time-frames
Comparison 1: Supplementation with vitamin B ₁₂ alone versus placebo (13 studies)									
Areekul 1979	RCT	East Asia & Pacific, South-East Asia, Thailand, hospital	8 to 10 years n = 8	5 weeks	Tablet	50 µg 3 times/day (150 µg/day)	Not reported	• Vitamin B₁₂ status ◦ Total vitamin B₁₂ concentrations: radioisotope dilution with coated charcoal technique (meta-analysis) ◦ Holotranscobalamin concentrations: serum fractionation (narrative summary)	Baseline, weekly, 5 weeks
Benjamin 1952	RCT	Europe & Central Asia, European Region, United Kingdom, open-air schools	Under 8 and 9, 10 and 11, 12 and over years n = 831 (n = 612; children < 12 years)	8 weeks (during school attendance days)	Tablet	10 µg/day, 5 days/week	Not reported	• Growth ◦ Height: not reported (narrative summary) • Morbidity: number of children absent from school due to a primary ailment (asthma, bronchitis, bronchiectasis) or a new condition (e.g. whooping cough, chicken pox, measles, mumps, acute upper respiratory infection, influenza, other respiratory diseases) (narrative summary)	Baseline, 4 weeks, 8 weeks
Chinnock 1952	Quasi-RCT	North America, Region of the Americas, United States of America, paediatric clinic	Full-term infants (age not reported) n = 90	8 weeks	Drops	5 µg/day for 30 or 60 days	Not reported	• Growth ◦ Height: not reported (narrative summary)	Baseline, 2 weeks, 4 weeks, 8 weeks
Craigmile 1956	Quasi-RCT	Europe & Central Asia,	1 to 2 years	6 months (during	Not reported	50 µg/day	Not reported	• Growth	Baseline, 6 months

Table 1. Overview of included studies and synthesis table illustrating key study characteristics sorted by comparison group^{a,b} (Continued)

		European Region, United Kingdom, day nurseries	n = 104	school attendance days)				◦ Height: not reported (narrative summary) • Health-related quality of life: assessment of appetite or behaviour reported by matrons of nurseries (time point not reported) (narrative summary) • Morbidity: the number of children absent due to colds during the trial (meta-analysis)	
Deshmukh 2010	Cluster-RCT	South Asia, South-East Asia, India, rural villages	Mean ± SD: 9.0 ± 0.2 years n = 119 families: 300 children and parents (n = 52; children < 12 years)	12 months	Capsule	2 µg/day and 10 µg/day	Cyanocobalamin	• Growth: ◦ Height: wall-mounted stadiometer (CMS Instruments, London, UK) (meta-analysis) • Cognitive function ◦ Visuospatial function: Malin's Intelligence Scale for Children (Block design score) (meta-analysis) ◦ Working memory: Malin's Intelligence Scale for Children (Digit Span Forward Test, Digit Span Backward Test) (meta-analysis) ◦ Intelligence quotient: Raven's Color Progressive Matrices (Color Progressive Matrices Score) (meta-analysis) ◦ Attention: Color Trail Test (Color Trail A and B (time in seconds)) (meta-analysis) • Vitamin B₁₂ status: ◦ Vitamin B₁₂ deficiency: vitamin B₁₂ concentration < 148 pmol/L determined by microbiologic assay (<i>Lactobacillus leichmannii</i>; plasma) (meta-analysis) ◦ Total vitamin B₁₂ concentrations: microbiologic assay (<i>Lactobacillus leichmannii</i>; plasma) (meta-analysis) ◦ Homocysteine concentrations: plasma fluorescence polarisation (Abbott IMx analyzer, Abbott Laboratories) (meta-analysis) • Anaemia and haemoglobin:	Baseline, 4 months, 12 months

Table 1. Overview of included studies and synthesis table illustrating key study characteristics sorted by comparison group^{a,b} (Continued)

								◦ Haemoglobin concentrations < 12 g/dL measured with a Beckman Coulter Analyzer (meta-analysis) • Megaloblastic anaemia: mean corpuscular volume greater than 97 fL (meta-analysis) • Body composition: percent body fat: dual-energy X-ray absorptiometry scans (paediatric mode) via a DPX-IQ 240 pencil beam machine (Lunar Corp., Madison, WI) (software version 4.7)	
Fukui 1959	Quasi-RCT	East Asia & Pacific, Western Pacific, Japan, primary school	10 to 11 years n = 117	6 months (during school attendance days)	Not reported	10 µg/day or 20 µg/day	Not reported	• Growth ◦ Height: not reported (narrative summary)	Baseline, 2 months, 4 months, 6 months
Kudo 1962	RCT	East Asia & Pacific, Western Pacific, Japan, rural prefecture	9 to 11 years n = 40	52 weeks	Tablet	25 µg/day	Cyanocobalamin	• Vitamin B₁₂ status ◦ Total vitamin B₁₂ concentrations: microbiologic assay (<i>Lactobacillus leichmannii</i>; whole blood) (meta-analysis)	Baseline, 10 weeks, 21 weeks, 33 weeks, 41.5 weeks, 52 weeks
Murakami 1962	RCT	East Asia & Pacific, Western Pacific, Japan, rural prefecture	9 to 11 years, n = 100	12 months	Tablet	25 µg/day	Cyanocobalamin	• Vitamin B₁₂ status ◦ Total vitamin B₁₂ concentrations: microbiologic assay (<i>Lactobacillus leichmannii</i>; whole blood) (meta-analysis) • Anaemia and haemoglobin: ◦ Haemoglobin concentrations: photoelectric colorimeter (Shigiya's method) (meta-analysis)	Baseline, 10 weeks, 21 weeks, 33 weeks, 41.5 weeks, 52 weeks
Scrimshaw 1953	Quasi-RCT	Latin America & Caribbean, Region of the Americas, El Salvador, urban	7 to 11 years n = 425 (across study sites)	El Salvador: 28 to 31 months during school days	El Salvador: not reported Guatemala: tablet	El Salvador: 20 µg/day Guatemala: 20 µg/day	El Salvador: not reported Guatemala: not reported	• Growth ◦ Height: not reported (narrative summary) • Anaemia and haemoglobin: ◦ Haemoglobin concentrations: not reported (narrative summary)	Baseline, monthly, 5 months, 6 months

Table 1. Overview of included studies and synthesis table illustrating key study characteristics sorted by comparison group^{a,b} (Continued)

		ban and rural schools		Guatemala: 11 to 28 months during school days				Megaloblastic anaemia: percent mean corpuscular volume greater than 94 cu μ (narrative summary)	
		Latin America & Caribbean, Region of the Americas, Guatemala, indigenous communities							
Scrimshaw 1959	RCT	Latin America & Caribbean, Region of the Americas, Guatemala, rural and highland villages	Preschool children (n = 50; age group not reported) and school-age children (n = 228; age group not reported) n = 278	18 months during the school days (intervention halted during school vacation)	Tablet	20 μ g/day, 6 days/week	Not reported	- Growth - Height: not reported (narrative summary)	Monthly, 6 months, 12 months, 18 months
Shimazono 1963	RCT	East Asia & Pacific, Western Pacific, Japan, elementary school, orphanages and nurseries	3 to 14 years (across study sites) n = 510 (n = 323; children < 12 years)	12 months	Tablet	25 μ g/day	Cyanocobalamin	- Growth - Height: not reported (narrative summary) - Vitamin B₁₂ status - Total vitamin B₁₂ concentrations: microbiologic assay (<i>Lactobacillus leichmannii</i>; whole blood) (meta-analysis (sensitivity analysis)) - Anaemia and haemoglobin: - Haemoglobin concentrations: not reported (meta-analysis)	Baseline, 2.5 months, 4 to 6 months, 7.5 months, 9.5 months, 12 months
Taneja 2013	RCT	South Asia, South-East Asia, India, urban communities	6 to 30 months n = 1000	6 months	Lipid-based nutritional supplement	1.8 μ g/day (children aged > 12 months); and 0.9	Cyanocobalamin	- Growth - Height-for-age z-score or length-for-age z-score: not reported (meta-analysis)	Baseline, 6 months (or 26 weeks), 6 to 9 years

Table 1. Overview of included studies and synthesis table illustrating key study characteristics sorted by comparison group^{a,b} (Continued)

	◦ Height: locally manufactured infantometers^c • Cognitive function ◦ Perceptual reasoning: WISC-IV^{INDIA} (meta-analysis) ◦ Verbal comprehension: CVS (meta-analysis) ◦ Visuospatial function: NEPSY-II (meta-analysis) ◦ Working memory: WISC-IV^{INDIA} (meta-analysis) ◦ Attention: WISC-IV^{INDIA} (meta-analysis) ◦ Processing speed: WISC-IV^{INDIA} (meta-analysis) • Development outcomes ◦ Motor skill development: NEPSY-II (meta-analysis) ◦ Motor skill development (fine motor): ASQ-3 (meta-analysis) ◦ Motor skill development (gross motor): ASQ-3 (meta-analysis) ◦ Socio-emotional and adaptive behaviour: NEPSY-II (meta-analysis) ◦ Language: NEPSY-II (meta-analysis) • Vitamin B₁₂ status ◦ Vitamin B₁₂ deficiency: microbiologic assay (<i>Lactobacillus leichmannii</i>) plasma, defined as vitamin B₁₂ concentration < 150 pmol/L (meta-analysis) ◦ Total vitamin B₁₂ concentrations: microbiologic assay (<i>Lactobacillus leichmannii</i>) plasma (meta-analysis) ◦ Homocysteine concentrations: plasma fluorescence polarisation (Abbott IMx analyzer, Abbott Laboratories) (meta-analysis) • Anaemia and haemoglobin: ◦ Haemoglobin: venous blood using HemoCue (HemoCue 201; Ångelholm) (meta-analysis)
--	---

Table 1. Overview of included studies and synthesis table illustrating key study characteristics sorted by comparison group^{a,b} (Continued)

• Morbidity (proportion of children with at least one reported illness) (meta-analysis)									
Yajnik 2021	RCT	South Asia, South-East Asia, India, rural school	10 to 13 years n = 164 (n = 164; children < 12 years)	120 days (consecutive school days)	Nutrient bar	2 µg/day	Cyanocobalamin	• Growth ◦ Height: stadiometer (CMS Instruments) (meta-analysis) • Vitamin B₁₂ status ◦ Vitamin B₁₂ deficiency: microbiologic assay (<i>Lactobacillus leichmannii</i>) plasma defined as vitamin B₁₂ concentration < 148 pmol/L (meta-analysis) ◦ Total vitamin B₁₂ concentrations: microbiologic assay (<i>Lactobacillus leichmannii</i>; plasma) (meta-analysis) ◦ Homocysteine concentrations: high-performance liquid chromatography using fluorescence detection (meta-analysis) • Anaemia and haemoglobin: ◦ Anaemia: not reported, defined as haemoglobin concentrations < 12g/dL (meta-analysis) ◦ Haemoglobin concentrations: not reported (meta-analysis) • Megaloblastic anaemia: mean corpuscular volume greater than 97 fL (meta-analysis)	Baseline, and 120 days
Comparison 2: Supplementation with vitamin B₁₂ alone versus no intervention (2 studies)									
Gutiérrez-Díaz 1959	Quasi-RCT	Europe & Central Asia, European Region, Spain (setting type not reported)	Female children (age not reported) n = 100	10 months	Syrup	10 µg/day	Not reported	• Growth ◦ Height: not reported (narrative summary) • Health-related quality of life: appetite rated by parents (narrative summary) • Morbidity: general, flu pandemic or childhood diseases (narrative summary)	2 months, and 10 months
Scaglione 1955	Quasi-RCT	Europe & Central Asia, European	2 days of age (new-born or	28 days	Drops	30 µg/day	Not reported	• Growth ◦ Height: not reported (narrative summary)	Baseline, every 2

Table 1. Overview of included studies and synthesis table illustrating key study characteristics sorted by comparison group^{a,b} (Continued)

		Region, Italy, obstetric clinic	premature or immature and low birth-weight infants)						days, 28 days
			n = 70						
Comparison 3: Supplementation with vitamin B₁₂ and other other micronutrients, compared to supplements with the same formulation without vitamin B₁₂ (3 studies)^a									
Strand 2020 ^d	RCT	South Asia, South-East Asia, Nepal, immunisation clinics or residing in the Bhaktapur municipality and surrounding peri-urban communities	6 to 11 months n = 600	12 months	Lipid-based paste	2 µg/day	Cyanocobalamin	- Growth - Height-for-age z-score or length-for-age z-score: WHO growth charts (meta-analysis) - Height: portable length board (model 417, Seca, California, USA) (meta-analysis) - Cognitive function - Verbal comprehension: WPPSI-IV (meta-analysis) - Visuospatial function: NEPSY-II (meta-analysis) - Working memory: WPPSI-IV (meta-analysis) - Attention: NEPSY-II (meta-analysis) - Intelligence quotient: WPPSI-IV (meta-analysis) - Development outcomes - Language: BSID-III (meta-analysis) - Language (receptive): BSID-III (meta-analysis) - Language (expressive): BSID-III (meta-analysis) - Socio-emotional and adaptive behaviour: BSID-III (meta-analysis) - Motor skill development: BSID-III (meta-analysis) - Motor skill development (fine motor): BSID-III (meta-analysis) - Motor skill development (gross motor): BSID-III (meta-analysis)	Baseline, 6 months, 12 months, 24 months

Table 1. Overview of included studies and synthesis table illustrating key study characteristics sorted by comparison group^{a,b} (Continued)

- Vitamin B₁₂ status
 - Total vitamin B₁₂ concentrations: microbiologic assay (*Lactobacillus leichmannii*; plasma) (meta-analysis)
 - Methylmalonic acid concentrations: gas chromatography-mass spectrometry (GC-MS/MS; Varian) (plasma) (meta-analysis)
 - Total homocysteine concentrations: gas chromatography-mass spectrometry (GC-MS/MS; Varian) (plasma) (meta-analysis)
- Anaemia and haemoglobin:
 - Haemoglobin: venous blood using HemoCue (HemoCue 201; Ångelholm) (meta-analysis)
- Any adverse effects: number of hospitalisations due to urinary tract infections, lower respiratory infections, seizure-related illness, constipation, or other illnesses (surgical, diarrhoea or sepsis) (meta-analysis)

^aDeshmukh 2010 and Taneja 2013 included more than one intervention arm and contributed data to Comparison 1 and Comparison 3. These two trials compared vitamin B₁₂ supplementation alone versus placebo and vitamin B₁₂ supplementation plus folic acid versus folic acid alone. For trials including children ≥ 12 years, only age-stratified data for children < 12 years were included in analyses.

^bAbbreviations: ASQ-3: Ages & Stages Questionnaire; BSID-III: Bayley Scale of Infant Development, Third Edition; CVS: Crichton Vocabulary Score; RCT: randomised controlled trial; NEPSY-II: NEuroPSYchological assessment, Second Edition; SD: standard deviation; WHO: World Health Organization; WISC-IV^{INDIA}: Wechsler Intelligence Scale for Children, Fourth Edition (India); WPPSI-IV: Wechsler Preschool and Primary Scale of Intelligence, Fourth Edition.

^cIn ancillary analyses [39], data were not reported in a format suitable for inclusion in either the meta-analysis or narrative summary. The results are presented as comparisons between a 'supplement group' (receiving either folic acid or vitamin B₁₂) and a 'placebo group', making it impossible to isolate the individual or combined effects of vitamin B₁₂ with or without other micronutrients.

^dTrial compared vitamin B₁₂ supplementation with other micronutrients (including calcium, phosphorus, potassium, magnesium, sodium, iron, zinc, copper, iodine, selenium, vitamin A, vitamin B₁, vitamin B₂, vitamin B₃, vitamin B₅, vitamin B₆, vitamin B₇, folate, vitamin C, vitamin D, vitamin E, and vitamin K), to supplements with the same micronutrient formulation without vitamin B₁₂.

Table 2. Overview of results of risk of bias assessment

Critical outcomes	Study	Risk of bias arising from the randomisation	Risk of bias due to deviations from the intended	Risk of bias due to missing outcome	Risk of bias in measurement of the	Risk of bias in selection of the report-	Overall risk of bias
-------------------	-------	---	--	-------------------------------------	------------------------------------	--	----------------------

Table 2. Overview of results of risk of bias assessment *(Continued)*

		process (Do- main 1-A)	interventions (effect of as- signment to intervention and effect of adhering to intervention) (Domain 2-B)	data (Domain 3-C)	outcome (Do- main 4-D)	ed result (Do- main 5-E)	
1. Growth							
Height/length (cm)							
Height/length (cm)	Deshmukh 2010	Low	Low	Low	Low	Low	Low
Height/length (cm)	Scaglione 1955	Some con- cerns	Low	Low	Some con- cerns	High	High
Height/length (cm)	Strand 2020	Low	Low	Low	Low	Low	Low
Height/length (cm)	Yajnik 2021	Low	Low	Low	Low	Low	Low
Height-for-age z-score (HAZ) or length-for-age z-score (LAZ) (WHO growth standards)							
Height-for-age z-score (HAZ) or length-for-age z-score (LAZ) (WHO growth standards)	Strand 2020	Low	Low	Low	Low	Low	Low
Height-for-age z-score (HAZ) or length-for-age z-score (LAZ) (WHO growth standards)	Taneja 2013	Low	Low	Low	Low	Low	Low
2. Cognitive function, using validated instruments, including general cognitive ability or cognitive domains							
Cognitive function, using validated instruments, including general cognitive ability or cognitive domains	Deshmukh 2010	Low	Low	Low	Low	Low	Low
Cognitive function, using validated instruments, including general cognitive ability or cognitive domains	Strand 2020	Low	Low	Low	Low	Low	Low

Table 2. Overview of results of risk of bias assessment (Continued)

Cognitive function, using validated instruments, including general cognitive ability or cognitive domains	Taneja 2013	Low	Low	Low	Low	Low	Low
3. Development outcomes (e.g. motor skill development, language, and socio-emotional and adaptive behaviour), assessed using validated instruments							
Development outcomes (e.g. motor skill development, language, and socio-emotional and adaptive behaviour), assessed using validated instruments	Strand 2020	Low	Low	Low	Low	Low	Low
Development outcomes (e.g. motor skill development, language, and socio-emotional and adaptive behaviour), assessed using validated instruments	Taneja 2013	Low	Low	Low	Low	Low	Low
4. Vitamin B₁₂ status							
<i>Vitamin B₁₂ deficiency, as defined by trial authors (e.g. < 148, < 221 pmol/L)</i>							
Vitamin B ₁₂ deficiency, as defined by trial authors (e.g. < 148, < 221 pmol/L)	Deshmukh 2010	Low	Low	Low	Low	Low	Low
Vitamin B ₁₂ deficiency, as defined by trial authors (e.g. < 148, < 221 pmol/L)	Strand 2020	Low	Low	Low	Low	Low	Low
Vitamin B ₁₂ deficiency, as defined by trial authors (e.g. < 148, < 221 pmol/L)	Taneja 2013	Low	Low	Low	Low	Low	Low
Vitamin B ₁₂ deficiency, as defined by trial authors (e.g. < 148, < 221 pmol/L)	Yajnik 2021	Low	Low	Low	Low	Low	Low
<i>Total vitamin B₁₂ concentrations (pmol/L)</i>							
Total vitamin B ₁₂ concentrations (pmol/L)	Areekul 1979	Some concerns	Some concerns	Low	Low	Some concerns	High
Total vitamin B ₁₂ concentrations (pmol/L)	Kudo 1962	Some concerns	Some concerns	Low	Low	Some concerns	High
Total vitamin B ₁₂ concentrations (pmol/L)	Murakami 1962	Some concerns	Some concerns	Low	Low	Some concerns	High

Table 2. Overview of results of risk of bias assessment (Continued)

Total vitamin B ₁₂ concentrations (pmol/L)	Shimazono 1963	Some concerns	High	Some concerns	Low	Some concerns	High
Total vitamin B ₁₂ concentrations (pmol/L)	Deshmukh 2010	Low	Low	Low	Low	Low	Low
Total vitamin B ₁₂ concentrations (pmol/L)	Strand 2020	Low	Low	Low	Low	Low	Low
Total vitamin B ₁₂ concentrations (pmol/L)	Taneja 2013	Low	Low	Low	Low	Low	Low
Total vitamin B ₁₂ concentrations (pmol/L)	Yajnik 2021	Low	Low	Low	Low	Low	Low
Methylmalonic acid (MMA) concentrations (μmol/L)							
Methylmalonic acid (MMA) concentrations (μmol/L)	Strand 2020	Low	Low	Low	Low	Low	Low
Holotranscobalamin concentrations (pmol/L)							
Holotranscobalamin concentrations (pmol/L)	Areekul 1979	Some concerns	Some concerns	Low	Low	Some concerns	High
Homocysteine concentrations (μmol/L)							
Homocysteine concentrations (μmol/L)	Deshmukh 2010	Low	Low	Low	Low	Low	Low
Homocysteine concentrations (μmol/L)	Strand 2020	Low	Low	Low	Low	Low	Low
Homocysteine concentrations (μmol/L)	Taneja 2013	Low	Low	Low	Low	Low	Low
Homocysteine concentrations (μmol/L)	Yajnik 2021	Low	Low	Low	Low	Some concerns	Some concerns
5. Anaemia and haemoglobin							
Anaemia, defined using age-specific cut-offs							
Anaemia, defined using age-specific cut-offs	Deshmukh 2010	Low	Low	Low	Low	Low	Low
Anaemia, defined using age-specific cut-offs	Yajnik 2021	Low	Low	Low	Low	Low	Low

Table 2. Overview of results of risk of bias assessment (Continued)

Haemoglobin (g/L) adjusted for altitude (and smoking if reported), when applicable

Haemoglobin (g/L) adjusted for altitude (and smoking if reported), when applicable	Murakami 1962	Some concerns	Some concerns	Low	Low	Some concerns	High
Haemoglobin (g/L) adjusted for altitude (and smoking if reported), when applicable	Shimazono 1963	Some concerns	High	Some concerns	Low	Some concerns	High
Haemoglobin (g/L) adjusted for altitude (and smoking if reported), when applicable	Deshmukh 2010	Low	Low	Low	Low	Low	Low
Haemoglobin (g/L) adjusted for altitude (and smoking if reported), when applicable	Strand 2020	Low	Low	Low	Low	Low	Low
Haemoglobin (g/L) adjusted for altitude (and smoking if reported), when applicable	Taneja 2013	Low	Low	Low	Low	Low	Low
Haemoglobin (g/L) adjusted for altitude (and smoking if reported), when applicable	Yajnik 2021	Low	Low	Low	Low	Low	Low
6. Any adverse effects							
Any adverse effects	Strand 2020	Low	Low	Low	Some concerns	Some concerns	Some concerns

Risk of bias assessments were reported for outcomes synthesised narratively due to inclusion of a single study (i.e. height/length (Scaglione 1955), vitamin B₁₂ status (i.e. methylmalonic acid (Strand 2020) or holotranscobalamin (Areekul 1979) concentrations), and any adverse effects (Strand 2020)).

Table 3. Overview of risk of bias assessments for outcomes including single studies

Study	Randomisation process	Deviations from intended interventions	Missing outcome data	Measurement of the outcome	Selection of the reported results	Overall risk of bias
	Authors' judgement Support for judgement	Authors' judgement Support for judgement	Authors' judgement Support for judgement	Authors' judgement Support for judgement	Authors' judgement Support for judgement	Authors' judgement Support for judgement
Growth						
Height/length (cm)						
Scaglione 1955	Some concerns Quote: "The infants were divided into 4 groups, 3 of which each consisted of 20 subjects, who were respectively treated with Goetsch vitamin T, vitamin B₁₂ and methylandrostendiol. The fourth group, comprising 10 untreated infants, was kept as a follow-up. We thought it appropriate that in the groups of newborns submitted to our observation, elements of identical weight should be represented, in order to better control the action of the various growth factors." Comment: No information is reported regarding allocation concealment prior to assignment. Limited information is provided on se-	Low risk of bias Quote: "The treatment was started from the second day of life and continued for a total period of 28 days. and more particularly the infants of the 1st group were given 5 drops 3 times a day of Goetsch vitamin T, the infants of the second group 5 drops 3 times a day of Vitamin B₁₂, to those of the III° group 3 drops 2 times a day of methylandrostendiol perlingually." "All the newborns, in the first 24 hours, were kept fasting, with small amounts of water, then fed with breast milk or, in the few cases in which this was not possible, with women's milk, administered at regular intervals and in suitable quantities, according to the common rules of childcare. In order to exclude the influence of the dietary factor on weight behaviour, particular care was taken to ensure that the breastfeeding arrangements were the same for everyone. The living environment of the newborns was also the same, hospitalized in rooms, kept at a constant temperature of 25° with	Low risk of bias Quote: "Our observations concern 70 immature infants of different sexes, all born at term of pregnancy and on the eutocic side in the Obstetric Clinic of the University of Rome, with birth weights ranging from a minimum of gr. 1570 to a maximum of gr. 2450. The infants were divided into 4 groups, 3 of which each consisted of 20 subjects, who were respectively treated with Goetsch vitamin T, vitamin	Some concerns Quote: "During this period, the weight of each newborn was checked every 2 days, at the same time, before feeding, and then the average of the values obtained was obtained; Height was also checked, at the beginning and at the end of the searches." Comment: No information is provided regarding the anthropometric method used in outcome ascertainment. There is no indication that methodology for measuring height differed between the different groups. However, limited information provided on process in outcome assessment or the use	High risk of bias Quote: "During this period, the weight of each newborn was checked every 2 days, at the same time, before feeding, and then the average of the values obtained was obtained; Height was also checked, at the beginning and at the end of the searches." Comment: Height measurement is reported at multiple time points. However, there is insufficient information regarding analysis of outcome as part of a pre-specified analysis plan. From Table 2 and Table 4, outcome measurements are reported at baseline, 14 days, and at 28 days. Justification for reporting or selecting three points as opposed to every 2 days as specified is not reported in sufficient detail. Outcome results were reported as raw data for each participant; no indi-	High risk of bias Outcome judged to be at high risk in at least one domain.

Table 3. Overview of risk of bias assessments for outcomes including single studies (Continued)

quence generation method used. Authors report minimal information on baseline characteristics such as socio-demographic characteristics among intervention groups at baseline.	70° hygrometry, by a central air conditioner: the assistance by specialized personnel allowed perfect hygienic conditions to be achieved." "The infants were divided into 4 groups, 3 of which each consisted of 20 subjects, who were respectively treated with Goetsch vitamin T, vitamin B₁₂ and methylandrostendiol. The fourth group, comprising 10 untreated infants, was kept as a follow-up. We thought it appropriate that in the groups of newborns submitted to our observation, elements of identical weight should be represented, in order to better control the action of the various growth factors." Comment: Limited information is provided regarding blinding of participants to intervention groups during the study and how blinding was maintained. However, participants were newborn infants and were not likely to be aware of intervention groups. The published report does not mention who is delivering the intervention nor their blinded status. Limited information is provided regarding blinding of trial investigators to intervention groups during the study and how blinding was maintained. The article details efforts to standardise diets and environments of the infants across the groups and reports balance in trial protocol, e.g. living environment and breastfeeding regimen, across intervention groups.	B₁₂ and methylandrostendiol. The fourth group, comprising 10 untreated infants, was kept as follow-up. We thought it appropriate that in the groups of newborns submitted to our observation, elements of identical weight should be represented, in order to better control the action of the various growth factors." Comment: Tables 2 and 4 both report 20 and 10 subjects in the intervention and control groups, respectively, indicating that all participants were included in an intervention group. Outcome data were available for all eligible and enrolled participants.	of a standard method, to permit judgement. No information is provided on blinding of outcome assessors. Outcome assessment involves an objective measurement in lieu of subjective outcome influenced by participant or investigator-led judgement, and less likely to be influenced by knowledge of the intervention received.	cation that numerical results were selected on the basis of results from multiple analyses.
--	--	--	---	--

Table 3. Overview of risk of bias assessments for outcomes including single studies (Continued)

No clear indication that protocol deviations may have affected the outcome. From Tables 2 and 4, all eligible participants enrolled were analysed with no indication of post-randomisation loss to follow-up.						
Vitamin B₁₂ status						
Methylmalonic acid (MMA) concentrations (µmol/L)						
Strand 2020	Low risk of bias Quote: "Randomization was concealed, and the study double-masked as the participants were only linked to the intervention through the identity number printed on the supplement labels. The list that linked this identity number to the randomization code was kept with the producers of the supplements and the independent scientist who generated it. None of the investigators had access to this list until the data collection and cleaning for the primary outcomes were completed." "Eligible children are allocated by block randomization (block sizes of 8) in a 1:1 ratio to one of the intervention groups. The vitamin B ₁₂ supplements and placebos have been	Low risk of bias Quote: "Randomization was concealed, and the study double-masked as the participants were only linked to the intervention through the identity number printed on the supplement labels. The list that linked this identity number to the randomization code was kept with the producers of the supplements and the independent scientist who generated it. None of the investigators had access to this list until the data collection and cleaning for the primary outcomes were completed." "During weekly visits to the homes, field workers asked the mothers about intake of the paste during the past 7 days and recorded in detail the amount of paste given to the children (i.e., half, one-third, three-fourths, or less)." "From April 2015 to February 2017, we screened 733 infants and randomized 600 into the study (Fig 1). A total of 26 infants dropped out due to refusal or migration. Two children were unable to complete the end-study activities, leaving 572 infants with complete neurodevelopmental as-	Low risk of bias Quote: "From April 2015 to February 2017, we screened 733 infants and randomized 600 into the study (Fig 1). A total of 26 infants dropped out due to refusal or migration. Two children were unable to complete the end-study activities, leaving 572 infants with complete neurodevelopmental as-	Low risk of bias Quote: "Plasma vitamin B12 and plasma folate concentration will be estimated by microbiological assays using a chloramphenicol-resistant strain of Lactobacillus casei and colistin sulfate-resistant strain of Lactobacillus leichmannii, respectively [36, 37]. We will also measure plasma concentrations of MMA and tHcy, which are good markers vitamin B12 status." Comment: Outcome measurement conforms to an appropriate and valid laboratory method. There is indication that standardised laboratory methods involving quality control measures were adopted in outcome assess-	Low risk of bias Quote: "Measure to what extent vitamin B12 administration alters plasma cobalamin, MMA, and total homocysteine (tHcy) concentrations." "The secondary outcome was the metabolic response measured by plasma concentrations of cobalamin, total homocysteine (tHcy), and methylmalonic acid (MMA)." "We will analyze the effect of vitamin B12 separately in the following subgroups: vitamin B12 status: based on cobalamin (cutoff 150 pmol/L); based on low tHcy (cutoff 10 µmol/L); based on low MMA (cutoff 0.26 µmol/L)." Comment: Recruitment for the study was from 2015-04-20 to 2024-12-28 and was registered on 2014-10-21. Based on the trial registry (NCT02272842), the outcome is pre-specified as a secondary outcome	Low risk of bias Outcome result judged to be at low risk of bias due to low risk in all domains.

Table 3. Overview of risk of bias assessments for outcomes including single studies (Continued)

produced specifically for this trial and are identical in taste and appearance. The supplements or placebos are offered in packages to the enrolled children according to a unique identification number provided by the company." "The baseline features were evenly distributed between the intervention groups. The baseline status of vitamin B₁₂ and its plasma biomarkers were also comparable between the intervention groups (S1 Table)." Comment: There is indication that a third-party scientist was responsible for generation of random sequence, suggesting the allocation sequence was concealed prior to assignment; in addition, vitamin B₁₂ and placebo supplements were similar in appearance and taste. Authors report a random number sequence generation method. There is indication from Table 1 that baseline characteristics were comparable be-	ssments. We were able to collect and analyze blood samples for biomarker assessments at end study from 567 children." Comment: Based on the trial registry (NCT02272842), blinding is reported as "Masking: Quadruple (Participant, Care Provider, Investigator, Outcomes Assessor)." In addition, there is indication from the published report that participants were unaware of assigned groups. Authors report weekly check-ins by field workers regarding supplementation intake and compliance. As a result, there is indication that the intervention was successful for ~95% of subjects. It appears that 26 infants dropped out for personal reasons and 2 participants were unable to complete study activities. It appears that reasons for loss to follow-up may be unrelated to study non-adherence.	Comment: Outcome data were available for ~95% of participants randomised at the end of the study.	ment, ensuring comparable methods across time points and treatment groups. There is indication from the trial registry that outcome assessors were blinded to treatment arms.	measure: "The Effect of Vitamin B₁₂ Supplementation on Markers of Vitamin B Status" with "Measure Description - We will draw a blood sample at study start and at end study to measure to what extent vitamin status is altered." Recruitment for the study was from 2015-04-20 to 2024-12-28 and was registered on 2014-10-21. Based on the trial registry (NCT02272842), the outcome is pre-specified as a secondary outcome measure: "The Effect of Vitamin B₁₂ Supplementation on Markers of Vitamin B Status" with "Measure Description - We will draw a blood sample at study start and at end study to measure to what extent vitamin status is altered." Outcome results reported at baseline and at 12 months appear to conform to outcome measurements specified in the study protocol and trial registry. There is minimal indication that outcome results are selected based on the results. Outcome results were analysed as mean change from baseline to endline and are consistent with the statistical analysis plan.
---	--	---	---	--

Table 3. Overview of risk of bias assessments for outcomes including single studies (Continued)

tween groups; differences in nutritional status of infants (anaemia, wasting, underweight) may be compatible with chance findings.

<i>Holotranscobalamin concentrations (pmol/L)</i>						
Areekul 1979	Some concerns	Some concerns	Low risk of bias	Low risk of bias	Some concerns	High risk of bias
	Comment: There is minimal information to permit judgement on whether the allocation sequence was concealed prior to allocation. Trial authors report no information on random sequence generation method used. There is limited information on demographic and other baseline characteristics to permit judgement on baseline differences between treatment groups.	Quote: "At the time of the study they were not suffering from any attack of malaria or other diseases. In a supplemented group which comprised of 3 children, vitamin B12 (50 µg/tablet) was given thrice a day. In the control group of 5 children, placebo was given instead of vitamin B12." Comment: Limited information on number of children randomised; thus, while there is insufficient information on attrition/exclusions, there is no clear indication that all eligible and randomised participants were analysed. Authors report no exclusion/attrition rates of eligible trial participants that may impact outcome result.	Quote: "At the time of the study they were not suffering from any attack of malaria or other diseases. In a supplemented group which comprised of 3 children, vitamin B12 (50 µg/tablet) was given thrice a day. In the control group of 5 children, placebo was given instead of vitamin B12." Comment: No information on N is provided, but since sample size was small, we could assume that results from all participants were reported. Limited	Quote: "TCs were fractionated and quantitatively measured by the method of Selhub et al. (1976. <i>Proc. Soc. Exp. Biol. Med.</i>, 152:161). The UBBC was calculated from the sum of TCI, TCII and TCIII." Comment: Laboratory method is consistent with a valid and appropriate outcome ascertainment method. The use of a standardised laboratory method ensures comparability across treatment groups and time points. There is insufficient information to permit judgement on whether outcome assessors were blinded to treatment groups.	Quote: "As the possibility of this serum vitamin B12 level from previously taking vitamin B12 could not be ruled out in some patients, the present study was therefore undertaken in order to determine serum vitamin B12 in order to determine serum vitamin B12 binding proteins in subjects giving vitamin B12 orally for 5 weeks." "Blood samples were collected every week for 6 weeks for determination of vitamin B12, unsaturated vitamin B12 binding capacity (UBBC) and vitamin B12 binding proteins (transcobalamin, TC)." Comment: No specific information is provided; however, there is indication of inclusion of outcome data in a pre-specified analysis plan as part of a follow-up of a prior study. Authors report weekly assessment of vitamin B₁₂ sta-	Outcome judged to be at high risk of bias in at least one domain.

Table 3. Overview of risk of bias assessments for outcomes including single studies (Continued)

			information on attrition/exclusion rates. However, trial authors report no missingness in the outcome suggesting that outcome data were analysed among all enrolled participants at the start of the study.	An objective laboratory method is employed. With limited influence from trial personnel-led judgement in outcome assessment, knowledge of the intervention received has minimal impact on the outcome result.	tus; however, outcome results are reported at baseline and at the end of the study. No justification is provided for reporting analysis at two time points only. Statistical analyses of outcome data are not reported. There is minimal information to permit judgement.
Any adverse effects					
Strand 2020	Low risk of bias Quote: "Randomization was concealed, and the study double-masked as the participants were only linked to the intervention through the identity number printed on the supplement labels. The list that linked this identity number to the randomization code was kept with the producers of the supplements and the independent scientist who generated it. None of the investigators had access to this list until the data collection and cleaning for the primary outcomes were completed."	Low risk of bias Quote: "Randomization was concealed, and the study double-masked as the participants were only linked to the intervention through the identity number printed on the supplement labels. The list that linked this identity number to the randomization code was kept with the producers of the supplements and the independent scientist who generated it. None of the investigators had access to this list until the data collection and cleaning for the primary outcomes were completed." "During weekly visits to the homes, field workers asked the mothers about intake of the paste during the past 7 days and recorded in detail the amount of paste given to the children (i.e., half, one-third, three-fourths, or less)."	Low risk of bias Quote: "From April 2015 to February 2017, we screened 733 infants and randomized 600 into the study (Fig 1). A total of 26 infants dropped out due to refusal or migration. Two children were unable to complete the end-study activities, leaving 572 infants with complete neurodevelopmental assessments. We were able	Some concerns Quote: "We obtain morbidity data on each child at our weekly visits. At the visits, mothers are asked about diarrheal illness (number and consistency of stools), symptoms of respiratory illness (cough, fast or difficult breathing), fever, and if there have been physician visits. Sick children who require treatment are referred to the study clinic for treatment." Trial registry (NCT02272842): "Masking: Quadruple (Participant, Care Provider, Investigator,	Some concerns Quote: "We obtain morbidity data on each child at our weekly visits. At the visits, mothers are asked about diarrheal illness (number and consistency of stools), symptoms of respiratory illness (cough, fast or difficult breathing), fever, and if there have been physician visits. Sick children who require treatment are referred to the study clinic for treatment." "For the other outcome variables (Bayley-III raw scores, growth, and hemoglobin concentrations), the effects of the intervention were estimated by comparing the changes from baseline to end study between the study groups.

Table 3. Overview of risk of bias assessments for outcomes including single studies (Continued)

"Eligible children are allocated by block randomization (block sizes of 8) in a 1:1 ratio to one of the intervention groups. The vitamin B12 supplements and placebos have been produced specifically for this trial and are identical in taste and appearance. The supplements or placebos are offered in packages to the enrolled children according to a unique identification number provided by the company." "The baseline features were evenly distributed between the intervention groups. The baseline status of vitamin B12 and its plasma biomarkers were also comparable between the intervention groups (S1 Table)." Comment: There is indication that a third-party scientist was responsible for generation of random sequence, suggesting the allocation sequence was concealed prior to assignment; in addition, vitamin B₁₂ and placebo supplements were similar in appearance and taste.	"From April 2015 to February 2017, we screened 733 infants and randomized 600 into the study (Fig 1). A total of 26 infants dropped out due to refusal or migration. Two children were unable to complete the end-study activities, leaving 572 infants with complete neurodevelopmental assessments. We were able to collect and analyze blood samples for biomarker assessments at end study from 567 children." Comment: Based on the trial registry (NCT02272842), blinding is reported as "Masking : Quadruple (Participant, Care Provider, Investigator, Outcomes Assessor)." In addition, there is indication from the published report that participants, investigators and carers delivering the intervention were unaware of assigned groups. Authors report weekly check-ins by field workers regarding supplementation intake and compliance. As a result, there is indication that the intervention was successful for ~95% of subjects. It appears that 26 infants dropped out for personal reasons and 2 participants were unable to complete study activities. It appears that reasons for loss to follow-up may be unrelated to study non-adherence.	to collect and analyze blood samples for biomarker assessments at end study from 567 children." Comment: Outcome data were available for ~95% of participants randomised at the end of the study.	Outcomes Assessor)." Comment: Adverse effects were reported as number of events resulting in hospitalisations (due to urinary tract infections, lower respiratory infections, seizure-related illness, constipation) via mothers' self-report report conforms to an appropriate method. Outcome data were collected via weekly self-report from mothers with field workers asking specific questions related to child's illness. This may result in comparable reports of illness across time points and treatment groups. There is indication from the trial registry that participants were blinded to treatment arms; however, there is limited information to ascertain whether mothers were unaware of diet groups. Since reporting of adverse effects involves some judgement, mothers' knowledge of the interven-	The precision of the effect estimates and corresponding values were calculated using the Student t test assuming equal variances." Comment: It is defined as an outcome and its methodology described in the published protocol (Strand 2017). Recruitment for the study was from 2015-04-20 to 2024-12-28 and was registered on 2014-10-21. Based on the trial registry (NCT02272842), there is minimal information regarding morbidity data in lieu of "secondary outcome measure - immediate adverse effects of the intervention". There is indication from the study protocol that adverse effects will be collected. Time frame is specified as 12 months. Outcome results reported at 12 months and appear to conform to outcome measurements specified in the study protocol and trial registry. There is minimal indication that outcome results are selected based on the results. There is limited information regarding the analysis of adverse effects (number of events resulting in hospitalisations).
--	--	--	--	---

Table 3. Overview of risk of bias assessments for outcomes including single studies *(Continued)*

Authors report a random number sequence generation method. There is indication from Table 1 that baseline characteristics were comparable between groups; differences in nutritional status of infants (anaemia, wasting, underweight) may be compatible with chance findings.	tion received could influence the severity of illness reported generally and for sick children. Morbidity data (supplementary information) suggests more frequent hospitalisations due to illness among those receiving vitamin B12 vs. placebo. However, there is no clear indication that mothers' knowledge of the intervention received influenced the frequency of illness reported or otherwise.
--	--

Risk of bias assessments were reported for outcomes synthesised narratively due to inclusion of a single study (i.e. height/length (Scaglione 1955), vitamin B₁₂ status (i.e. methylmalonic acid (Strand 2020) or holotranscobalamin (Areekul 1979) concentrations), and any adverse effects (Strand 2020)).

Table 4. Sensitivity analysis of effect estimates adjusted for intracluster correlation (ICC) in Comparison 1

Comparison 1: Supplementation with vitamin B ₁₂ alone versus placebo	ICC	Effective sample size (intervention)	Effective sample size (control)	Number of studies	Number of children	Measure of effect (95% CI)	P value for overall effect	Chi ²	I ²
1.1 Growth: height/length (cm) (ALL)									
Original sample size (unadjusted)	N/A	34	18	2	160	MD -0.16 (-2.27 to 1.96)	0.89	0.03	0%
Low ICC adjusted sample size	0.01	23	12	2	143	MD -0.19 (-2.44 to 2.06)	0.87	0.02	0%
High ICC adjusted sample size	0.05	10	5	2	123	MD -0.25 (-2.70 to 2.21)	0.85	0.01	0%
1.2 Cognitive function, using validated instruments, including general cognitive ability or cognitive domains: visuospatial function (ALL)									
Original sample size (unadjusted)	N/A	34	18	2	455	SMD -0.14 (-0.32 to 0.05)	0.15	0.90	0%
Low ICC adjusted sample size	0.01	23	12	2	438	SMD -0.13 (-0.32 to 0.06)	0.18	0.61	0%
High ICC adjusted sample size	0.05	10	5	2	418	SMD -0.12 (-0.31 to 0.08)	0.24	0.24	0%
1.3 Cognitive function, using validated instruments, including general cognitive ability or cognitive domains: working memory (ALL)									
Original sample size (unadjusted)	N/A	34	18	2	455	SMD 0.01 (-0.18 to 0.19)	0.93	1.18	15%
Low ICC adjusted sample size	0.01	23	12	2	438	SMD 0.00 (-0.19 to 0.19)	0.98	0.81	0%
High ICC adjusted sample size	0.05	10	5	2	418	SMD -0.02 (-0.21 to 0.18)	0.87	0.33	0%
1.4 Cognitive function, using validated instruments, including general cognitive ability or cognitive domains: attention (ALL)									
Original sample size (unadjusted)	N/A	34	18	2	455	SMD 0.08 (-0.11 to 0.26)	0.40	1.58	37%
Low ICC adjusted sample size	0.01	23	12	2	438	SMD 0.09 (-0.10 to 0.28)	0.34	1.08	7%

Table 4. Sensitivity analysis of effect estimates adjusted for intracluster correlation (ICC) in Comparison 1 (Continued)

High ICC adjusted sample size	0.05	10	5	2	418	SMD 0.11 (-0.08 to 0.30)	0.27	0.45	0%
Cognitive function, using validated instruments, including general cognitive ability or cognitive domains: intelligence quotient									
Original sample size (unadjusted)	N/A	34	18	1	52	N/A	N/A	N/A	N/A
Low ICC adjusted sample size	0.01	23	12	1	35	N/A	N/A	N/A	N/A
High ICC adjusted sample size	0.05	10	5	1	15	N/A	N/A	N/A	N/A
1.5 Vitamin B₁₂ status: vitamin B₁₂ deficiency, as defined by trial authors (e.g. < 148, < 221 pmol/L) (ALL)									
Original sample size (unadjusted)	N/A	34	18	3	659	RR 0.27 (0.11 to 0.69)	0.006	3.01	33%
Low ICC adjusted sample size	0.01	23	12	3	642	RR 0.27 (0.11 to 0.69)	0.006	3.02	34%
High ICC adjusted sample size	0.05	10	5	3	622	RR 0.23 (0.09 to 0.61)	0.003	1.66	40%
1.7 Vitamin B₁₂ status: total vitamin B₁₂ concentrations (pmol/L) (ALL)									
Original sample size (unadjusted)	N/A	34	18	7	722	MD 67.52 (41.18 to 93.85)	< 0.001	16.42	63%
Low ICC adjusted sample size	0.01	23	12	7	705	MD 68.53 (41.83 to 95.23)	< 0.001	16.08	63%
High ICC adjusted sample size	0.05	10	5	7	685	MD 70.45 (43.07 to 97.84)	< 0.001	15.67	62%
1.11 Vitamin B₁₂ status: homocysteine concentrations (μmol/L) (ALL)									
Original sample size (unadjusted)	N/A	12	12	3	262	MD -4.02 (-6.15 to -1.89)	< 0.001	7.09	72%
Low ICC adjusted sample size	0.01	8	8	3	254	MD -3.97 (-6.16 to -1.78)	< 0.001	6.72	70%
High ICC adjusted sample size	0.05	3	3	3	244	MD -3.85 (-6.17 to -1.53)	0.001	6.24	68%
1.13 Anaemia and haemoglobin: anaemia, defined using age-specific cut-offs (ALL)									

Table 4. Sensitivity analysis of effect estimates adjusted for intracluster correlation (ICC) in Comparison 1 *(Continued)*

Original sample size (unadjusted)	N/A	34	18	2	161	RR 1.37 (0.46 to 4.11)	0.58	0.66	0%
Low ICC adjusted sample size	0.01	23	12	2	144	RR 1.48 (0.41 to 5.27)	0.55	0.60	0%
High ICC adjusted sample size	0.05	10	5	2	124	RR 1.71 (0.36 to 8.04)	0.50	0.50	0%
1.17 Megaloblastic anaemia, as defined by trial authors (ALL)									
Original sample size (unadjusted)	N/A	34	18	2	161	N/A	N/A	N/A	N/A
Low ICC adjusted sample size	0.01	23	12	2	144	N/A	N/A	N/A	N/A
High ICC adjusted sample size	0.05	10	5	2	124	N/A	N/A	N/A	N/A
Body composition, as defined by trial authors									
Original sample size (unadjusted)	N/A	33	18	1	51	N/A	N/A	N/A	N/A
Low ICC adjusted sample size	0.01	22	12	1	34	N/A	N/A	N/A	N/A
High ICC adjusted sample size	0.05	10	5	1	15	N/A	N/A	N/A	N/A

Abbreviations

CI: confidence interval; **ICC:** intracluster correlation coefficient; **MD:** mean difference; **N/A:** not applicable; **RR:** risk ratio; **SMD:** standardised mean difference.

Table 5. Sensitivity analysis of effect estimates adjusted for intracluster correlation (ICC) in Comparison 3

Comparison 3: Supplementation with vitamin B ₁₂ and other micronutrients, compared to supplements with the same formulation without vitamin B ₁₂	ICC	Effective sample size (intervention)	Effective sample size (control)	Number of studies	Number of children	Measure of effect (95% CI)	P value for overall effect	Chi ²	I ²
2.1 Growth: height/length (cm) (ALL)									
Original sample size (unadjusted)	N/A	35	18	2	626	MD -0.24 (-0.66 to 0.19)	0.27	1.75	43%
Low ICC adjusted sample size	0.01	23	12	2	608	MD -0.22 (-0.65 to 0.20)	0.30	1.16	14%

Table 5. Sensitivity analysis of effect estimates adjusted for intracluster correlation (ICC) in Comparison 3 *(Continued)*

High ICC adjusted sample size	0.05	10	5	2	588	MD -0.21 (-0.64 to 0.21)	0.33	0.50	0%
2.4 Cognitive function, using validated instruments, including general cognitive ability or cognitive domains: visuospatial function (ALL)									
Original sample size (unadjusted)	N/A	35	18	3	974	SMD 0.00 (-0.12 to 0.13)	0.98	0.48	0%
Low ICC adjusted sample size	0.01	23	12	3	956	SMD 0.00 (-0.12 to 0.13)	0.95	0.41	0%
High ICC adjusted sample size	0.05	10	5	3	936	SMD 0.01 (-0.12 to 0.13)	0.92	0.34	0%
2.7 Cognitive function, using validated instruments, including general cognitive ability or cognitive domains: working memory (ALL)									
Original sample size (unadjusted)	N/A	35	18	3	974	SMD -0.17 (-0.29 to -0.04)	0.009	3.72	46%
Low ICC adjusted sample size	0.01	23	12	3	956	SMD -0.17 (-0.30 to -0.04)	0.01	3.72	46%
High ICC adjusted sample size	0.05	10	5	3	936	SMD -0.17 (-0.30 to -0.04)	0.01	3.71	46%
2.10 Cognitive function, using validated instruments, including general cognitive ability or cognitive domains: attention (ALL)									
Original sample size (unadjusted)	N/A	35	18	3	974	SMD 0.00 (-0.13 to 0.12)	0.97	0.88	0%
Low ICC adjusted sample size	0.01	23	12	3	956	SMD 0.00 (-0.13 to 0.13)	0.98	0.63	0%
High ICC adjusted sample size	0.05	10	5	3	936	SMD 0.01 (-0.12 to 0.13)	0.92	0.34	0%
2.13 Cognitive function, using validated instruments, including general cognitive ability or cognitive domains: intelligence quotient (ALL)									
Original sample size (unadjusted)	N/A	35	18	2	586	SMD -0.07 (-0.23 to 0.10)	0.43	0.04	0%
Low ICC adjusted sample size	0.01	23	12	2	568	SMD -0.07 (-0.23 to 0.10)	0.42	0.03	0%

Table 5. Sensitivity analysis of effect estimates adjusted for intracluster correlation (ICC) in Comparison 3 (Continued)

High ICC adjusted sample size	0.05	10	5	2	548	SMD -0.07 (-0.24 to 0.10)	0.42	0.01	0%
2.19 Vitamin B₁₂ status: vitamin B₁₂ deficiency, as defined by trial authors (e.g. < 148, < 221 pmol/L) (ALL)									
Original sample size (unadjusted)	N/A	35	18	2	552	RR 0.96 (0.15 to 6.19)	0.96	0.73	0%
Low ICC adjusted sample size	0.01	23	12	2	534	RR 0.77 (0.12 to 5.16)	0.79	0.35	0%
High ICC adjusted sample size	0.05	10	5	2	514	RR 0.78 (0.12 to 5.15)	0.80	0.36	0%
2.20 Vitamin B₁₂ status: total vitamin B₁₂ concentrations (pmol/L) (ALL)									
Original sample size (unadjusted)	N/A	35	18	3	749	MD 32.15 (-35.51 to 99.82)	0.35	11.27	82%
Low ICC adjusted sample size	0.01	23	12	3	731	MD 34.32 (-31.13 to 99.77)	0.30	8.81	77%
High ICC adjusted sample size	0.05	10	5	3	711	MD 42.21 (-18.50 to 102.91)	0.17	5.54	64%
2.23 Vitamin B₁₂ status: homocysteine concentrations (μmol/L) (ALL)									
Original sample size (unadjusted)	N/A	12	12	3	724	MD -1.41 (-2.74 to -0.08)	0.04	9.18	78%
Low ICC adjusted sample size	0.01	8	8	3	716	MD -1.35 (-2.68 to -0.01)	0.05	8.86	77%
High ICC adjusted sample size	0.05	3	3	3	706	MD -1.25 (-2.60 to 0.09)	0.07	8.45	76%
Anaemia and haemoglobin: anaemia, defined using age-specific cut-offs									
Original sample size (unadjusted)	N/A	35	18	1	53	N/A	N/A	N/A	N/A
Low ICC adjusted sample size	0.01	23	12	1	35	N/A	N/A	N/A	N/A
High ICC adjusted sample size	0.05	10	5	1	15	N/A	N/A	N/A	N/A
Megaloblastic anaemia, as defined by trial authors									

Table 5. Sensitivity analysis of effect estimates adjusted for intracluster correlation (ICC) in Comparison 3 *(Continued)*

Original sample size (unadjusted)	N/A	35	18	1	53	N/A	N/A	N/A	N/A
Low ICC adjusted sample size	0.01	23	12	1	35	N/A	N/A	N/A	N/A
High ICC adjusted sample size	0.05	10	5	1	15	N/A	N/A	N/A	N/A
Body composition, as defined by trial authors									
Original sample size (unadjusted)	N/A	35	17	1	52	N/A	N/A	N/A	N/A
Low ICC adjusted sample size	0.01	23	11	1	34	N/A	N/A	N/A	N/A
High ICC adjusted sample size	0.05	10	5	1	15	N/A	N/A	N/A	N/A

Abbreviations

CI: confidence interval; **ICC:** intracluster correlation coefficient; **MD:** mean difference; **N/A:** not applicable; **RR:** risk ratio; **SMD:** standardised mean difference.

Table 6. Sensitivity analysis using a fixed-effect model, or excluding studies with high risk of bias or cluster RCT(s): Outcome 1.7 (Vitamin B₁₂ status: total vitamin B₁₂ concentrations (pmol/L) (ALL))

Comparison 1: Supplementation with vitamin B ₁₂ alone versus placebo	Number of studies	Number of children	Mean difference (95% CI)	P value for overall effect	Chi ²	I ²
1.7 Vitamin B₁₂ status: total vitamin B₁₂ concentrations (pmol/L)						
All studies	7	705	68.53 (41.83 to 95.23)	< 0.001	16.08	63%
Excluding studies with high risk of bias	3	273	76.62 (37.64 to 115.60)	0.25	2.80	29%
Using a fixed-effect model	7	705	58.82 (46.02 to 71.63)	0.01	16.08	63%

Abbreviations

CI: confidence interval; MD: mean difference; RCT(s): randomised controlled trial(s).

Table 7. Sensitivity analysis using a fixed-effect model or excluding cluster RCT(s): Outcome 2.20 (Vitamin B₁₂ status: total vitamin B₁₂ concentrations (pmol/L) (ALL))

Comparison 3: Supplementation with vitamin B ₁₂ and other micronutrients, compared to supplements with the same formulation without vitamin B ₁₂	Number of studies	Number of children	Mean difference (95% CI)	P value for overall effect	Chi ²	I ²
2.20 Vitamin B₁₂ status: total vitamin B₁₂ concentrations (pmol/L) (ALL)						
All studies	3	731	MD 34.32 (-31.13 to 99.77)	0.30	8.81	77%
Using a fixed-effect model	3	731	MD 35.90 (10.12 to 61.67)	0.006	8.81	77%

Abbreviations

CI: confidence interval; MD: mean difference; RCT(s): randomised controlled trial(s).